JD9366TC

User Guide

1600RGB x 2560 dots, 16.7M color,
with MIPI interface,TFT LCD One/Two Chip Driver with Touch

Preliminary Version 0.01
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1. Revision History

Version	Date	Description of modification
0.00	2021/04/13	New setup
0.01	2022/09/16	Add OTP program Flow in P.251~252 .

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2. Jadard Command List

2.1. Page 1 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
40	DUM_REG	W/R	dummy_00[7:0]								00
		W/R	dummy_01[7:0]								00
		W/R	dummy_02[7:0]								00
		W/R	dummy_03[7:0]								00
		W/R	dummy_04[7:0]								00
		W/R	dummy_05[7:0]								00
		W/R	dummy_06[7:0]								00
		W/R	dummy_07[7:0]								00
		W/R	dummy_08[7:0]								00
		W/R	dummy_09[7:0]								00
41	DUM_REG	W/R	dummy_10[7:0]								00
		W/R	dummy_11[7:0]								00
		W/R	dummy_12[7:0]								00
		W/R	dummy_13[7:0]								00
		W/R	dummy_14[7:0]								00
		W/R	dummy_15[7:0]								00
		W/R	dummy_16[7:0]								00
		W/R	dummy_17[7:0]								00
		W/R	dummy_18[7:0]								00
		W/R	dummy_19[7:0]								00
42	DUM_REG	W/R	dummy_20[7:0]								00
		W/R	dummy_21[7:0]								00
		W/R	dummy_22[7:0]								00
		W/R	dummy_23[7:0]								00
		W/R	dummy_24[7:0]								00
		W/R	dummy_25[7:0]								00
		W/R	dummy_26[7:0]								00
		W/R	dummy_27[7:0]								00
		W/R	dummy_28[7:0]								00
		W/R	dummy_29[7:0]								00
43	DUM_REG	W/R	dummy_30[7:0]								00
		W/R	dummy_31[7:0]								00
		W/R	dummy_32[7:0]								00
		W/R	dummy_33[7:0]								00
		W/R	dummy_34[7:0]								00
		W/R	dummy_35[7:0]								00
		W/R	dummy_36[7:0]								00
		W/R	dummy_37[7:0]								00

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		W/R	dummy_38[7:0]							00
		W/R	dummy_39[7:0]							00
44	DUM_REG	W/R	dummy_40[7:0]							00
		W/R	dummy_41[7:0]							00
		W/R	dummy_42[7:0]							00
		W/R	dummy_43[7:0]							00
		W/R	dummy_44[7:0]							00
		W/R	dummy_45[7:0]							00
		W/R	dummy_46[7:0]							00
		W/R	dummy_47[7:0]							00
		W/R	dummy_48[7:0]							00
		W/R	dummy_49[7:0]							00
45	DUM_REG	W/R	SA_Code1[7:0]							00
		W/R	SA_Code2[7:0]							00
		W/R	ddb0[7:0]							00
		W/R	ddb0[7:0]							00
		W/R	ddb0[7:0]							00
		W/R	ddb0[7:0]							00
46		W/R	-	-	-	-	-	-	r_crc_interface_en	00
47		R	crc_interface_o[7:0]							00
70		W/R	touch_id[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
		W/R	lot_id1[7:0]							00
71		W/R	-	-	-	-	-	-	otp_rd_req	00
72		W/R	-	-	-	-	-	-	otp_auto_prog	00
73		W/R	-	-	-	-	-	-	spi_addrx_bit7	00
78		W/R	reg_key1[7:0]							00
		W/R	reg_key2[7:0]							00
		W/R	reg_key3[7:0]							00
		W/R	reg_key41[7:0]							00

Table 2.1: Page 1 command list

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2.2. Page 2 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	ckv_width[3;0]				stv-width[3:0]				22
32		W/R	r_flc_single_cnt_en	r_force_flc_inv_en	r_learn_len_sel	phase[4:0]					0C
33		W/R	rst_width[3:0]				grst_dly_sft[3:0]				30
3B		W/R	ckv_blkpulse_stvsp_fall_dis[3:0]				ckv_blkpulse_stvsp_rise_dis[3:0]				3C
3C		W/R	ckv_blkpulse_phase_sel	ckv_blkpulse_stagger_sel[1:0]		dummy_sel[4:0]					00
3D		W/R	gflc[7:0]								08
3E		W/R	flc_trans_stvsp_dis [3:0]				flc_stag_trans_stvsp_dis[3:0]				67
3F		W/R	flc_hole_vgl_dly_sel[1:0]		gip_updn_en	r_learn_type	blk_post_dmy_num[3:0]				60
40		W/R	sub_flc_dis[1:0]		eq_rise[2:0]			eq_fall[2:0]			00
41		W/R	-	gstv_rise_o[6:0]							00
		W/R	-	gstv_rise_e[6:0]							00
		W/R	-	gstv_fall_o[6:0]							20
		W/R	-	gstv_fall_e[6:0]							20
		W/R	-	gckv_rise_o[6:0]							00
		W/R	-	gckv_rise_e[6:0]							00
		W/R	-	gckv_fall_o[6:0]							20
		W/R	-	gckv_fall_e[6:0]							20
		W/R	-	grst_rise_o[6:0]							00
		W/R	-	grst_rise_e[6:0]							00
		W/R	-	grst_fall_o[6:0]							20
		W/R	-	grst_fall_e[6:0]							20
42		W/R	esd_clr_lastline_en	tianma_sel	dummy_en	en_ckv_all	sft_goe_scale	goa_fall_sft_drct	goa_rise_sft_drct	flc_stagger_en	80
43		W/R	r_stv_ckv_dis_ext[1:0]		stv_abn_clr	ckv_abn_clr	esd_clr_goa_en	rst_cp_tianma[2:0]			30
44		W/R	esd_rst_sel	r_eq_stv_rstb_enb	stv_ckv_dis	stv_ckv_select	ckv_blkpulse_en	goa_msk_sel[1:0]		learn_info_sel	01
45		W/R	ag_act_length[7:0]								80
46		W/R	Dummy[2:0]			gip_probe_ext_en	ag_act_length[11:8]				07
47		W/R	ag_vblk_length[7:0]								37
48		W/R	vblk_line_learn_num[3:0]				ag_vblk_length[11:8]				30
49		W/R	r_vblk_grst_msk	r_post_dmy_msk[3:0]				r_grst_adv_sft[2:0]			84
4A		W/R	poff_all_gate_on_en	gip_rising_eq_type	gip_sel_eq_en_r[1:0]		gip_sel_eq_en_l[1:0]		gip_pweq_en_r	gip_pweq_en_l	BF
4B		W/R	gip_dr2_r[1:0]		gip_dr2_l[1:0]		gip_dr1_r[1:0]		gip_dr1_l[1:0]		AA
4C		W/R	Dummy	gip_tp_sen_l[6:0]							00
		W/R	Dummy	gip_tp_sen_r[6:0]							00
4E		W/R	Dummy	gip_lpwg_idle_l[6:0]							00
		W/R	Dummy	gip_lpwg_idle_r[6:0]							00
50		W/R	Dummy	gip_lpwg_sen_l[6:0]							00
		W/R	Dummy	gip_lpwg_sen_r[6:0]							00
55		W/R	Dummy	gip_lpwg_xon_gnd_l[6:0]							00
		W/R	Dummy	gip_lpwg_xon_gnd_r[6:0]							00

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56		W/R	Dummy	gip_lpwg_xon_hl_l[6:0]							**00**
		W/R	Dummy	gip_lpwg_xon_hl_r[6:0]							**00**
57		W/R	Dummy	gip_slpin_gnd_l[6:0]							**00**
		W/R	Dummy	gip_slpin_gnd_r[6:0]							**00**
58		W/R	Dummy	gip_slpin_hl_l[6:0]							**7F**
		W/R	Dummy	gip_slpin_hl_r[6:0]							**7F**
59		W/R	flc_hole_vgh_dly_sel[1:0]	gip_igzo_intsw_r	gip_igzo_intsw_l	gip_igzo_intdis_r	gip_igzo_intdis_l	gip_igzo_en_r	gip_igzo_en_l		**30**
5A		W/R	-	-	forward_goutl1_sel[5:0]						**23**
		W/R	-	-	forward_goutl2_sel[5:0]						**23**
		W/R	-	-	forward_goutl3_sel[5:0]						**23**
		W/R	-	-	forward_goutl4_sel[5:0]						**23**
		W/R	-	-	forward_goutl5_sel[5:0]						**23**
		W/R	-	-	forward_goutl6_sel[5:0]						**23**
		W/R	-	-	forward_goutl7_sel[5:0]						**23**
		W/R	-	-	forward_goutl8_sel[5:0]						**23**
		W/R	-	-	forward_goutl9_sel[5:0]						**23**
		W/R	-	-	forward_goutl10_sel[5:0]						**23**
		W/R	-	-	forward_goutl11_sel[5:0]						**23**
5B		W/R	-	-	forward_goutl12_sel[5:0]						**23**
		W/R	-	-	forward_goutl13_sel[5:0]						**23**
		W/R	-	-	forward_goutl14_sel[5:0]						**23**
		W/R	-	-	forward_goutl15_sel[5:0]						**23**
		W/R	-	-	forward_goutl16_sel[5:0]						**23**
		W/R	-	-	forward_goutl17_sel[5:0]						**23**
		W/R	-	-	forward_goutl18_sel[5:0]						**23**
		W/R	-	-	forward_goutl19_sel[5:0]						**23**
		W/R	-	-	forward_goutl20_sel[5:0]						**23**
		W/R	-	-	forward_goutl21_sel[5:0]						**23**
		W/R	-	-	forward_goutl22_sel[5:0]						**23**
5C		W/R	-	-	forward_goutr1_sel[5:0]						**23**
		W/R	-	-	forward_goutr2_sel[5:0]						**23**
		W/R	-	-	forward_goutr3_sel[5:0]						**23**
		W/R	-	-	forward_goutr4_sel[5:0]						**23**
		W/R	-	-	forward_goutr5_sel[5:0]						**23**
		W/R	-	-	forward_goutr6_sel[5:0]						**23**
		W/R	-	-	forward_goutr7_sel[5:0]						**23**
		W/R	-	-	forward_goutr8_sel[5:0]						**23**
		W/R	-	-	forward_goutr9_sel[5:0]						**23**
		W/R	-	-	forward_goutr10_sel[5:0]						**23**
		W/R	-	-	forward_goutr11_sel[5:0]						**23**
5D		W/R	-	-	forward_goutr12_sel[5:0]						**23**

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		W/R	-	-	forward_goutr13_sel[5:0]	23
		W/R	-	-	forward_goutr14_sel[5:0]	23
		W/R	-	-	forward_goutr15_sel[5:0]	23
		W/R	-	-	forward_goutr16_sel[5:0]	23
		W/R	-	-	forward_goutr17_sel[5:0]	23
		W/R	-	-	forward_goutr18_sel[5:0]	23
		W/R	-	-	forward_goutr19_sel[5:0]	23
		W/R	-	-	forward_goutr20_sel[5:0]	23
		W/R	-	-	forward_goutr21_sel[5:0]	23
		W/R	-	-	forward_goutr22_sel[5:0]	23
5E		W/R	-	-	backward_goutl1_sel[5:0]	23
		W/R	-	-	backward_goutl2_sel[5:0]	23
		W/R	-	-	backward_goutl3_sel[5:0]	23
		W/R	-	-	backward_goutl4_sel[5:0]	23
		W/R	-	-	backward_goutl5_sel[5:0]	23
		W/R	-	-	backward_goutl6_sel[5:0]	23
		W/R	-	-	backward_goutl7_sel[5:0]	23
		W/R	-	-	backward_goutl8_sel[5:0]	23
		W/R	-	-	backward_goutl9_sel[5:0]	23
		W/R	-	-	backward_goutl10_sel[5:0]	23
		W/R	-	-	backward_goutl11_sel[5:0]	23
5F		W/R	-	-	backward_goutl12_sel[5:0]	23
		W/R	-	-	backward_goutl13_sel[5:0]	23
		W/R	-	-	backward_goutl14_sel[5:0]	23
		W/R	-	-	backward_goutl15_sel[5:0]	23
		W/R	-	-	backward_goutl16_sel[5:0]	23
		W/R	-	-	backward_goutl17_sel[5:0]	23
		W/R	-	-	backward_goutl18_sel[5:0]	23
		W/R	-	-	backward_goutl19_sel[5:0]	23
		W/R	-	-	backward_goutl20_sel[5:0]	23
		W/R	-	-	backward_goutl21_sel[5:0]	23
		W/R	-	-	backward_goutl22_sel[5:0]	23
60		W/R	-	-	backward_goutr1_sel[5:0]	23
		W/R	-	-	backward_goutr2_sel[5:0]	23
		W/R	-	-	backward_goutr3_sel[5:0]	23
		W/R	-	-	backward_goutr4_sel[5:0]	23
		W/R	-	-	backward_goutr5_sel[5:0]	23
		W/R	-	-	backward_goutr6_sel[5:0]	23
		W/R	-	-	backward_goutr7_sel[5:0]	23
		W/R	-	-	backward_goutr8_sel[5:0]	23
		W/R	-	-	backward_goutr9_sel[5:0]	23

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		W/R	-	-	backward_goutr10_sel[5:0]	**23**
		W/R	-	-	backward_goutr11_sel[5:0]	**23**
61		W/R	-	-	backward_goutr12_sel[5:0]	**23**
		W/R	-	-	backward_goutr13_sel[5:0]	**23**
		W/R	-	-	backward_goutr14_sel[5:0]	**23**
		W/R	-	-	backward_goutr15_sel[5:0]	**23**
		W/R	-	-	backward_goutr16_sel[5:0]	**23**
		W/R	-	-	backward_goutr17_sel[5:0]	**23**
		W/R	-	-	backward_goutr18_sel[5:0]	**23**
		W/R	-	-	backward_goutr19_sel[5:0]	**23**
		W/R	-	-	backward_goutr20_sel[5:0]	**23**
		W/R	-	-	backward_goutr21_sel[5:0]	**23**
		W/R	-	-	backward_goutr22_sel[5:0]	**23**
62		W/R	gcl_mask_num[3:0]		hole_ckv_front_dmy[3:0]	**00**
63		W/R			hole_ckv_back_dmy[3:0]	**00**
64		W/R	00000000			**00**
65		W/R	11111111			**FF**

Table 2.2: Page 2 command list

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2.3. Page 3 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	probe_sel[7:0]								00
3A		R	core_state[7:0]								00
3B		W/R	r_lb_mbist_w_setup_sel	r_lb_mbist_r_setup_sel	r_lb_mbist_r_hold_sel[1:0]		r_lb_debug_sub_ram_sel[3:0]				D0
3C		W/R	treg[7:0]								00
3D		W/R	-	-	-	-	-	-	-	r_lb_mbist_auto_en	00
3E		W/R	-	-	-	r_lb_mbist_pattern_sel[12:8]					00
		W/R	r_lb_mbist_pattern_sel[7:0]								00
3F		W/R	r_lb_mbist_ext_pat[47:40]								00
		W/R	r_lb_mbist_ext_pat[39:32]								00
		W/R	r_lb_mbist_ext_pat[31:24]								00
		W/R	r_lb_mbist_ext_pat[23:16]								00
		W/R	r_lb_mbist_ext_pat[15:8]								00
		W/R	r_lb_mbist_ext_pat[7:0]								00
40		W/R	-	-	-	r_lb_debug_adx_sel[4:0]					00
41		W/R	-	-	r_lb_debug_ady_sel[7:0]						00
42		W/R	-	-	-	-	-	-	r_lb_debug_read_trig	r_lb_debug_write_trig	00
43		R	repaire_wline_num_0[7:0]								00
44		R	repaire_wline_num_1[7:0]								00
45		R	debug_sram_rd_data_a[47:40]								00
		R	debug_sram_rd_data_a[39:32]								00
		R	debug_sram_rd_data_a[31:24]								00
		R	debug_sram_rd_data_a[23:16]								00
		R	debug_sram_rd_data_a[15:8]								00
		R	debug_sram_rd_data_a[7:0]								00
46		R	debug_sram_rd_data_b[47:40]								00
		R	debug_sram_rd_data_b[39:32]								00
		R	debug_sram_rd_data_b[31:24]								00
		R	debug_sram_rd_data_b[23:16]								00
		R	debug_sram_rd_data_b[15:8]								00
		R	debug_sram_rd_data_b[7:0]								00
47		R	debug_sram_rd_data_c[47:40]								00
		R	debug_sram_rd_data_c[39:32]								00
		R	debug_sram_rd_data_c[31:24]								00
		R	debug_sram_rd_data_c[23:16]								00
		R	debug_sram_rd_data_c[15:8]								00
		R	debug_sram_rd_data_c[7:0]								00
48		R	debug_sram_rd_data_d[47:40]								00
		R	debug_sram_rd_data_d[39:32]								00

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		R	debug_sram_rd_data_d[31:24]								00
		R	debug_sram_rd_data_d[23:16]								00
		R	debug_sram_rd_data_d[15:8]								00
		R	debug_sram_rd_data_d[7:0]								00
49		W/R	-	-	-	r_replace_probe_en	r_gpio_probe_sel_1[1:0]	r_gpio_probe_sel_0[1:0]			04
4A		W/R	r_te1_probe_sel[3:0]				r_te_probe_sel[3:0]				00
4B		W/R	-	-	-	-	r_pwm_probe[3:0]				00
55		W/R	-	-	-	-	-	cmd_wr_s_en	cmd_wr_m_en	spi_rd_s	00
56		W/R	otp_mask[7:0]								00
57		W/R	-	-	-	-	-	otp_pa[10:8]			00
		W/R	otp_pa[7:0]								00
58		W/R	otp_all_wr	otp_pgm_bit[1:0]		otp_single_rd	otp_single_wr	otp_load_dis	otp_int_vpp	0	00
59		R	otp_single_rd_data[7:0]								00
5A		W/R	otp_single_wr_data[7:0]								00
5B		W/R	reg_ptr	reg_ptc	reg_pwe	reg_pclk	reg_pprog	reg_pce	reg_pdstb	reg_pdin	00
5C		W/R	otp_sw_sel	0	sw_pprog	reg_paio[2:0]			reg_pas[1:0]		00
5D		W/R	otp_pwr_rdy_bypass	otp_tvps_sel[1:0]		otp_pg_speedup	reg_ptm[3:0]				00
5E		W/R	pg_width_sel[1:0]		rd_clk_width[2:0]			otp_sw	standby_reload_en	0	58

Table 2.3: Page 3 command list

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2.4. Page 4 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	-	rg_edg_sp_prot	rg_sat_sp_prot	rg_hue_en	ce_sat_mode[1:0]		ce_edg_mode_en_pre	ce_sat_mode_en_pre	7C
32		W/R	rg_ce_SG016[3:0]				rg_ce_SG000[3:0]				A0
		W/R	rg_ce_SG048[3:0]				rg_ce_SG032[3:0]				FE
		W/R	rg_ce_SG080[3:0]				rg_ce_SG064[3:0]				EE
		W/R	rg_ce_SG112[3:0]				rg_ce_SG096[3:0]				CD
		W/R	rg_ce_SG144[3:0]				rg_ce_SG128[3:0]				BC
		W/R	rg_ce_SG176[3:0]				rg_ce_SG160[3:0]				9A
		W/R	re_ce_SG208[3:0]				rg_ce_SG192[3:0]				67
		W/R	rg_ce_SG240[3:0]				rg_ce_SG224[3:0]				34
		W/R	-	-	-	-	rg_ce_SG255[3:0]				00
33		W/R	-	-	-	rg_ce_THCOLOR2[2:0]			rg_col_en	rg_ce_vv_en	01
		W/R	-	-	rg_ce_THREF[2:0]			rg_ce_THSAT[2:0]			25
		W/R	-	-	rg_ce_THGREY[2:0]			rg_ce_THCOLOR[2:0]			04
		W/R	rg_ce_SAT_OFFS[7:0]								08
		W/R	rg_ce_REF_OFFS[7:0]								0A
		W/R	rg_ce_COL_OFFs[7:0]								00
3B		W/R	-	-	rg_ce_HUE_TBL00[5:0]						1D
		W/R	-	-	rg_ce_HUE_TBL01[5:0]						1F
		W/R	-	-	rg_ce_HUE_TBL02[5:0]						1F
		W/R	-	-	rg_ce_HUE_TBL03[5:0]						20
		W/R	-	-	rg_ce_HUE_TBL04[5:0]						21
		W/R	-	-	rg_ce_HUE_TBL05[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL06[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL07[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL08[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL09[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL10[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL11[5:0]						22
3C		W/R	-	-	rg_ce_HUE_TBL12[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL13[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL14[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL15[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL16[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL17[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL18[5:0]						22
		W/R	-	-	rg_ce_HUE_TBL19[5:0]						20
		W/R	-	-	rg_ce_HUE_TBL20[5:0]						20
		W/R	-	-	rg_ce_HUE_TBL21[5:0]						20
		W/R	-	-	rg_ce_HUE_TBL22[5:0]						1F

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		W/R	-	-	rg_ce_HUE_TBL23[5:0]						1E
3D		W/R	rg_ce_EDG_FAC2[3:0]				rg_ce_EDG_FAC1[3:0]				85
3E		W/R	-	-	-	rg_xy_fac_mode	rg_roi_auto	rg_roi_en	rg_ce_dim_reset	rg_ce_sat_dim_en	09
3F		W/R	rg_ce_sat_DLT_FD[3:0]				rg_ce_sat_FPS_FD[3:0]				22
40		W/R	rg_bk_color[7:0]								00
		W/R	rg_bk_color[15:8]								00
		W/R	rg_bk_color[23:16]								00
41		W/R	rg_EN_TC_AMS_DET	rg_EN_LE_AMS_DE	rg_EN_CE_AMS_DET	0	rg_PAT_MODE_SEL[3:0]				EF
		W/R	rg_SP_SATDIFF_TH[7:0]								05
		W/R	rg_SP_WHITE_TH[7:0]								FA
		W/R	rg_SP_COLOR_TH[7:0]								05
		W/R	rg_SP_GREY_TH[7:0]								05
		W/R	rg_SP_SWCGP_NUM_TH[7:0]								FB
42		W/R	-	-	-	-	-	rg_AMMD_SC_CNT_RESET	rg_AMMD_SMP_FNUM_RESET	rg_AMS_PREF_VMAX_RESET	00
		W/R	rg_AMMD_SMP_FNUM[7:0]								05
		W/R	-	-	rg_AMMD_SC_CNT_UBTH[5:0]						08
		W/R	-	-	rg_AMMD_SC_CNT_LBTH[5:0]						05
		W/R	rg_AMMD_SC_DEC_STEP[3:0]				rg_AMMD_SC_INC_STEP[3:0]				46
		W/R	rg_AMMD_VMAX_SUM_TH[7:0]								01
		W/R	rg_AMMD_VMAX_SUM_TH[15:8]								00
43		W/R	rg_x_fac_m1[7:0]								00
		W/R	rg_y_fac_m1[7:0]								00
44		W/R	rg_ce_gating_en[7:0]								FF
45		W/R	rg_ce_gate_selA[7:0]								FF
46		W/R	rg_ce_gate_selB[7:0]								FF
47		W/R	rg_ce_gate_selC[7:0]								FF
48		W/R	rg_ce_gate_selD[7:0]								FF
49		W/R	rg_ce_ctrl_sel[7:0]								1B
4A		W/R	rg_BinF_IP_BYPASS	0	rg_dbg_ce_en	rg_dbg_mux_sel[4:0]					00
4B		W/R	-	-	-	-	rg_LE_DIM_DBIT[1:0]	rg_LE_MAP_FUN[1:0]			01
		W/R	rg_LE_L_BOUND[7:0]								00
		W/R	rg_LE_U_BOUND[7:0]								FF
4C		W/R	00000000								00
4E		W/R	00000000								00
50		R	ce_top_in_chksum1[7:0]								00
		R	ce_top_out_chksum1[7:0]								00
		R	ColEng2LE_TP_type[1:0]	ro_ColEng_AMS_DET_idx[5:0]							00
		R	ColEng2LE_mean[7:0]								00
		R	ce_top_in_chksum2[7:0]								00
		R	ce_top_out_chksum2[7:0]								00

Table 2.4: Page 4 command list

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2.5. Page 5 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	-	-	RG_BL	RG_DD	RG_BCTRL	RG_2FRM_DIMO_ENB	RG_USER_PWM_WT	RG_SLR_AMB_CTL	39
32		W/R	RG_WRCABCMB[7:0]								00
33		W/R	RG_WRDISBV[7:0]								FF
3B		W/R	RG_AMB_IN[31:24]								00
		W/R	RG_AMB_IN[23:16]								00
		W/R	RG_AMB_IN[15:8]								00
		W/R	RG_AMB_IN[7:0]								00
3C		W/R	-	-	-	RG_BE_STR000[4:0]					00
		W/R	-	-	-	RG_BE_STR064[4:0]					00
		W/R	-	-	-	RG_BE_STR128[4:0]					00
		W/R	-	-	-	RG_BE_STR192[4:0]					00
		W/R	-	-	-	RG_BE_STR255[4:0]					00
3D		W/R	-	RG_CABC_InHouse_	RG_CABC_TP_BLK_	RG_SLRH_UI_REF	RG_SLRH_REF[1:0]		RG_SLRH_EN	RG_BE_EN	3C
3E		W/R	-	RG_SLRH_STR[6:0]							00
3F		W/R	RG_CABC_CRV_REF	RG_CABC_BLK_CUT_SEL	RG_CABC_PWM_SYNC_FD	RG_CABC_CUT_TH[4:0]					E0
40		W/R	RG_CABC_PWM_WT_UI[7:0]								80
		W/R	RG_CABC_PWM_WT_PIC[7:0]								80
		W/R	RG_CABC_PWM_WT_MOV[7:0]								80
41		W/R	RG_CABC_MIN_PWM[7:0]								00
42		W/R	RG_CABC_MAX_PWM[7:0]								80
43		W/R	RG_CABC_TP_BLK_PWM[7:0]								00
44		W/R	-	RG_CABC_CRV_TURN_TH[6:0]							46
45		W/R	-	-	rg_LE_IP_BYPASS	RG_LE_TP_EN[4:0]					1F
46		W/R	RG_LE_CUT2PWM_INP0[7:0]								00
		W/R	RG_LE_CUT2PWM_INP1[7:0]								01
		W/R	RG_LE_CUT2PWM_INP2[7:0]								06
		W/R	RG_LE_CUT2PWM_INP3[7:0]								0E
		W/R	RG_LE_CUT2PWM_INP4[7:0]								1C
		W/R	RG_LE_CUT2PWM_INP5[7:0]								2D
		W/R	RG_LE_CUT2PWM_INP6[7:0]								44
		W/R	RG_LE_CUT2PWM_INP7[7:0]								60
47		W/R	RG_LE_PWM2CUT_INP0[7:0]								00
		W/R	RG_LE_PWM2CUT_INP1[7:0]								87
		W/R	RG_LE_PWM2CUT_INP2[7:0]								BA
		W/R	RG_LE_PWM2CUT_INP3[7:0]								DF
48		W/R	RG_LE_DIM_DLT[3:0]				RG_LE_DIM_FPS[3:0]				11
49		W/R	-	-	RG_LE_CHKSUM_E	RG_LE_DIM_MAD_E	RG_LE_DIM_EN	RG_LE_DIM_MAD[2:0]			09
4A		W/R	-	ECO_AE_GATING_SOT123	ECO_AE_GATING_HS_CMD	00000					00

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		W/R	RG_LE_L_BOUND[7:0]							00
4B		W/R	-	-	-	-	-	-	RG_LE_MAP_FUN[1:0]	01
4C		W/R	00000000							00
4E		W/R	-	-	RG_AMB_CABC_PWM0[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM1[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM2[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM3[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM4[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM5[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM6[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM7[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM8[5:0]					00
		W/R	-	-	RG_AMB_CABC_PWM9[5:0]					00
50		W/R	-	-	-	-	RG_AMB_SLRH_STR0[3:0]			00
		W/R	-	-	-	-	RG_AMB_SLRH_STR1[3:0]			00
		W/R	-	-	-	-	RG_AMB_SLRH_STR2[3:0]			00
		W/R	-	-	-	-	RG_AMB_SLRH_STR3[3:0]			00
		W/R	-	-	-	-	RG_AMB_SLRH_STR4[3:0]			04
		W/R	-	-	-	-	RG_AMB_SLRH_STR5[3:0]			08
		W/R	-	-	-	-	RG_AMB_SLRH_STR6[3:0]			0A
		W/R	-	-	-	-	RG_AMB_SLRH_STR7[3:0]			0C
		W/R	-	-	-	-	RG_AMB_SLRH_STR8[3:0]			0E
		W/R	-	-	-	-	RG_AMB_SLRH_STR9[3:0]			0F
55		W/R	RG_LEDPWM_POL	RG_LEDON_POL	RG_LEDON_EN	RG_PWM_OFFSET[4:0]				C0
56		W/R	RG_SEL_CLK_DIV[7:0]							00
57		W/R	RG_N_VAL[7:0]							00
58		W/R	RG_PWM_SYNC2VS	RG_DIM_FD_EN	RG_INC_STP_FT[2:0]			RG_DEC_STP_FT[2:0]		49
59		W/R	RG_INC_FPS_FT[3:0]				RG_DEC_FPS_FT[3:0]			11
5A		W/R	RG_DEC_DLT_FD[7:0]							40
		W/R	RG_INC_DLT_FD[7:0]							40
5B		W/R	RG_INC_FPS_FD[3:0]				RG_DEC_FPS_FD[3:0]			11
5C		W/R	RG_DIM_RESET	RG_DIM_RESET_MODE	RG_CABC_PWM_SEL	RG_CABC_PWM_BUND	RG_PWM_MIN_MODE	RG_DIM_CTRL_MODE	RG_DIM_CTRL_IMD	RG_DIM_DEC_CTRL	20
5D		W/R	RG_DIM_RESET_VAL[7:0]							00
5E		W/R	RG_PWM_FRAC[7:0]							00
5F		W/R	RG_CABC_PWM[7:0]							00
60		W/R	RG_MANU_RESET_VAL[7:0]							00
61		W/R	RG_LE_GATING_EN[7:0]							FE
62		W/R	RG_BC_GATING_EN[7:0]							FE
63		W/R	-	RG_EXT_PWMI_SEL	RG_PWM8B14B_SEL	RG_BE_AMB_CTRL	RG_BE_MODE[1:0]		RG_SEL_DBG_PWM[1:0]		00
64		W/R	RG_dbg_mux_sel[3:0]				dbg_mux_en	RG_LE_DBG_SEL[2:0]			08

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65		W/R	RG_SEL_PWMI_DET_CLK_DIV[7:0]	**00**
66		R	ACC0_FINAL[7:0]	**00**
		R	ACC1_FINAL[7:0]	**00**
		R	ACC2_FINAL[7:0]	**00**
		R	ACC3_FINAL[7:0]	**00**
		R	ACC4_FINAL[7:0]	**00**
		R	ACC5_FINAL[7:0]	**00**
		R	ACC6_FINAL[7:0]	**00**
67		R	GAIN0_M32[7:0]	**00**
		R	GAIN1_M32[7:0]	**00**
		R	GAIN2_M32[7:0]	**00**
		R	GAIN3_M32[7:0]	**00**
		R	GAIN4_M32[7:0]	**00**
		R	GAIN5_M32[7:0]	**00**
		R	GAIN6_M32[7:0]	**00**
68		R	Ygry_avg[7:0]	**00**
69		R	- | slope_cond0 | hv_cond1 | hv_cond0 | u2u4_cond1 | u2u4_cond0 | u4[14:13]	**00**
6A		R	target_cutpt[7:0]	**00**
		R	output_cutpt[7:0]	**00**
		R	target_PWM[7:0]	**00**
		R	output_PWM[7:0]	**00**

Table 2.5: Page 5 command list

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2.6. Page 6 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	r_contrast[7:0]								80
32		W/R	g_contrast[7:0]								80
33		W/R	b_contrast[7:0]								80
3B		W/R	-	-	dither_en	d_gma_en	dgma_ext_pol_en	dgma_ext_pol_enb	contrast_en	bright_en	08
3C		W/R	r_bright[7:0]								40
3D		W/R	g_bright[7:0]								40
3E		W/R	b_bright[7:0]								40
3F		W/R	r_point_1[7:0]								00
		W/R	r_point_2[7:0]								04
		W/R	r_point_3[7:0]								0C
		W/R	r_point_4[7:0]								1C
		W/R	r_point_5[7:0]								2C
		W/R	r_point_6[7:0]								3C
		W/R	r_point_7[7:0]								5C
		W/R	r_point_8[7:0]								7C
		W/R	r_point_4[9:8]		r_point_3[9:8]		r_point_2[9:8]		r_point_1[9:8]		00
		W/R	r_point_8[9:8]		r_point_7[9:8]		r_point_6[9:8]		r_point_5[9:8]		00
40		W/R	r_point_9[7:0]								BC
		W/R	r_point_10[7:0]								FC
		W/R	r_point_11[7:0]								7C
		W/R	r_point_12[7:0]								FC
		W/R	r_point_13[7:0]								00
		W/R	r_point_14[7:0]								80
		W/R	r_point_15[7:0]								00
		W/R	r_point_16[7:0]								40
		W/R	r_point_12[9:8]		r_point_11[9:8]		r_point_10[9:8]		r_point_9[9:8]		50
		W/R	r_point_16[9:8]		r_point_15[9:8]		r_point_14[9:8]		r_point_13[9:8]		FA
41		W/R	r_point_17[7:0]								80
		W/R	r_point_18[7:0]								A0
		W/R	r_point_19[7:0]								C0
		W/R	r_point_20[7:0]								D0
		W/R	r_point_21[7:0]								E0
		W/R	r_point_22[7:0]								F0
		W/R	r_point_23[7:0]								F8
		W/R	r_point_24[7:0]								FC
		W/R	r_point_20[9:8]		r_point_19[9:8]		r_point_18[9:8]		r_point_17[9:8]		FF
		W/R	r_point_24[9:8]		r_point_23[9:8]		r_point_22[9:8]		r_point_21[9:8]		FF
42		W/R	g_point_1[7:0]								00

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		W/R	g_point_2[7:0]				04
		W/R	g_point_3[7:0]				0C
		W/R	g_point_4[7:0]				1C
		W/R	g_point_5[7:0]				2C
		W/R	g_point_6[7:0]				3C
		W/R	g_point_7[7:0]				5C
		W/R	g_point_8[7:0]				7C
		W/R	g_point_4[9:8]	g_point_3[9:8]	g_point_2[9:8]	g_point_1[9:8]	00
		W/R	g_point_8[9:8]	g_point_7[9:8]	g_point_6[9:8]	g_point_5[9:8]	00
43		W/R	g_point_9[7:0]				BC
		W/R	g_point_10[7:0]				FC
		W/R	g_point_11[7:0]				7C
		W/R	g_point_12[7:0]				FC
		W/R	g_point_13[7:0]				00
		W/R	g_point_14[7:0]				80
		W/R	g_point_15[7:0]				00
		W/R	g_point_16[7:0]				40
		W/R	g_point_12[9:8]	g_point_11[9:8]	g_point_10[9:8]	g_point_9[9:8]	50
		W/R	g_point_16[9:8]	g_point_15[9:8]	g_point_14[9:8]	g_point_13[9:8]	FA
44		W/R	g_point_17[7:0]				80
		W/R	g_point_18[7:0]				A0
		W/R	g_point_19[7:0]				C0
		W/R	g_point_20[7:0]				D0
		W/R	g_point_21[7:0]				E0
		W/R	g_point_22[7:0]				F0
		W/R	g_point_23[7:0]				F8
		W/R	g_point_24[7:0]				FC
		W/R	g_point_20[9:8]	g_point_19[9:8]	g_point_18[9:8]	g_point_17[9:8]	FF
		W/R	g_point_24[9:8]	g_point_23[9:8]	g_point_22[9:8]	g_point_21[9:8]	FF
45		W/R	b_point_1[7:0]				00
		W/R	b_point_2[7:0]				04
		W/R	b_point_3[7:0]				0C
		W/R	b_point_4[7:0]				1C
		W/R	b_point_5[7:0]				2C
		W/R	b_point_6[7:0]				3C
		W/R	b_point_7[7:0]				5C
		W/R	b_point_8[7:0]				7C
		W/R	b_point_4[9:8]	b_point_3[9:8]	b_point_2[9:8]	b_point_1[9:8]	00
		W/R	b_point_8[9:8]	b_point_7[9:8]	b_point_6[9:8]	b_point_5[9:8]	00
46		W/R	b_point_9[7:0]				BC

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		W/R	b_point_10[7:0]					FC
		W/R	b_point_11[7:0]					7C
		W/R	b_point_12[7:0]					FC
		W/R	b_point_13[7:0]					00
		W/R	b_point_14[7:0]					80
		W/R	b_point_15[7:0]					00
		W/R	b_point_16[7:0]					40
		W/R	b_point_12[9:8]	b_point_11[9:8]	b_point_10[9:8]	b_point_9[9:8]		50
		W/R	b_point_16[9:8]	b_point_15[9:8]	b_point_14[9:8]	b_point_13[9:8]		FA
47		W/R	b_point_17[7:0]					80
		W/R	b_point_18[7:0]					A0
		W/R	b_point_19[7:0]					C0
		W/R	b_point_20[7:0]					D0
		W/R	b_point_21[7:0]					E0
		W/R	b_point_22[7:0]					F0
		W/R	b_point_23[7:0]					F8
		W/R	b_point_24[7:0]					FC
		W/R	b_point_20[9:8]	b_point_19[9:8]	b_point_18[9:8]	b_point_17[9:8]		FF
		W/R	b_point_24[9:8]	b_point_23[9:8]	b_point_22[9:8]	b_point_21[9:8]		FF
48		W/R	esd_bright_en	esd_bright_step[1:0]	esd_bright[4:0]			00
49		W/R	00000000					00
4A		W/R	11111111					FF
56		W/R	-	-	0	r_detect_127	h_table_en	dith_tabset_sel	hfrc_sel[1:0]	0A
		W/R	-	-	-	-	-	ext_dith_tab_en	h_table_sel[1:0]	00
		W/R	ext_dith_tab1_0[7:0]					00
		W/R	ext_dith_tab1_1[7:0]					00
		W/R	ext_dith_tab1_2[7:0]					00
		W/R	ext_dith_tab1_3[7:0]					00
		W/R	ext_dith_tab1_4[7:0]					00
		W/R	ext_dith_tab1_5[7:0]					00
		W/R	ext_dith_tab1_6[7:0]					00
		W/R	ext_dith_tab1_7[7:0]					00

Table 2.6: Page 6 command list

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2.7. Page 7 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	r_lp11_check_enb	r_din_eq_z_sel[1:0]		r_hs_settle[4:0]					40
32		W/R	-	-	-	r_esd_prot	r_wc_esd_det_en	r_en_cmd_queue	r_hster_delay[1:0]		05
33		W/R	r_crc_esd_clr	r_cdcnt[1:0]		r_ta_sure[4:0]					44
3B		W/R	r_tx_eotp	r_tx_pnsw	r_lptx_bpn[1:0]		r_t_lpx_rl[3:0]				D8
3C		W/R	-	-	-	-	r_t_lpx[3:0]				08
3D		W/R	-	-	r_non_gated_dclk	r_non_gated_mipiclk	r_dynamic_lane0_en	r_en_dynamic_terr	r_dynamic_de_adj_sel	r_back_chip_non_gat	08
3E		W/R	-	-	r_mipi_dclk_inv	r_non_gated_ret_oscc	r_en_glb_dynamic	r_mipi_overlap[1:0]		r_extra_dummy	00
3F		W/R	-	r_wr_parainit[6:0]							00
40		W/R	-	-	r_tm_hster_switch	r_tm_hster_en[4:0]					00
41		W/R	-	-	-	r_en_prbs	r_prbs_bit_err_lansel[1:0]		r_prbs_err_clr	r_err_sts_cnt_clr	00
42		W/R	r_cd_en	r_false_ctrl_err_rpt	r_periph_tout_err_rpt	r_lp_sync_err_rpt	r_esc_entry_err_rpt	r_eot_sync_err_rpt	r_sot_sync_err_rpt	r_sot_err_rpt	00
43		W/R	r_mipi_err_rpt_en	r_protol_violation	r_invld_transmit_length	r_invld_vcid_err_rpt	r_invld_dtype_err_rpt	r_en_crc_det	r_ecc_on	r_ecc_1bit_err_rpt	80
44		W/R	index_pg7_dummy				r_hsrx_ff_ter_cnt		r_hsrx_ff_en_ck_sel	r_dcs_idle_mode_sel	02
45		W/R	index_pg7_dummy								00

Table 2.4: Page 7 command list

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2.8. Page 8 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	r_off_num[3:0]				start_num[3:0]				68
32		W/R	r_regen_delay_num[3:0]				r_waiteof_num[1:0]		freerun_num[1:0]		4A
33		W/R	-	-	-	r_nokia_flow_enb	r_all_gate_on_sel	r_tx_check_en	r_slp_in_skip_vs_f	r_skip_str2fr_en	07
3B		W/R	-	-	-	r_oneFtenL	0	tm_skip_all	0	for_sim	00
3C		W/R	-	r_flicker_tgl_enb	r_pixel_inv	r_chessboard_sel	border_sel[2:0]			r_pattern_keep	00
3D		W/R	r_gray_pat_sel_r[7:0]								80
		W/R	r_gray_pat_sel_g[7:0]								80
		W/R	r_gray_pat_sel_b[7:0]								80
3E		W/R	r_pattern_sel[7:0]								FF
		W/R	r_pattern_sel[15:8]								FF
3F		W/R	r_pattern_frm_sel[7:0]								78
		W/R	-	-	-	-	-	-	-	r_pattern_frm_sel[8]	00
40		W/R	r_hbp[7:0]								50
41		W/R	r_hfp[7:0]								DA
42		W/R	r_vbp[7:0]								0A
43		W/R	r_vfp[7:0]								14
44		W/R	r_hfp_least[7:0]								0A
45		W/R	r_hfp_free[7:0]								DA
46		W/R	r_vfp_free[7:0]								75
47		W/R	-	r_hole_time_sel_doze[2:0]			r_hole_time_sel[3:0]				03
48		W/R	r_vs_clr_hole_qsync_e	r_scan_hole_in_vbp_e	r_vs_clr_hole_cnt_en	r_hole_num[4:0]					0C
49		W/R	r_hole_sft_front[7:0]								00
4A		W/R	r_hole_sft_back[7:0]								00
4B		W/R	r_hole_dly_4_source[1:0]		r_rescan_num_gip[2:0]		r_rescan_num_sd[2:0]				18
4C		W/R	r_hole_dis_1st_longh[7:0]								96
		W/R	-	-	-	-	-	-	-	r_hole_dis_1st_longh[8]	00
4E		W/R	r_hole_dis_2nd_longh[7:0]								96
		W/R	-	-	-	-	-	-	-	r_hole_dis_2nd_longh[8]	00
50		W/R	r_hole_dis_1st_longv[7:0]								04
51		W/R	r_line_shift_step[3:0]				r_line_shift_range[3:0]				00
		W/R	r_te_hblk_range_inv	r_line_shift_en	r_line_shift_direction	r_line_shift_mode	r_line_shift_rate[3:0]				00
52		W/R	r_te1_active_high_en	r_te1_enable	r_hole_dly_4_gip[3:0]				r_pump_clk_en	r_set_by_vs_en	83
53		W/R	r_te1_position[7:0]								00
		W/R	-	-	-	-	r_te1_position[11:8]				00
54		W/R	r_te1_pulse_width[7:0]								00
		W/R	-	-	-	-	r_te1_pulse_width[11:8]				00
55		W/R	r_hole_dis_1st_qsync[7:0]								A6

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		W/R	-	-	-	r_hole_dis_1st_qsync[12:8]					04
56		W/R	r_hole_dis_2nd_qsync[7:0]								A6
		W/R	-	-	-	r_hole_dis_2nd_qsync[12:8]					04
57		W/R					r_regen_hs_width[3:0]				05
58		W/R	r_vs_width[3:0]				r_vs_r_adj[3:0]				25
59		W/R	r_ll_width[3:0]				r_ll_r_adj[3:0]				71
5A		W/R	r_h_res_set[7:0]								00
5B		W/R	r_v_res_set[7:0]								00
5C		W/R	r_v_res_set[11:8]				r_h_res_set[11:8]				00
5D		W/R	r_masl_sel_reg	0	r_mode_sel_reg	r_bist_en_sel_reg	r_lansel_sel_reg[1:0]		r_pnsw_sel_reg	r_masloc	01
5E		W/R	r_bist_en	r_pnsw	r_casc_enb_sel_reg	r_dswap_sel_reg[1:0]		r_dual_sel_reg	r_dswap[1:0]		00
5F		W/R	0	0	r_dual	r_masl	0	r_mode	r_lansel[1:0]		00
60		W/R	r_led_en_reg	reg_wr_protect2	reg_wr_protect1	0	r_nbw	0	r_all_sel_pin	r_casc_enb	E0
61		W/R	-	-	-	-	-	-	vs_inv_sel	hs_inv_sel	00
62		W/R	-	r_rescan_in_blk_en	r_doze_long_v_en	0	r_sync_to_disp_vs_en	r_cas_slave_sync_en	r_qsync_en	r_scan_long_h_en	05
63		W/R	-	-	-	-	-	rst_deglitch_sel	0	0	04
64		W/R	slpout_reload_en	0	r_bist_gating_en	r_clk_pol	r_non_gated_en	r_lvds_skew_sel[2:0]			A0
65		W/R	r_disp_pll_clk_sel	clkdet_en	r_oscd_sel[1:0]		r_dat_gating_en	r_gip_gating_en	1	r_datapath_gating_en	7F
66		W/R	0	0	r_masl_rd_sel	r_8to10_sel	hfrc	r_ext_pll_clk_en	r_ext_osct_clk_en	r_ext_oscd_clk_en	08
67		W/R	-	-	-	-	-	-	-	pin_dg_enb	00
68		W/R	-	-	-	r_resend_en	r_sync_slp_out_en	r_sync_disp_on_en	r_time_out_sel[1:0]		00
69		W/R	r_repaire_wline_0[7:0]								FF
6A		W/R	r_repaire_wline_1[7:0]								FF
6D		W/R	r_sh2c_sft_sel[3:0]				r_rd2lc_sft_sel[3:0]				00
6E		W/R	-	-	-	r_int_read_latch_sel	r_new_de_sft_sel[3:0]				00
6F		W/R	-	-	-	r_pll_refint_sel[1:0]		r_pll_ref_sel[1:0]		r_disp_gating_en	01
70		W/R	-	-	-	-	-	-	-	r_tm_reg_key_en	00
72		W/R	tp_afe_trim1_left[7:0]								3F
		W/R	tp_afe_trim2_left[7:0]								08
		W/R	tp_afe_trim3_left[7:0]								00
		W/R	tp_afe_trim4_left[7:0]								00
		W/R	tp_afe_trim1_right[7:0]								3F
		W/R	tp_afe_trim2_right[7:0]								08
		W/R	tp_afe_trim3_right[7:0]								00
		W/R	tp_afe_trim4_right[7:0]								00
73		W/R	-	-	-	-	-	r_crc_mux_sel[1:0]		r_crc_mux_en	00
74		W/R	crc_internal_o[7:0]								00
75		W/R	index_pg8_dummy								00
76		W/R	index_pg8_dummy								00
77		W/R	index_pg8_dummy								00

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78		W/R	index_pg8_dummy	**00**

Table 2.8: Page 8 command list

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2.9. Page 9 command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
5D		W/R	cp_version_msb[7:0]								00
5E		W/R	cp_version_lsb[7:0]								00
5F		R	version_id[7:0]								11
60		R	index_gip	index_ce	index_le	index_dgamma_set	index_r_dgamma	index_g_dgamma	index_b_dgamma	index_dither	00
61		R	index_mipi	index_pg7_dummy	index_corestate	index_regen	index_glue	index_framing	index_group	index_esd	00
62		R	index_sync_ctrl	index_repaire	index_sram_ctrl	index_clk_gen	index_tm	index_pw	index_gas	index_tsen	00
63		R	index_pw_tp	index_dphy_ana	index_misc1	index_misc2	index_misc3	index_misc4	index_vddd	index_vcipr	00
64		R	index_pw_ana1	index_dcdc_ana1	index_sd_ana	index_gam	index_sram	index_afe	index_trimming1	index_trimming2	00
65		R	index_sacode	index_icinfo	index_gamma	index_sd_dmy	index_ana_dmy	index_trimming3	index_trimming4	index_trimmingtp	00
66		R	-	-	-	index_vcom_5	index_vcom_4	index_vcom_3	index_vcom_2	index_vcom_1	00
67		R	-	-	-	index_vcomr_5	index_vcomr_4	index_vcomr_3	index_vcomr_2	index_vcomr_1	00
68		R	-	index_cp_ver	index_dabc	index_04	index_vcom	index_pg8_dummy	index_vcom2	index_cp_id	00
70		W/R	index_gip	index_ce	index_le	index_dgamma_set	index_r_dgamma	index_g_dgamma	index_b_dgamma	index_dither	00
71		W/R	index_mipi	index_pg7_dummy	index_corestate	index_regen	index_glue	index_framing	index_group	index_esd	00
72		W/R	index_sync_ctrl	index_repaire	index_sram_ctrl	index_clk_gen	index_tm	index_pw	index_gas	index_tsen	00
73		W/R	index_pw_tp	index_dphy_ana	index_misc1	index_misc2	index_misc3	index_misc4	index_vddd	index_vcipr	00
74		W/R	index_pw_ana1	index_dcdc_ana1	index_sd_ana	index_gam	index_sram	index_afe	index_trimming1	index_trimming2	00
75		W/R	index_sacode	index_icinfo	index_gamma	index_sd_dmy	index_ana_dmy	index_trimming3	index_trimming4	index_trimmingtp	00
76		W/R	-	-	-	index_vcom_5	index_vcom_4	index_vcom_3	index_vcom_2	index_vcom_1	00
77		W/R	-	-	-	index_vcomr_5	index_vcomr_4	index_vcomr_3	index_vcomr_2	index_vcomr_1	00
78		W/R	-	index_cp_ver	index_dabc	index_04	index_vcom	index_pg8_dummy	index_vcom2	index_cp_id	00

Table 2.9: Page 9 command list

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2.10. Page A command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	SD_SLPOUT	SD_GND	SD_EN	SD_OP_EN	0	SD_OUT_FIX	SD_POL_LEVEL_FIX[1:0]		B0
32		W/R	SD_OP_EN_SEL[7:0]								FF
33		W/R	-	SD_SDT[6:0]							08
3B		W/R	SD_CRT[7:0]								00
3C		W/R	SD_EQT[7:0]								0C
3D		W/R	SD_PCT[7:0]								0C
3E		W/R	SD_PCST[7:0]								18
3F		W/R	SD_POLT[1:0]		SD_DRVT[1:0]		0	SD_OVDT[2:0]			53
40		W/R	SD_TP_PS_SEL[1:0]		SD_CHOP_FRAME_OPT[1:0]		0	SD_CHOP_LINE_OPT[2:0]			10
41		W/R	SD_CP_TEST_EN	SD_CP_TEST_PHASE1[2:0]			0	0	SD_CP_TEST_PHASE2[1:0]		00
42		W/R	-	SD_BIAS_SEL_SRE[2:0]			0	SD_BIAS_SEL_N[2:0]			33
43		W/R	-	SD_BIAS_SEL_TP[2:0]			0	SD_BIAS_SEL_VPH[2:0]			33
44		W/R	-	SD_COMPR[2:0]			0	SD_BIAS_SEL_SRE_TD[2:0]			70
45		W/R	-	-	-	SD_PS_EN	SD_CTOT_TD[1:0]		SD_CTOT_LT[1:0]		15
46		W/R	-	-	-	-	SD_TP_OUT_SEL[1:0]		SD_TP_OUT_DLY[1:0]		08
47		W/R	-	SD_GAS_OPT	SD_DLY_SEL[1:0]		SD_DRV_SEL	SD_SLPIN_OPT	SD_V_PORCH_OPT[1:0]		20
48		W/R	zdata[7:0]								80
49		W/R	sd_test_en	r_frm_line_tog_clr	invsel[2:0]			zz_2pl	zz_lr	zz_eo	13
4A		W/R	-	-	-	-	-	nbw_inv	data_clr_blk	sd_dls_always_on	00
4B		W/R	00000000								00
4C		W/R	11111111								F0
4D		W/R	00000000								00

Table 2.10: Page A command list

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2.11. Page B command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	-	-	-	-	-	-	-	DSTBY	00
32		W/R	OTP_INT VPP	VPP_VGHO_S[6:0]							8B
33		W/R	-	-	-	-	-	-	-	VCOM_SEL[8]	00
		W/R	VCOM_SEL[7:0]								64
3C		W/R	-	-	-	-	-	-	-	VCOMR_SEL[8]	00
		W/R	VCOMR_SEL[7:0]								64
3D		W/R	-	-	-	-	-	-	-	VCOM2_SEL[8]	00
		W/R	VCOM2_SEL[7:0]								64
3E		W/R	-	GA_VHP0[6:0]							00
		W/R	-	GA_VHP1[6:0]							04
		W/R	-	GA_VHP2[6:0]							0B
		W/R	-	GA_VHP3[6:0]							11
		W/R	-	GA_VHP4[6:0]							19
3F		W/R	GA_VMP0[7:0]								2A
		W/R	GA_VMP1[7:0]								42
		W/R	GA_VMP2[7:0]								4A
		W/R	GA_VMP3[7:0]								53
		W/R	GA_VMP4[7:0]								51
		W/R	GA_VMP5[7:0]								6E
		W/R	GA_VMP6[7:0]								77
		W/R	GA_VMP7[7:0]								81
		W/R	GA_VMP8[7:0]								93
		W/R	GA_VMP9[7:0]								95
		W/R	GA_VMP10[7:0]								A0
		W/R	GA_VMP11[7:0]								AA
		W/R	GA_VMP12[7:0]								BD
		W/R	GA_VMP13[7:0]								BB
40		W/R	-	GA_VLP0[6:0]							59
		W/R	-	GA_VLP1[6:0]							5B
		W/R	-	GA_VLP2[6:0]							5F
		W/R	-	GA_VLP3[6:0]							65
		W/R	-	GA_VHN0[6:0]							00
		W/R	-	GA_VHN1[6:0]							04
		W/R	-	GA_VHN2[6:0]							0B
		W/R	-	GA_VHN3[6:0]							11
		W/R	-	GA_VHN4[6:0]							19
41		W/R	GA_VMN0[7:0]								2A

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		W/R	GA_VMN1[7:0]								42
		W/R	GA_VMN2[7:0]								4A
		W/R	GA_VMN3[7:0]								53
		W/R	GA_VMN4[7:0]								51
		W/R	GA_VMN5[7:0]								6E
		W/R	GA_VMN6[7:0]								77
		W/R	GA_VMN7[7:0]								81
		W/R	GA_VMN8[7:0]								93
		W/R	GA_VMN9[7:0]								95
		W/R	GA_VMN10[7:0]								A0
		W/R	GA_VMN11[7:0]								AA
		W/R	GA_VMN12[7:0]								BD
		W/R	GA_VMN13[7:0]								BB
42		W/R	-	GA_VLN0[6:0]							5E
		W/R	-	GA_VLN1[6:0]							67
		W/R	-	GA_VLN2[6:0]							70
		W/R	-	GA_VLN3[6:0]							73
43		W/R	VGM_EN_M	VGM_EN_S	VGMP_S[5:0]						AB
44		W/R	-	-	VGMN_S[5:0]						2B
45		W/R	-	VGHO_S[6:0]							4D
46		W/R	-	VGLO_S[6:0]							2F
47		W/R	-	-	-	-	VCL_EN	VCI_EN	VGL_EN	VGH_EN	0F
48		W/R	-	VGH_S[6:0]							2D
49		W/R	-	-	VGL_S[5:0]						19
4A		W/R	VGH_NC	VGL_NC	VGH_RT[2:0]			VGL_RT[2:0]			1B
4B		W/R	GAS_DEBOUNCE_SEL[1:0]		GAS_DELAY_OPT[1:0]		GAS_EN	GAS_VDDI_EN	GAS_VSP_EN	GAS_VSN_EN	0F
4C		W/R	GAS_SEL_VDDI[1:0]		GAS_SEL_VSP[2:0]			GAS_SEL_VSN[2:0]			9B
		W/R	-	-	-	-	-	-	-	rst_hw_gas	00
4E		W/R	-	-	TEMP_VCOM_EN	TEMP_VCOMR_EN	TEMP_VGHO_EN	TEMP_VGLO_EN	TEMP_VGH_EN	TEMP_VGL_EN	3F
4F		W/R	THH[7:0]								96
50		W/R	THL[7:0]								55
55		W/R	-	-	TEMP_IS[5:0]						20
56		W/R		HT_VCOM_OS[6:0]							40
57		W/R		LT_VCOM_OS[6:0]							40
58		W/R		HT_VCOMR_OS[6:0]							40
59		W/R		LT_VCOMR_OS[6:0]							40
5A		W/R		TSENH_VGHO_S[6:0]							4D
5B		W/R		TSENL_VGHO_S[6:0]							4D
5C		W/R		TSENH_VGLO_S[6:0]							2F
5D		W/R		TSENL_VGLO_S[6:0]							2F

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5E		W/R	-	-	TSENH_VGL_S[5:0]			**19**
5F		W/R	-	-	TSENL_VGL_S[5:0]			**19**
60		W/R	-	TSENH_VGH_S[6:0]				**2D**
61		W/R	-	TSENL_VGH_S[6:0]				**2D**
62		W/R	H10000_TEMP_Q[7:0]					**00**
63		W/R	H11111_TEMP_Q[7:0]					**00**
64		W/R	00000000					**00**
65		W/R	11111111					**FF**
66		W/R	-	VPP_VGH_S[6:0]				**07**
67		W/R	-	TSENH_VGH_RT[2:0]		-	TSENL_VGH_RT[2:0]	**33**
68		W/R	-	TSENH_VGL_RT[2:0]		-	TSENL_VGL_RT[2:0]	**33**

Table 2.1: Ppage B command list

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2.12. Page C command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	DPHY_HSRX_RT[2:0]			DPHY_HSRX_RT_HALF	0	DPHY_HSRX_PRE_ISEL	DPHY_HSRX_PST_ISEL	DPHY_HSRX_D2S_ISEL	A0
32		W/R	-	DPHY_HSRX_ALL_ISEL[2:0]			-	DPHY_FF_1ST_EN_SEL	DPHY_HSRX_STL[1:0]		12
33		W/R	-	DPHY_HSRX_SKW_D3[2:0]			-	DPHY_HSRX_SKW_D2[2:0]			00
3B		W/R	-	DPHY_HSRX_SKW_D1[2:0]			-	DPHY_HSRX_SKW_D0[2:0]			00
3C		W/R	-	DPHY_HSRX_SKW_CK[2:0]			-	DPHY_HSRX_DLY_D3[2:0]			00
3D		W/R	-	DPHY_HSRX_DLY_D2[2:0]			-	DPHY_HSRX_DLY_D1[2:0]			00
3E		W/R	-	DPHY_HSRX_DLY_D0[2:0]			-	DPHY_HSRX_DLY_CK[2:0]			00
3F		W/R	-	DPHY_LPCD_ISEL	DPHY_DSI_LDO_BIAS_EN	DPHY_BIAS_EN	DPHY_RX_LDO_EN	DPHY_RX_LDO2_EN	DPHY_DSI_LDO_STBY	DPHY_TX_LDO_VDDH	0C
40		W/R	DPHY_RX_LDO_SEL[2:0]			DPHY_TX_LDO_SEL[2:0]			01		6D
41		W/R	DPHY_LPCD_VT[1:0]		DPHY_LPRX_VT[1:0]		DPHY_LPRXCD_LPF[1:0]		DPHY_LPTX_SR[1:0]		59
42		W/R	BGR_TEST_SEL[1:0]		BGR_PRE_EN	BGR_BUF_EN	BGR_LPWG_OPTION	BGR_SHORT_EN	BGR_CAS_EN_M	BGR_CAS_EN_S	31
43		W/R	BGR_AP_IO[2:0]			BGR_BUF1_EN	BGR_BUF2_EN	BGR_BUF3_EN	BGR_BUFTP1_EN	BGR_BUFTP2_EN	7F
44		W/R	-	BGR_AP[2:0]			BGR_TEST_EN	BGR_APT[2:0]			33
45		W/R	oscl_always_on	OSCD_ENH	OSCL_BIAS_EN	OSCL_EN	OSCT_BIAS_EN	OSCT_EN	OSCD_BIAS_EN	OSCD_EN	7F
46		W/R	-	OSCD_TEMP[2:0]			-	OSCD_ADJ[2:0]			44
47		W/R	-	OSCT_TEMP[2:0]			-	OSCT_ADJ[2:0]			44
48		W/R	-	OSCL_TEMP[2:0]			-	OSCL_ADJ[2:0]			44
49		W/R	TEMP_FRAME[1:0]		TEMP_AVG[1:0]		TEMP_EN_ALWAYS	TEMP_TM	TSEN_PTAT_TEN_M	TSEN_PTAT_TEN_S	88
4A		W/R	THH_HYS_S[1:0]		THL_HYS_S[1:0]		TEMP_EN_BLANK	TEMP_DIM[2:0]			5B
4B		W/R	TSEN_EN_M	TSEN_EN_S	TEMP_RST2CLK[1:0]		TEMP_DLY_LINE[3:0]				13
4C		W/R	TSEN_VCCEN_M	TSEN_VCCEN_S	TEMP_CLK[2:0]		TEMP_CLK_H_N[2:0]				C9
4F		W/R	TEMP_EN2RST[1:0]			TEMP_RST_W[1:0]	TEMP_TEST_OUT_SEL[2:0]				68
50		W/R	TEMPM_Q[7:0]								73
55		W/R	TEMP_REF[7:0]								73
56		W/R	R10000_TEMP_Q[7:0]								73
57		W/R	R11111_TEMP_Q[7:0]								73
58		W/R	TEMP_TEST_IN[7:0]								73
59		W/R	TEMP_TEST_OUT[7:0]								00
5A		W/R	TEMP_Q[7:0]								00
5B		W/R	-	-	-	-	-	-	-	TEMP_TEST_OUT[8]	00
5C		W/R	M_PLL_LDO_EN	S_PLL_LDO_EN	PLL_ISEL[2:1]		PLL_VDDD_S[2:0]			-	F6
5D		W/R	-	TEST_PLLREG_EN	TEST_PLLREG_SEL[1:0]		PLL_NOUT_S[1:0]		M_PLL_EN	S_PLL_EN	07
5E		W/R	VCOM_GND_HIZ_GAS	VCOM2_EN	PLL_TMCKO[1:0]		PLL_TMVCTRL	PLL_CKSEL	PLL_DIVM[1:0]		80
5F		W/R	PLL_PFD_BYPS	PLL_DIVN[6:0]							6A
60		W/R	VCOM_EN_M	VCOM_EN_S	VCOM_SSTB_TIME	VCOM_TEST	VCOM_EXTSEL	VCOM_FBSEL	VCOM_FB_SHORT_M	VCOM_FB_SHORT_	80
61		W/R	VCOM_DA[1:0]		VCOM_HD_LPC[2:0]			VCOM_LPC[2:0]			C0
62		W/R	VCOM2_GROUP_OFF[3:0]				VCOM_HD	VCOM_R_SEN	VCOM_GND_HIZ_M	VCOM_GND_HIZ_S	43

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63		W/R	VDDD_EXTRA_VDDD_EN	VDDD_VDDD_S[2:0]			VDDD_SRE_EN	VDDD_MILLER_EN	VDDD_POUT_S[1:0]		**35**
64		W/R	-	VDDD_VDDD_S_R[2:0]			VDDD_ISEL_R[1:0]		VDDD_NOUT_S_R[1:0]		**3D**
65		W/R	-	-	VDDD_VDDD_IOVCC_EN_L	VDDD_EXTRA_VDDD_EN_L	VDDD_DSI_ON_EN_L	VDDD_VDDD_IOVCC_EN_R	VDDD_EXTRA_VDDD_EN_R	VDDD_DSI_ON_EN_R	**2D**
66		W/R	-	VDDD_VDDD_S_L[2:0]			VDDD_ISEL_L[1:0]		VDDD_NOUT_S_L[1:0]		**3D**
67		W/R	VGLO_DISCH_EN	VGLO_EN	VGHO_DISCH_EN	VGHO_EN	VCIPR_SLPIN_EN	VCIPR_S[2:0]			**53**
68		W/R	LPWG_SOFT1[7:0]								**80**
69		W/R	SOFT0L[3:0]				SOFT0H[3:0]				**71**
6A		W/R	CPCLK_TP[2:0]			CHG2AVDD_EH	VGL_POS_DISCH	VGH_POS_DISCH	VCI_NC	VCL_NC	**7C**
6B		W/R	VGH_TP_OTA_EN	VGL_TP_OTA_EN	DCDC_VGH_GAS_EN	DCDC_VGL_GAS_EN	VCI_S[1:0]		VCL_S[1:0]		**C5**
6C		W/R	-	VGHO_TP_PWR_SWP_EN	VGHO_SOFT_LPWG	VGHO_GND_LPWG	VGHO_GND_SLPI	VGHO_SHT_SLPO	VGHO_SHT_LPWG	VGHO_SHT_SLPI	**43**
6D		W/R	-	VGLO_TP_PWR_SWP_EN	VGLO_SOFT_LPWG	VGLO_GND_LPWG	VGLO_GND_SLPI	VGLO_SHT_SLPO	VGLO_SHT_LPWG	VGLO_SHT_SLPI	**43**
6E		W/R	GAS_DISCH_SEL	VGH_TP_CSW	VGH_TP_SLPIN	VGH_TP_S[1:0]		VGH_TP_REG_HIZ	VGH_MSF_EN	VGH_PWS_POS	**3E**
6F		W/R	OTP_DISCH_T_SEL	VGL_TP_CSW	VGL_TP_SLPIN	VGL_TP_S[1:0]		VGL_TP_REG_HIZ	VGH_PWS	VGH_PWS_GND	**3E**
70		W/R	DCDC_SOFTL[3:0]				OTP_VGH_AUTO_EN	DCDC_VGL_TP_SEL	DCDC_VGH_TP_SEL	VCL2VDDN_SEL	**89**
71		W/R	-	VGH_CPCLK[2:0]			-	VGL_CPCLK[2:0]			**33**
72		W/R	VCI_TP_SEL	VCI_CPCLK[2:0]			VCL_TP_SEL	VCL_CPCLK[2:0]			**33**
73		W/R	ON_GROUP1[3:0]				OFF_GROUP1[3:0]				**17**
74		W/R	ON_GROUP2[3:0]				OFF_GROUP2[3:0]				**23**
75		W/R	ON_GROUP3[3:0]				OFF_GROUP3[3:0]				**53**
76		W/R	ON_GROUP4[3:0]				OFF_GROUP4[3:0]				**24**
77		W/R	ON_GROUP5[3:0]				OFF_GROUP5[3:0]				**71**
78		W/R	pumb_clk_sync_enb	cpclk_tp[2:0]			oscd_off_time[1:0]		osct_always_on	oscd_always_on	**28**

Table 2.12: Page C command list

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2.13. Page D command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
47		W/R	-	-	-	-	GA_EN	GA_EXT	GA_OUT_TEST_EN	GA_IN_TEST_EN	08
48		W/R	GA_OUT_TEST_SEL[3:0]				GA_IN_TEST_SEL[3:0]				00
49		W/R	-	GA_BIAS_SEL_N[2:0]			-	GA_BIAS_SEL_TP[2:0]			33
4A		W/R	GA_SYNC_EN	GA_CHOP_BUF_EN	GA_CHOP_FRAME_OPT[1:0]		-	GA_CHOP_LINE_OPT[2:0]			24
4B		W/R	VGMP_EN	VGMN_EN	VGMP_BUF_EN	VGMN_BUF_EN	VGM_TEST	VGM_BIAS_SEL[2:0]			F3
4C		W/R	-	VGM_MC_EN	VGM_BUF_EN_SEL_TP[1:0]		VGM_TIE_SEL_M[1:0]		VGM_TIE_SEL_S[1:0]		34
4E		W/R	SRAM_R_ERDSOPT[1:0]		SRAM_F_ERDSOPT[1:0]		SRAM_BLKOPT[1:0]		SRAM_CHGOPT[1:0]		85
4F		W/R	-	-	SRAM_F_EWRSOPT[1:0]		0	SRAM_R_EWRSOPT[2:0]			04
50		W/R	-	-	SRAM_F_IRDSOPT[1:0]		0	SRAM_R_IRDSOPT[2:0]			04
55		W/R	SRAM_TOPT[3:0]				SRAM_WRENH	SRAM_PRIOR_SEL	SRAM_DEGLITCH_EXT_WE[1:0]		41

Table 2.13: Page D command list

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2.14. Page F command

Index	Operation Code	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Default
31		W/R	-	-	-	VREF_1D45V_T[4:0]					10
32		W/R	BGR_IBIAS1_2U_T[3:0]				BGR_IBIAS2_2U_T[3:0]				88
33		W/R	BGR_BG1_T[3:0]				BGR_BG2_T[3:0]				88
3B		W/R	-	-	BGR_VREF_T_SLPIN[5:0]						20
3C		W/R	-	-	BGR_VREF_T_SLPOUT[5:0]						20
3D		W/R	-	VDDD_VDDD_T_R[2:0]			-	VDDD_VDDD_T[2:0]			44
3E		W/R	-	PLL_VDDD_T[2:0]			-	VDDD_VDDD_T_L[2:0]			44
3F		W/R	-	-	OSCD_TRIM[5:0]						20
40		W/R	-	-	OSCT_TRIM[5:0]						20
41		W/R	-	-	OSCL_TRIM[5:0]						20
42		W/R	PLL_FREE RUN	PLL_INI[2:0]			PLL_ICGM[1:0]		PLL_LPFC[1:0]		00
43		W/R	-	-	PLL_PFDTM[1:0]		PLL_CIS[1:0]		PLL_CPI[1:0]		00
44		W/R	-	GAS_TRIM_VSP[2:0]			-	GAS_TRIM_VSN[2:0]			33
45		W/R	-	-	-	-	-	GAS_TRIM_VDDI[2:0]			03
46		W/R	TSEN_TEMP_T[4:0]				TSEN_TEMP_TSR[2:0]				84
47		W/R	-	GA_TRIM_VGMP[2:0]			-	GA_TRIM_VP27[2:0]			00
48		W/R	-	GA_TRIM_VP39[2:0]			-	GA_TRIM_VP51[2:0]			00
49		W/R	-	GA_TRIM_VP75[2:0]			-	GA_TRIM_VP99[2:0]			00
4A		W/R	-	GA_TRIM_VP127[2:0]			-	GA_TRIM_VP155[2:0]			00
4B		W/R	-	GA_TRIM_VP179[2:0]			-	GA_TRIM_VP203[2:0]			00
4C		W/R	-	GA_TRIM_VP215[2:0]			-	GA_TRIM_VP227[2:0]			00
4E		W/R	-	GA_TRIM_VGMN[2:0]			-	GA_TRIM_VN27[2:0]			00
4F		W/R	-	GA_TRIM_VN39[2:0]			-	GA_TRIM_VN51[2:0]			00
50		W/R	-	GA_TRIM_VN75[2:0]			-	GA_TRIM_VN99[2:0]			00
55		W/R	-	GA_TRIM_VN127[2:0]			-	GA_TRIM_VN155[2:0]			00
56		W/R	-	GA_TRIM_VN179[2:0]			-	GA_TRIM_VN203[2:0]			00
57		W/R	-	GA_TRIM_VN215[2:0]			-	GA_TRIM_VN227[2:0]			00
58		W/R	VGMP_T1[3:0]				VGMN_T1[3:0]				88
59		W/R	VGMP_T2[3:0]				VGMN_T2[3:0]				00
5A		W/R	-	-	-	VCOM_TRIM[4:0]					00
5B		W/R	VGLO_T[3:0]				VGHO_T[3:0]				77
5C		W/R	-	-	DCDC_VGL_T[2:0]		DCDC_VGH_T[2:0]				23
5D		W/R	-	-	DPHY_RX_LDO_TRIM[2:0]		DPHY_RX_LDO2_TRIM[2:0]				1B
5E		W/R	-	-	TP_ITRIM_L[5:0]						3F
5F		W/R	-	-	TP_ITRIM_R[5:0]						3F
60		W/R	TP_VTRIM_L[3:0]				TP_VTRIM_R[3:0]				88
61		W/R	-	-	-	-	-	TP_TPVDD_VTRIM[2:0]			00

Table 2.14: Page F command list

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3. Page 1 Command Description

3.1. SET_CUS_ID_INF: Set Customer ID Information (Page1-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	dummy_00[7:0]								00
Parameter 2	**R/W**	dummy_01[7:0]								00
Parameter 3	**R/W**	dummy_02[7:0]								00
Parameter 4	**R/W**	dummy_03[7:0]								00
Parameter 5	**R/W**	dummy_04[7:0]								00
Parameter 6	**R/W**	dummy_05[7:0]								00
Parameter 7	**R/W**	dummy_06[7:0]								00
Parameter 8	**R/W**	dummy_07[7:0]								00
Parameter 9	**R/W**	dummy_08[7:0]								00
Parameter 10	**R/W**	dummy_09[7:0]								00
Description	This command is used to set customer ID information and 1 time OTP.									
Restriction	-									

3.2. SET_CUS_ID_INF: Set Customer ID Information (Page1-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	dummy_10[7:0]								00
Parameter 2	**R/W**	dummy_11[7:0]								00
Parameter 3	**R/W**	dummy_12[7:0]								00
Parameter 4	**R/W**	dummy_13[7:0]								00
Parameter 5	**R/W**	dummy_14[7:0]								00
Parameter 6	**R/W**	dummy_15[7:0]								00
Parameter 7	**R/W**	dummy_16[7:0]								00
Parameter 8	**R/W**	dummy_17[7:0]								00
Parameter 9	**R/W**	dummy_18[7:0]								00
Parameter 10	**R/W**	dummy_19[7:0]								00
Description	This command is used to set customer ID information and 1 time OTP.									
Restriction	-									

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3.3. SET_CUS_ID_INF: Set Customer ID Information (Page1-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	dummy_20[7:0]								00
Parameter 2	**R/W**	dummy_21[7:0]								00
Parameter 3	**R/W**	dummy_22[7:0]								00
Parameter 4	**R/W**	dummy_23[7:0]								00
Parameter 5	**R/W**	dummy_24[7:0]								00
Parameter 6	**R/W**	dummy_25[7:0]								00
Parameter 7	**R/W**	dummy_26[7:0]								00
Parameter 8	**R/W**	dummy_27[7:0]								00
Parameter 9	**R/W**	dummy_28[7:0]								00
Parameter 10	**R/W**	dummy_29[7:0]								00
Description	This command is used to set customer ID information and 1 time OTP.									
Restriction	-									

3.4. SET_CUS_ID_INF: Set Customer ID Information (Page1-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	dummy_30[7:0]								00
Parameter 2	**R/W**	dummy_31[7:0]								00
Parameter 3	**R/W**	dummy_32[7:0]								00
Parameter 4	**R/W**	dummy_33[7:0]								00
Parameter 5	**R/W**	dummy_34[7:0]								00
Parameter 6	**R/W**	dummy_35[7:0]								00
Parameter 7	**R/W**	dummy_36[7:0]								00
Parameter 8	**R/W**	dummy_37[7:0]								00
Parameter 9	**R/W**	dummy_38[7:0]								00
Parameter 10	**R/W**	dummy_39[7:0]								00
Description	This command is used to set customer ID information and 1 time OTP.									
Restriction	-									

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3.5. SET_CUS_ID_INF: Set Customer ID Information (Page1-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	dummy_40[7:0]								00
Parameter 2	**R/W**	dummy_41[7:0]								00
Parameter 3	**R/W**	dummy_42[7:0]								00
Parameter 4	**R/W**	dummy_43[7:0]								00
Parameter 5	**R/W**	dummy_44[7:0]								00
Parameter 6	**R/W**	dummy_45[7:0]								00
Parameter 7	**R/W**	dummy_46[7:0]								00
Parameter 8	**R/W**	dummy_47[7:0]								00
Parameter 9	**R/W**	dummy_48[7:0]								00
Parameter 10	**R/W**	dummy_49[7:0]								00
Description	This command is used to set customer ID information and 1 time OTP.									
Restriction	-									

3.6. SET_CUS_INI_INF: Set Customer Initial Code Information (Page1-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	SA_Code1[7:0]								00
Parameter 2	**R/W**	SA_Code2[7:0]								00
Parameter 3	**R/W**	ddb0[7:0]								00
Parameter 4	**R/W**	ddb1[7:0]								00
Parameter 5	**R/W**	ddb2[7:0]								00
Parameter 6	**R/W**	ddb3[7:0]								00
Description	This command is used to set customer initial code information and 1 time OTP. **SA_Code1/2[7:0]:** Set customer initial code version. **ddb0/1[7:0]:** Set initial information and it can read from Page0_A1h. **ddb2/3[7:0]:** Set initial information and it can read from Page0_A8h.									
Restriction	-									

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3.7. SET_TX_CRC_ENB: Set TX CRC Enable (Page1-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	-	-	-	-	-	-	-	r_crc_interface_en	00
Description	This command is used to set TX CRC enable. **r_crc_interface_en:** TX crc calculate enable. //0: disable. //1: enable.									
Restriction	-									

3.8. RD_TX_CRC_INF: Read TX CRC Information (Page1-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R**	crc_interface_o[7:0]								00
Description	This command is used to read TX CRC information. **crc_interface_o[7:0]:** TX crc calculate result. Valuable only when r_crc_interface_en=1.									
Restriction	-									

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3.9. SET_LOT_ID_INF: Set Lot ID Information (Page1-R70h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	0	70
Parameter 1	**R/W**	touch_id[7:0]								00
Parameter 2	**R/W**	lot_id1[7:0]								00
Parameter 3	**R/W**	lot_id1[7:0]								00
Parameter 4	**R/W**	lot_id1[7:0]								00
Parameter 5	**R/W**	lot_id1[7:0]								00
Parameter 6	**R/W**	lot_id1[7:0]								00
Parameter 7	**R/W**	lot_id1[7:0]								00
Parameter 8	**R/W**	lot_id1[7:0]								00
Parameter 9	**R/W**	lot_id1[7:0]								00
Description	This command is used to set touch ID and Lot ID information. **touch_id[7:0]:** Set touch ID information. **lot_id1[7:0]:** Set lot ID information.									
Restriction	-									

3.10. SET_OTP_RD_REQ: (Page1-R71h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	1	71
Parameter 1	**R/W**	-	-	-	-	-	-	-	otp_rd_req	00
Description	This command is used to set **t otp_rd_req:**									
Restriction	-									

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3.11. SET_OTP_PRM: Set OTP Program (Page1-R72h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	1	0	72
Parameter 1	**R/W**	-	-	-	-	-	-	-	otp_auto_prog	00
Description	This command is used to perform OTP program. **t otp_rd_req:** perform OTP program. //0: disable. //1: enable OTP program.									
Restriction	-									

3.12. SET_OTP_RD_REQ: (Page1-R73h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	1	1	73
Parameter 1	**R/W**	-	-	-	-	-	-	-	spi_addrx_bit7	00
Description	This command is used to set spi_addrx_bit7:									
Restriction	-									

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3.13. SET_ENG_CMD: Set Engineer Command (Page1-R78h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	1	0	0	0	78
Parameter 1	**R/W**	reg_key1[7:0]								00
Parameter 2	**R/W**	reg_key2[7:0]								00
Parameter 3	**R/W**	reg_key3[7:0]								00
Parameter 4	**R/W**	reg_key41[7:0]								00
Description	This command is used to enter Fiti command. **reg_key1[7:0]**: 49h. **reg_key2[7:0]**: 61h. **reg_key3[7:0]**: 02h. **reg_key4[7:0]**: 00h.									
Restriction	-									

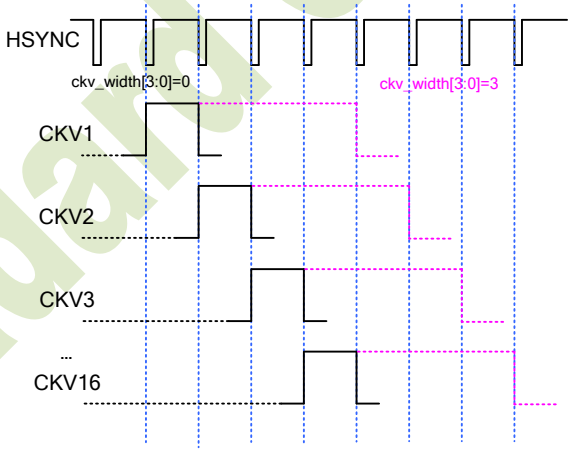
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4. Page 2 Command Description

4.1. SET_GIP_CKV/STV WIDTH: Set GIP Information (Page2-R31h)

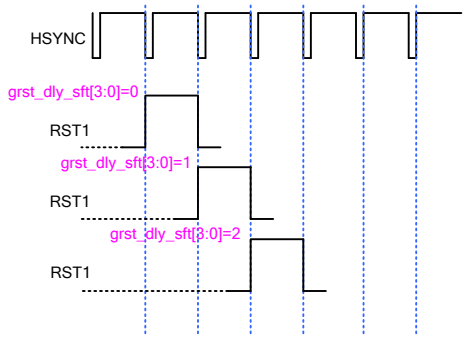
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	ckv_width[3:0]				stv-width[3:0]				22
Description	This command is used to set GIP and 1 time OTP. **stv-width[3:0]**: gSTV width setting. //2'b000: 2 line width. //2'b001: 3 line width. //2'b010: 4 line width. //2'b011: 5 line width. **ckv_width[3;0]**: gCKV width setting. //2'b000: 2 line width. //2'b001: 3 line width. //2'b010: 4 line width. //2'b011: 5 line width.									
Restriction	-									

4.2. SET_GIP_CKV PHASE NUM: Set GIP Information (Page2-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	r_flc_single_cnt_en	r_force_flc_inv_en	r_learn_len_sel	phase[4:0]					0C
Description	This command is used to set GIP and 1 time OTP. **phase[4:0]**: gCKV phase setting.(Max 16 phase) //2'b00001: 1phase //2'b00010: 2phase //2'b00011: 3phase ... //2'b10000: 16phase  **r_learn_len_sel**: Select learning H length area. //0: learning h length from v-blaking N line. //1: learning h length from last active line. **r_force_flc_inv_en**: Force flca signal invert after waiteof setate. //0: disable. //1: enable. **r_flc_single_cnt_en**: flcB counter set signal follow flcA counter. //0: disable. //1: enable.									
Restriction	-									

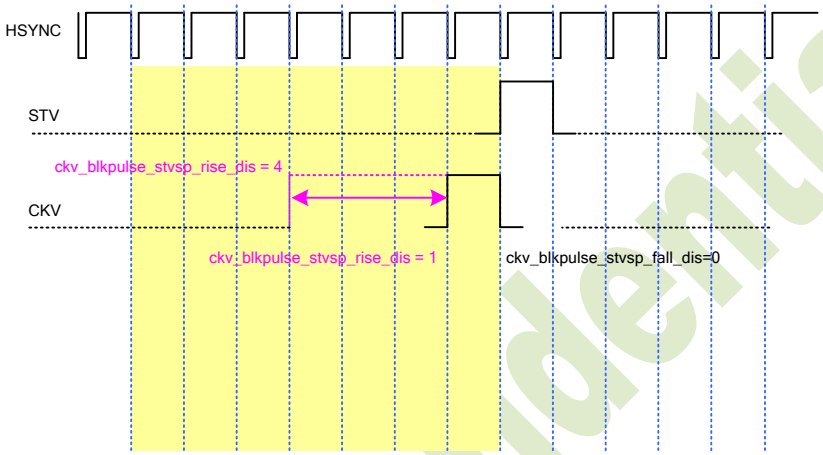
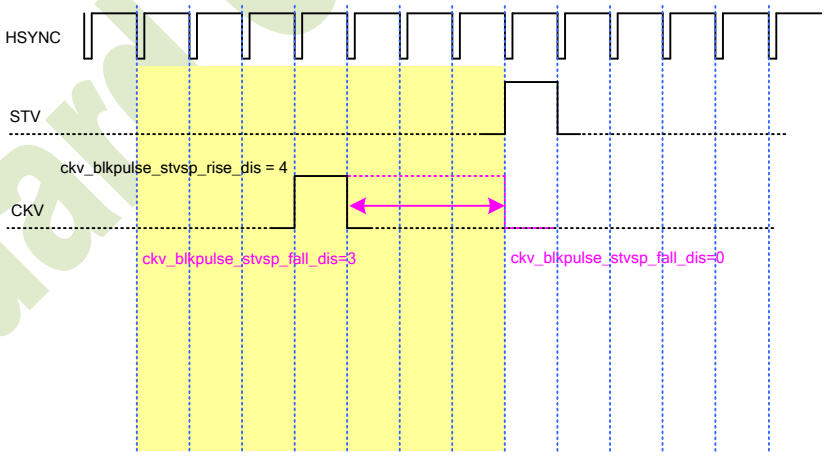
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4.3. SET_GIP_RST TIMING: Set GIP Information (Page2-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	rst_width[3:0]				grst_dly_sft[3:0]				30
Description	This command is used to set GIP and 1 time OTP. **grst_dly_sft[3:0]**: gRST location shift delay. (Line base) //2'b0000: whitout shift //2'b0001: shift 1H //2'b0010: shift 2H //2'b0011: shift 3H  **rst_width[3:0]**: gRST width setting. //2'b0000: 1 line width. //2'b0001: 2 line width. //2'b0010: 3 line width. //2'b0011: 4 line width.									
Restriction	-									

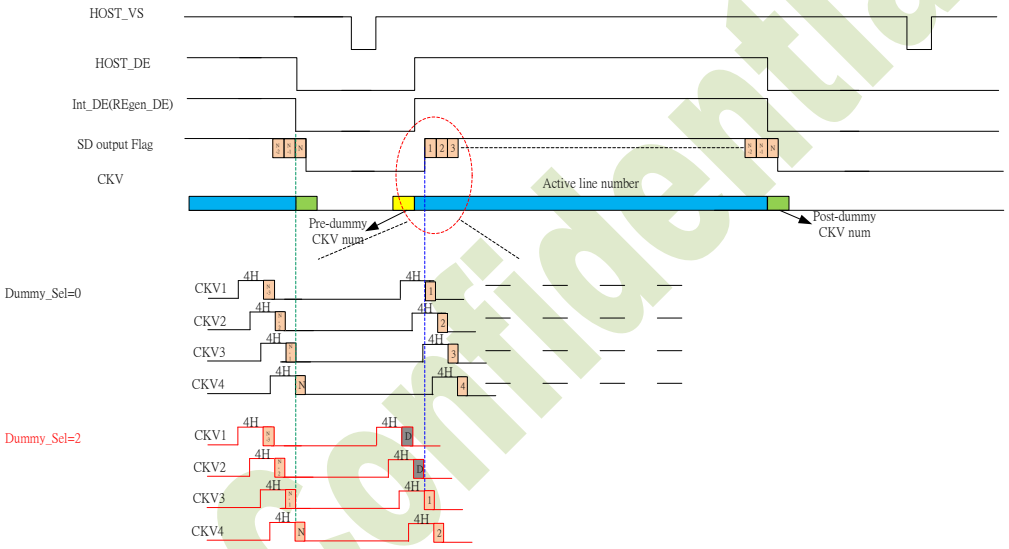
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4.4. SET_GIP_ALL GAE ON TIMING: Set GIP Information (Page2-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	ckv_blkpulse_stvsp_fall_dis[3:0]				ckv_blkpulse_stvsp_rise_dis[3:0]				3C
Description	This command is used to set GIP and 1 time OTP. **ckv_blkpulse_stvsp_rise_dis[3:0]**: Distance from vblking gCKV all on rising edge to gSTV rising edge. //2'b0000: 0H //2'b0001: 1H //2'b0010: 2H //2'b0011: 3H  V Blanking **ckv_blkpulse_stvsp_fall_dis[3:0]**: Distance from vblking gCKV all on falling edge to gSTV rising edge. //2'b0000: 0H //2'b0001: 1H //2'b0010: 2H //2'b0011: 3H  V Blanking									
Restriction	-									

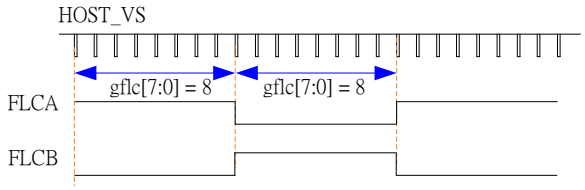
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4.5. SET_GIP_CKV PRE DMY CYCLE: Set GIP Information (Page2-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	ckv_blkpulse_phase_sel	ckv_blkpulse_stagger_sel[1:0]		dummy_sel[4:0]					00
Description	This command is used to set GIP and 1 time OTP. **dummy_sel[4:0]**: Shift gCKV pulse forward to V-porch region. (Line base) //2'b00000: non-pre dummy gCKV. //2'b00001: 1 : pre-gCKV 1 pulse. //2'b00010: 2: pre-gCKV 1 pulse. //2'b00011: 3 : pre-gCKV 1 pulse. Example  **ckv_blkpulse_stagger_sel[1:0]**: Phase of gCKV all on stagger distance. //2'b0000: No stagger. //2'b0001: shift 0.5 us. //2'b0010: shift 1 us. //2'b0011: shift 1.5 us. **ckv_blkpulse_phase_sel**: gCKV all on stagger type select. //0: gCKV pulse stagger by 1 2 -> 3 4 -> 5 6 -> 7 8 -> 9 10 -> 11 12. //1: gCKV pulse stagger by odd/even.									
Restriction	-									

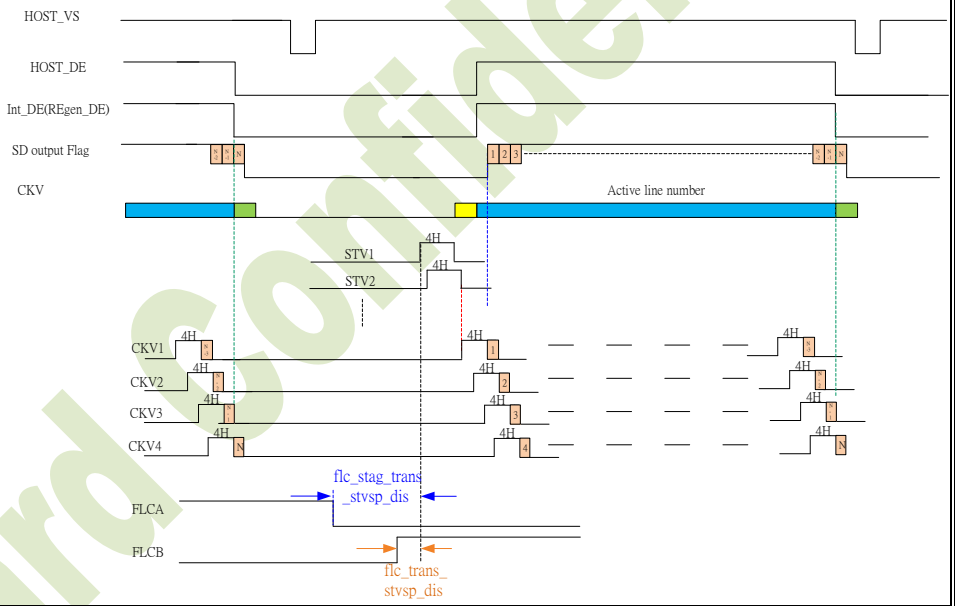
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4.6. SET_GIP_FLC_FRAM_CYCLE: Set GIP Information (Page2-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	gflc[7:0]								08
Description	This command is used to set GIP and 1 time OTP. **gflc[7:0]**: flc High / Low change by N frame. //2'b00000000: non-frame flc change. //2'b00000001: 1 frame flc change. //2'b00000010: 1 frame flc change. //2'b00000011: 1 frame flc change. Example  									
Restriction	-									

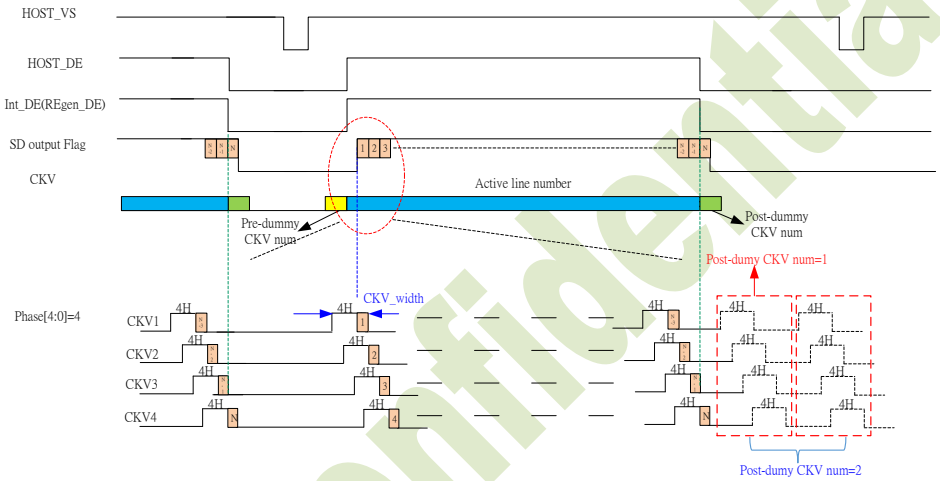
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4.7. SET_GIP_FLC TIMING: Set GIP Information (Page2-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	flc_trans_stvsp_dis [3:0]				flc_stag_trans_stvsp_dis[3:0]				67
Description	This command is used to set GIP and 1 time OTP. **flc_stag_trans_stvsp_dis[3:0]**: Distance between the falling edge of flc and the risingg edge of gSTV1. //2'b0000: 0H //2'b0001: 1H //2'b0010: 2H //2'b0011: 3H **flc_trans_stvsp_dis [3:0]**: Distance between the rising edge of flc and the rising edge of gSTV1. //2'b0000: 0H //2'b0001: 1H //2'b0010: 2H //2'b0011: 3H Example  									
Restriction	flc_stagger_en = 1 (Page 02 , R42[0])									

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4.8. SET_GIP_CKV_DMY_NUM: Set GIP Information (Page2-R3Fh)

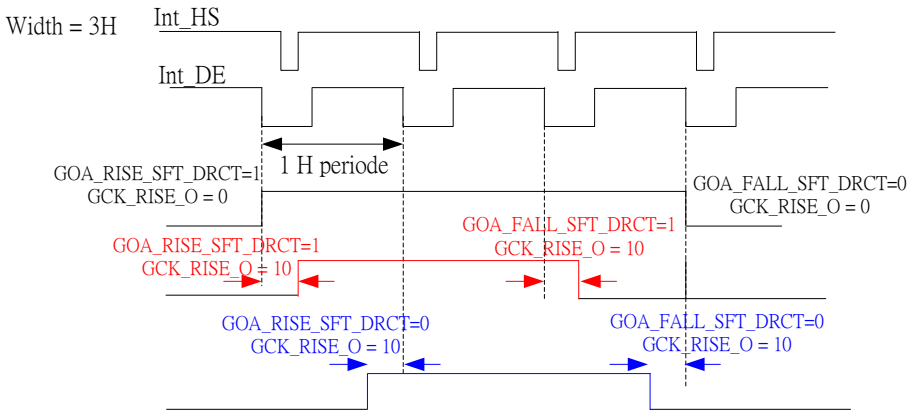
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	flc_hole_vgl_dly_sel[1:0]		gip_updn_en	r_learn_type	blk_post_dmy_num[3:0]				60
Description	This command is used to set GIP and 1 time OTP. **blk_post_dmy_num[3:0]**: V-blanking dummy gCKV pulse group. (after active area) //2'b0000: non-post dummy. //2'b0001: 1 gCKV of N phase pulse. //2'b0010: 2 gCKV of N phase pulse. //2'b0011: 3 gCKV of N phase pulse. **Note: Phase group is define by R32h[4:0] (Phase[4:0])** Example:  **r_learn_type**: Select learning H-length follow conter directly or follow total length +1 / -. //0: flollow osc_h_len_cnt number. //1: inc 1 / dec 1. **gip_updn_en**: Enable GIP control by UPDN sginal.. //0: Both forward gout mapping and backward gout mapping are refer R5Ah. //1: Forward gout mapping refer R5Ah, backward gout mapping refer R5Bh.. **flc_hole_vgl_dly_sel[1:0]**: flc pull to VGL delay timing select. //2'b00: non-delay. //2'b01: delay 16T. 1T= 25ns //2'b10: delay 171T. 1T= 25ns //2'b11: delay 255T. 1T= 25ns									
Restriction	-									

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4.9. SET_GIP_EQ_TIME: Set GIP Information (Page2-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	sub_flc_dis[1:0]		eq_rise[2:0]			eq_fall[2:0]			00
Description	This command is used to set GIP and 1 time OTP. **eq_fall[2:0]**: EQ timing select for falling edge. //2'b000: non-EQ. //2'b001: 0.5us. //2'b010: 1us. //2'b011: 1.5us. **eq_rise[2:0]**: EQ timing select for rising edge. //2'b000: non-EQ. //2'b001: 0.5us. //2'b010: 1us. //2'b011: 1.5us. **sub_flc_dis[1:0]**: Sub flca and flcb stagger distance. //2'b00: 0H //2'b01: 1H //2'b10: 2H //2'b11: 3H									
Restriction	-									

4.10. SET_GIP_RISE_FALL_TIME: Set GIP Information (Page2-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	-	gstv_rise_o[6:0]							**00**
Parameter 2	**R/W**	-	gstv_rise_e[6:0]							**00**
Parameter 3	**R/W**	-	gstv_fall_o[6:0]							**20**
Parameter 4	**R/W**	-	gstv_fall_e[6:0]							**20**
Parameter 5	**R/W**	-	gckv_rise_o[6:0]							**00**
Parameter 6	**R/W**	-	gckv_rise_e[6:0]							**00**
Parameter 7	**R/W**	-	gckv_fall_o[6:0]							**20**
Parameter 8	**R/W**	-	gckv_fall_e[6:0]							**20**
Parameter 9	**R/W**	-	grst_rise_o[6:0]							**00**
Parameter 10	**R/W**	-	grst_rise_e[6:0]							**00**
Parameter 11	**R/W**	-	grst_fall_o[6:0]							**20**
Parameter 12	**R/W**	-	grst_fall_e[6:0]							**20**
Description	This command is used to set GIP and 1 time OTP. The rise/ fall time need refer R42h[2:1](goa_fall_sft_drct/ goa_rise_sft_drct) to judge shift direction.  									

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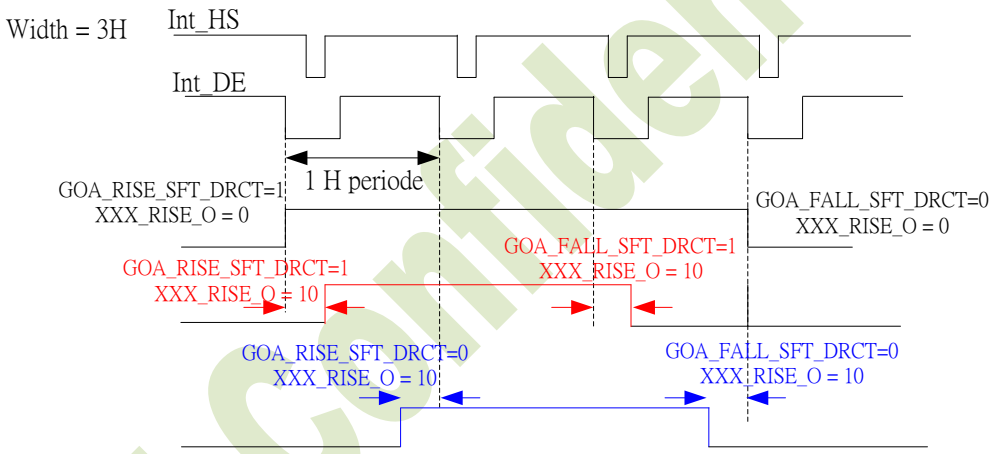
	gstv_rise_o[6:0]: Odd gSTV rising edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gstv_rise_e[6:0]**: Even gSTV rising edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gstv_fall_o[6:0]**: Odd gSTV falling edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gstv_fall_e[6:0]**: Even gSTV falling edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gckv_rise_o[6:0]**: Odd gCKV rising edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gckv_rise_e[6:0]**: Even gCKV rising edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gckv_fall_o[6:0]**: Odd gCKV falling edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **gckv_fall_e[6:0]**: Even gCKV falling edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **grst_rise_o[6:0]**: Odd gRST rising edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns

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	//2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **grst_rise_e[6:0]**: Even gRST rising edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **grst_fall_o[6:0]**: Odd gRST falling edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us **grst_fall_e[6:0]**: Even gRST falling edge fine tune. //2'b000000: 0T //2'b000001: 1T : 25ns //2'b000010: 2T : 50ns //2'b000011: 3T : 75ns ... //2'b111111: 127T : 3.2us
Restriction	-

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4.11. SET_GIP_RISE_FALL_DIR: Set GIP Information (Page2-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	0	1	0	0	0	0	1	0	42
Parameter 1	R/W	esd_clr_lastline_en	tianma_sel	dummy_en	en_ckv_all	sft_goe_scale	goa_fall_sft_drct	goa_rise_sft_drct	flc_stagger_en	82
Description	This command is used to set GIP and 1 time OTP. **flc_stagger_en**: Enable flca and flcb stagger. //0: disable. //1: enable. **goa_rise_sft_drct**: Direction of rising edge fine tune for gSTV / gCKV / gRST. //0: backward. //1: forward. **goa_fall_sft_drct**: Direction of falling edge fine tune for gSTV / gCKV / gRST. //0: backward. //1: forward. Example:  **sft_goe_scale**: Scale of gSTV and gCKV width fine tune. //0: osc period x 8. //1: osc period x 16. **en_ckv_all**: Enable gCKV all on at power-on / waiteof. //0: disable. //1: enable. **dummy_en**: enable pre gCKV dummy pulse. //0: disable. //1: enable. **tianma_sel**: gRST generate before gSTV for tianma pannel. //0: disable. //1: enable. **esd_clr_lastline_en**: esd occure last-line clear ckv_sft_enable aviod signal too long. //0: disable. //1: enable.									
Restriction	-									

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4.12. SET_GIP_INF: Set GIP Information (Page2-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	r_stv_ckv_dis [2:1]		stv_abn _clr	ckv_ab n_clr	esd_clr _goa_e n	rst_cp_tianma[2:0]			30
Description	This command is used to set GIP and 1 time OTP. **rst_cp_tianma[2:0]**: Shift gRST rising edge for Tianma pannel timing. (line base) //2'b000: 0H //2'b001: 1H //2'b010: 2H //2'b011: 3H ... //2'b111: 7H **esd_clr_goa_en**: Clear GOA enable when ESD detected. //0: disable. //1: enable. **ckv_abn_clr**: Enable gCKV abnormal keep high clear. //0: disable. //1: enable. **stv_abn_clr**: Enable gSTV abnormal keep high clear. //0: disable. //1: enable. **r_stv_ckv_dis[2:1]**: Extend shift gSTV1 and gCKV rising edge. (MSB 2 bits, gSTV and gCKV shift total 3 bits) //2'b00: 0H //2'b01: 2H //2'b10: 4H //2'b11: no use									
Restriction	-									

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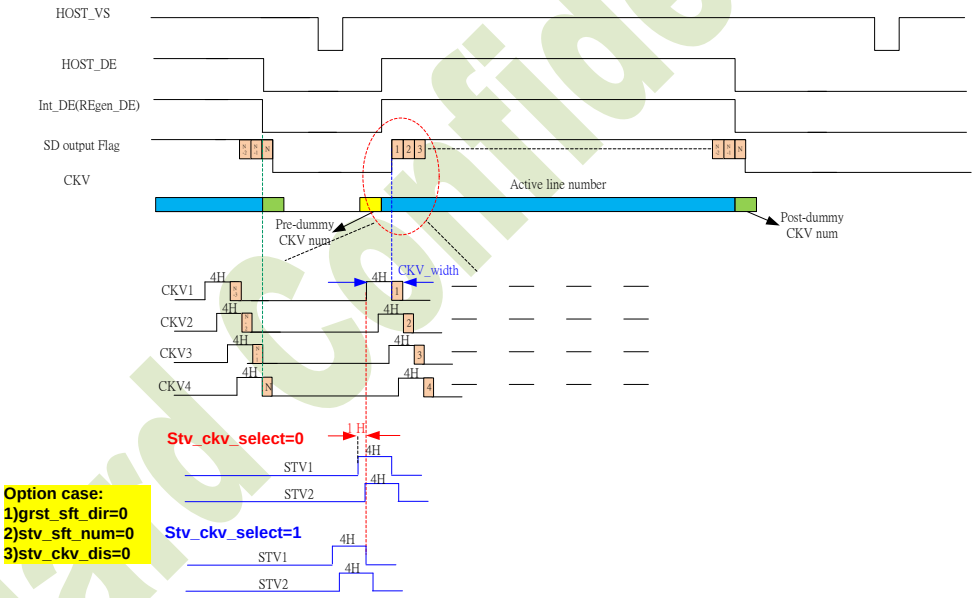
4.13. SET_GIP_STV_CKV: Set GIP Information (Page2-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	0	1	0	0	0	1	0	0	44
Parameter 1	R/W	esd_rst_sel	r_eq_stv_rstb_enb	stv_ckv_dis[0]	stv_ckv_select	ckv_blkpulse_en	goa_msk_sel [1:0]		learn_info_sel	01

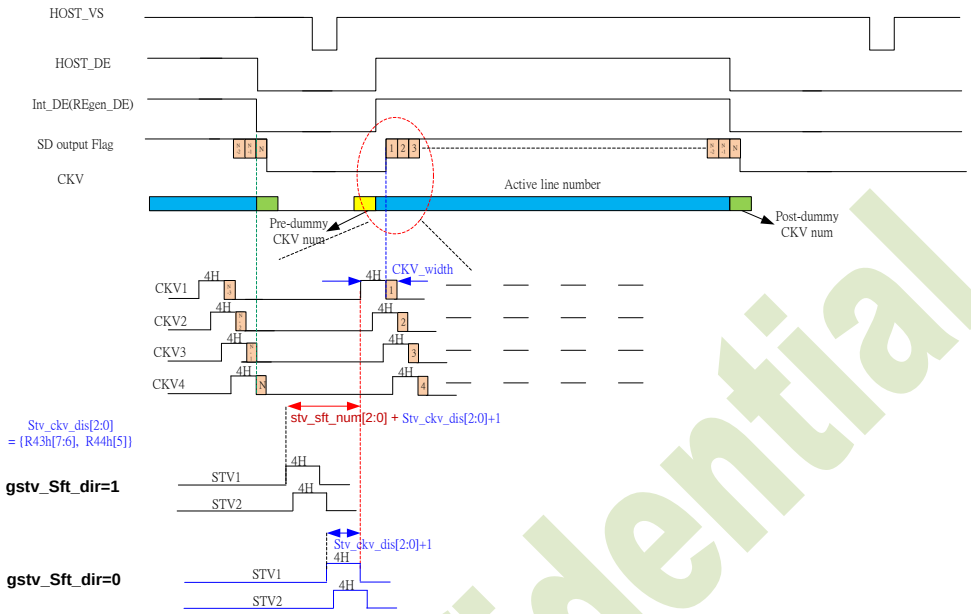
Description

This command is used to set GIP and 1 time OTP.

learn_info_sel: learning information selection.
//0: select length from reginster.
//1: select length from de_regen.
goa_msk_sel[1:0]: GOA esd mask finish line selection.
//2'b00: no use
//2'b01: first line.
//2'b10: no use
//2'b11: last line.
ckv_blkpulse_en: Enable gCKV all on before active area.
//0: disable.
//1: enable.
stv_ckv_select: gSTV and gCKV pulse releation.
//0: overlap.
//1: stagger.



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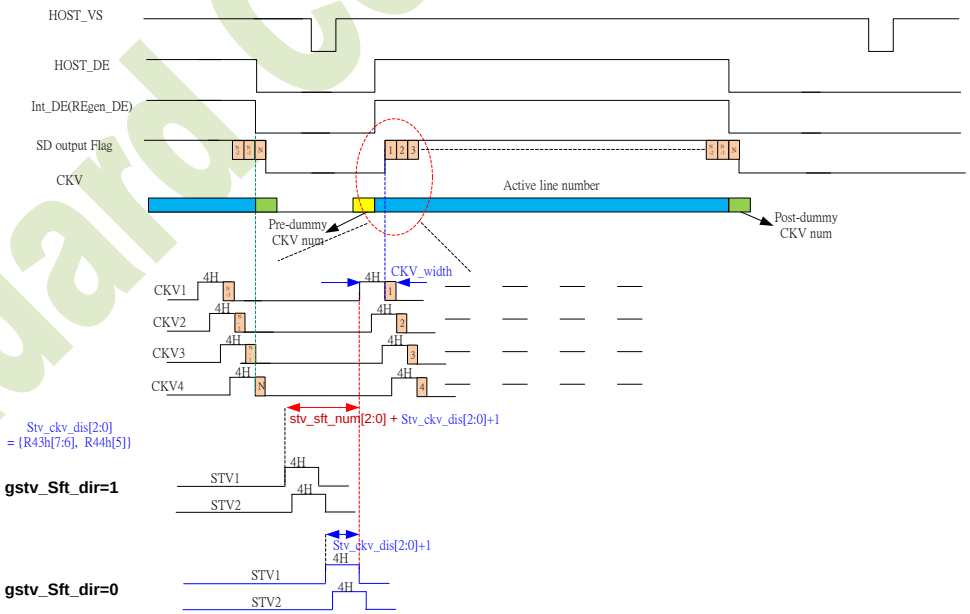
	stv_ckv_dis[0]: Shift gSTV and gCKV rising edge. (LSB bit, gSTV and gCKV shift total 3 bits) //0: 0H //1: 1H  **r_eq_stv_rstb_enb**: Enable normal state to freerun state EQ signal go back to logic High. //0: disable. //1: enable. **esd_rst_sel**: Enable mask gRST signal when ESD detected. //0: disable mask gRST. //1: enable mask gRST.
Restriction	-

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4.14. SET_GIP_INF: Set GIP Information (Page2-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	ag_act_length[7:0]								80
Description	This command is used to set GIP and 1 time OTP. **ag_act_length[7:0]**: active line number at agen mode. (LSB 7 bits) //2'b00000000: 0H //2'b00000001: 1H //2'b00000010: 2H //2'b00000011: 3H ... //2'b11111111: 255H									
Restriction	-									

4.15. SET_GIP_STV_SFT: Set GIP Information (Page2-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	gstv_sft_dir	gstv_sft_num[2:0]			ag_act_length[11:8]				07
Description	This command is used to set GIP and 1 time OTP. **gstv_sft_dir**: Enable STV1 shfit forward to V-porch region. **gstv_sft_num[2:0]**: To set shfit line number. (0~7 lines). Case 1: gstv_sft_dir=0, the distance is fixed 1H between STV1 and CKV1. Case 2: gstv_sft_dir=1, the distance is defined by stv_sft_num between STV1 and CKV1.  **ag_act_length[11:8]**: active line number at agen mode. (MSB 4 bits) //2'b0000: 255H //2'b0001: 256H //2'b0010: 257H //2'b0011: 258H									
Restriction	-									

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4.16. SET_GIP_INF: Set GIP Information (Page2-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	ag_vblk_length[7:0]								37
Description	This command is used to set GIP and 1 time OTP. **ag_vblk_length[7:0]**: vblanking line length number at agen mode. (LSB 7 bits) //2'b00000000: 0H //2'b00000001: 1H //2'b00000010: 2H //2'b00000011: 3H ... //2'b11111111: 255H									
Restriction	-									

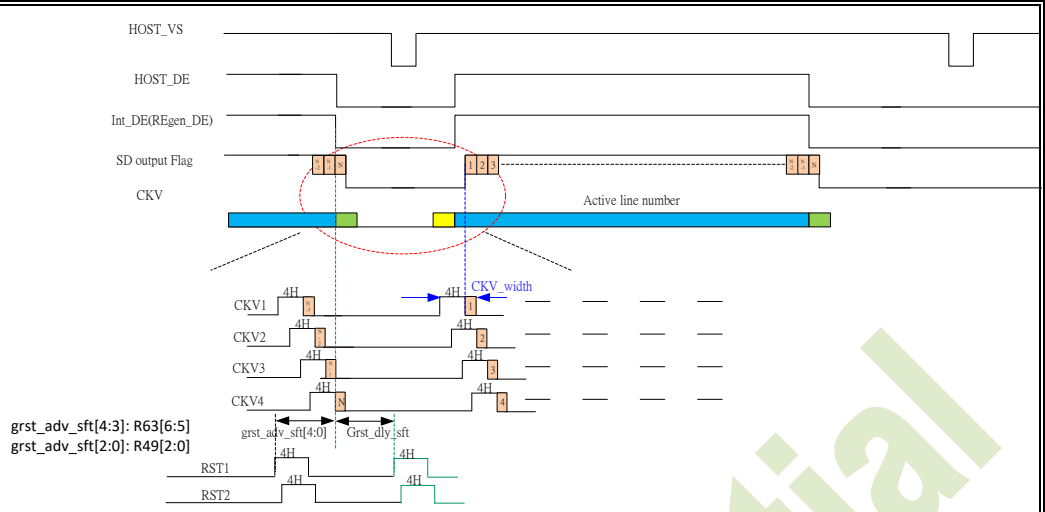
4.17. SET_GIP_INF: Set GIP Information (Page2-R48h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	vblk_line_learn_num[3:0]				ag_vblk_length[11:8]				30
Description	This command is used to set GIP and 1 time OTP. **ag_vblk_length[11:8]**: vblanking line length number at agen mode. (MSB 4 bits) //2'b0000: 255H //2'b0001: 256H //2'b0010: 257H //2'b0011: 258H **vblk_line_learn_num[3:0]**: select vblk line number for learning H length.									
Restriction	-									

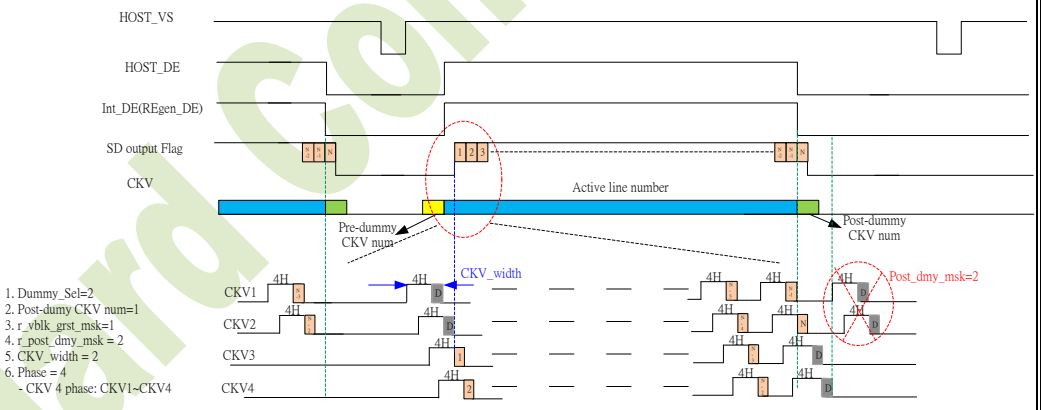
4.18. SET_GIP_CKV_MASK/RST_ADV: Set GIP Information (Page2-R49h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	r_vblk_grst_msk	post_dmy_msk[3:0]			grst_adv_sft[2:0]				84
Description	This command is used to set GIP and 1 time OTP. **grst_adv_sft[2:0]**: gRST advance shift line follow last data. //2'b000: no shift //2'b001: forward 1H //2'b010: forward 2H //2'b011: forward 3H ... //2'b011: forward 15H Example									

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post_dmy_msk[3:0]: post dummy gCKV pulse group maske.
//2'b000: no mask
//2'b001: mask last ckv
//2'b010: mask second last ckv
//2'b011: mask third last ckv
...
//2'b011: mask seventh last ckv
r_vblk_grst_msk: Enable mask for active length less then 64 lines.
//0: disable.
//1: enable.
Example



Restriction	-

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4.19. SET_GIP_EQF: Set GIP Information (Page2-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	poff_all_gate_on_en	gip_rising_eq_type	gip_sel_eq_en_r [1:0]		gip_sel_eq_en_l [1:0]		gip_pweq_en_r	gip_pweq_en_l	BF
Description	This command is used to set GIP and 1 time OTP. **gip_pweq_en_l**: GOA left side anallog EQ enable. //0: disable. //1: enable. **gip_pweq_en_r**: GOA right side anallog EQ enable. //0: disable. //1: enable. **gip_sel_eq_en_l[1:0]**: select GOA left side EQ1 / EQ2 powet-type. //2'b00: floating/floating. //2'b01: AVDD/floating. //2'b10: floating/GND. //2'b11: AVDD/GND. **gip_sel_eq_en_r[1:0]**: select GOA right side EQ1 / EQ2 powet-type. //2'b00: floating/floating. //2'b01: AVDD/floating. //2'b10: floating/GND. //2'b11: AVDD/GND. **gip_rising_eq_type**: Select eq power-type as GND or VSP. //0: GND. //1: VSP. **poff_all_gate_on_en**: Enable power-off all gate on. //0: disable. //1: enable.									
Restriction	-									

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4.20. SET_GIP_Driving ability: Set GIP Information (Page2-R4Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	0	1	4B
Parameter 1	**R/W**	gip_dr2_r[1:0]		gip_dr2_l[1:0]		gip_dr1_r[1:0]		gip_dr1_l[1:0]		AA
Description	This command is used to set GIP and 1 time OTP. **gip_dr1_l[1:0]**: GOA left side dr1 strength. **gip_dr1_r[1:0]**: GOA right side dr1 strength. **gip_dr2_l[1:0]**: GOA left side dr2 strength. **gip_dr2_r[1:0]**: GOA right side dr2 strength.									
Restriction	-									

4.21. SET_GIP_Nor_SEN: Set GIP Information (Page2-R4Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	0	0	4C
Parameter 1	**R/W**	gip_tp_sen_l[7:0]								00
Parameter 2	**R/W**	gip_tp_sen_r[7:0]								00

Description

This command is used to set GIP and 1 time OTP.

gip_tp_sen_l[7:0]: during touch sensing at slpout + display on.
GIP_TP_SEN_L[x]=1 GOUT group keep VGHO
GIP_TP_SEN_L[x]=0 GOUT group keep VGLO

gip_tp_sen_r[7:0]: during touch sensing at slpout + display on.
GIP_TP_SEN_R[x]=1 GOUT group keep VGHO
GIP_TP_SEN_R[x]=0 GOUT group keep VGLO

Function	Setting	Group definition	Output state
gip_tp_sen_l/r[7]	0	ETV1~4	VGLO
	1		VGHO
gip_tp_sen_l/r[6]	0	VGL	VGLO
	1		VGHO
gip_tp_sen_l/r[5]	0	VGH	VGLO
	1		VGHO
gip_tp_sen_l/r[4]	0	GCL/ GCH	VGLO
	1		VGHO
gip_tp_sen_l/r[3]	0	STV0~8	VGLO
	1		VGHO
gip_tp_sen_l/r[2]	0	CKV1~16	VGLO
	1		VGHO
gip_tp_sen_l/r[1]	0	RST0~10	VGLO
	1		VGHO
gip_tp_sen_l/r[0]	0	FLCA/ FLCB/ Sub_FLCA/ Sub_FLCB	VGLO
	1		VGHO

Restriction: -

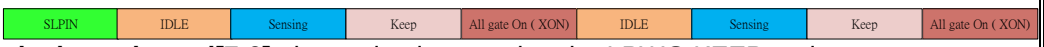
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4.22. SET_GIP_LPWG_KEEP: Set GIP Information (Page2-R4Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	1	0	4E
Parameter 1	**R/W**	gip_lpwg_keep_l[7:0]								00
Parameter 2	**R/W**	gip_lpwg_keep_r[7:0]								00

Description

This command is used to set GIP and 1 time OTP.



gip_lpwg_keep_l[7:0]: the setting is control under LPWG KEEP region.
GIP_LPWG_KEEP_L[x]=1 GOUT group keep VGHO
GIP_LPWG_KEEP_L [x]=0 GOUT group keep VGLO

gip_lpwg_keep_r[7:0]: the setting is control under LPWG KEEP region.
GIP_LPWG_KEEP_R [x]=1 GOUT group keep VGHO
GIP_LPWG_KEEP_R [x]=0 GOUT group keep VGLO

Function	Setting	Group definition	Output state
gip_lpwg_keep_l/r[7]	0	ETV1~4	VGLO
	1		VGHO
gip_lpwg_keep_l/r[6]	0	VGL	VGLO
	1		VGHO
gip_ lpwg_keep _l/r[5]	0	VGH	VGLO
	1		VGHO
gip_ lpwg_keep _l/r[4]	0	GCL/ GCH	VGLO
	1		VGHO
gip_ lpwg_keep _l/r[3]	0	STV0~8	VGLO
	1		VGHO
gip_ lpwg_keep _l/r[2]	0	CKV1~16	VGLO
	1		VGHO
gip_ lpwg_keep _l/r[1]	0	RST0~10	VGLO
	1		VGHO
gip_ lpwg_keep _l/r[0]	0	FLCA/ FLCB/ Sub_FLCA/ Sub_FLCB	VGLO
	1		VGHO

Restriction	-

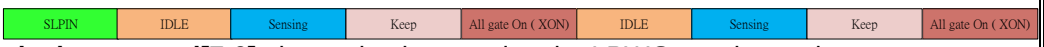
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4.23. SET_GIP_LPWG_SEN: Set GIP Information (Page2-R50h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	0	0	50
Parameter 1	**R/W**	gip_lpwg_sen_l[7:0]								00
Parameter 2	**R/W**	gip_lpwg_sen_r[7:0]								00

Description

This command is used to set GIP and 1 time OTP.



gip_lpwg_sen_l[7:0]: the setting is control under LPWG sensing region.
GIP_LPWG_SEN_L[x]=1 GOUT group keep VGHO
GIP_LPWG_ SEN_L [x]=0 GOUT group keep VGLO

gip_lpwg_sen_r[7:0]: the setting is control under LPWG sensing region.
GIP_LPWG_SEN_R [x]=1 GOUT group keep VGHO
GIP_LPWG_SEN_R [x]=0 GOUT group keep VGLO

Function	Setting	Group definition	Output state
gip_lpwg_sen_l/r[7]	0	ETV1~4	VGLO
	1		VGHO
gip_lpwg_sen_l/r[6]	0	VGL	VGLO
	1		VGHO
gip_ lpwg_sen_l/r[5]	0	VGH	VGLO
	1		VGHO
gip_ lpwg_ sen_l/r[4]	0	GCL/ GCH	VGLO
	1		VGHO
gip_ lpwg_ sen_l/r[3]	0	STV0~8	VGLO
	1		VGHO
gip_ lpwg_ sen_l/r[2]	0	CKV1~16	VGLO
	1		VGHO
gip_ lpwg_ sen_l/r[1]	0	RST0~10	VGLO
	1		VGHO
gip_ lpwg_ sen_l/r[0]	0	FLCA/ FLCB/ Sub_FLCA/ Sub_FLCB	VGLO
	1		VGHO

Restriction: -

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4.24. SET_GIP_XON1: Set GIP Information (Page2-R55h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	0	1	0	1	0	1	0	1	55
Parameter 1	R/W	gip_lpwg_xon_gnd_l[7:0]								00
Parameter 2	R/W	gip_lpwg_xon_gnd_r[7:0]								00
Description	This command is used to set GIP and 1 time OTP.  **gip_lpwg_xon_gnd_l[7:0]**: the setting is control under LPWG all gate on(XON) region. **gip_lpwg_xon_gnd_r[7:0]**: the setting is control under LPWG all gate on(XON) region. Please refer R56h for detail.									
Restriction	-									

4.25. SET_GIP_XON2: Set GIP Information (Page2-R56h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	0	1	0	1	0	1	1	0	56
Parameter 1	R/W	gip_lpwg_xon_hl_l[7:0]								00
Parameter 2	R/W	gip_lpwg_xon_hl_r[7:0]								00

Description

This command is used to set GIP and 1 time OTP.



gip_lpwg_xon_hl_l[7:0]: the setting is control under LPWG all gate on(XON) region.
gip_lpwg_xon_hl_r[7:0]: the setting is control under LPWG all gate on(XON) region.

GIP_CTRL[1:0] = {gip_lpwg_xon_gnd, gip_lpwg_xon_hl}
- 2'b00: VSP
- 2'b01: GND
- 2'b10: VGHO
- 2'b11: VGLO

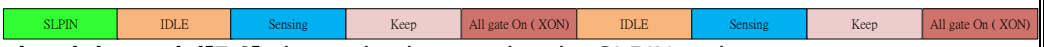
Function	Setting	Group definition	Output state
{R55h[7], R56h[7]} gip_lpwg_xon_gnd_l/r[7] gip_lpwg_xon_hl_l/r[7]	00	ETV1~4	VSP
	01		GND
	10		VGHO
	11		VGLO
{R55h[6], R56h[6]} gip_lpwg_xon_gnd_l/r[6] gip_lpwg_xon_hl_l/r[6]	00	VGL	VSP
	01		GND
	10		VGHO
	11		VGLO
{R55h[5], R56h[5]} gip_lpwg_xon_gnd_l/r[5] gip_lpwg_xon_hl_l/r[5]	00	VGH	VSP
	01		GND
	10		VGHO
	11		VGLO
{R55h[4], R56h[4]} gip_lpwg_xon_gnd_l/r[4] gip_lpwg_xon_hl_l/r[4]	00	GCL/ GCH	VSP
	01		GND
	10		VGHO
	11		VGLO
{R55h[3], R56h[3]} gip_lpwg_xon_gnd_l/r[3] gip_lpwg_xon_hl_l/r[3]	00	STV0~8	VSP
	01		GND
	10		VGHO
	11		VGLO
{R55h[2], R56h[2]} gip_lpwg_xon_gnd_l/r[2] gip_lpwg_xon_hl_l/r[2]	00	CKV1~CKV16	VSP
	01		GND
	10		VGHO
	11		VGLO

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	{R55h[1], R56h[1]} gip_lpwg_xon_gnd_l/r[1] gip_lpwg_xon_hl_l/r[1]	00	RST0~10	VSP
		01		GND
		10		VGHO
		11		VGLO
	{R55h[0], R56h[0]} gip_lpwg_xon_gnd_l/r[0] gip_lpwg_xon_hl_l/r[0]	00	FLCA/ FLCB/ Sub_FLCA/ Sub_FLCB	VSP
		01		GND
		10		VGHO
		11		VGLO
Restriction	-			

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4.26. SET_GIP_SLPIN1: Set GIP Information (Page2-R57h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	0	1	0	1	0	1	1	1	57
Parameter 1	R/W	gip_slpin_gnd_l[7:0]								00
Parameter 2	R/W	gip_slpin_gnd_r[7:0]								00
Description	This command is used to set GIP and 1 time OTP.  **gip_slpin_gnd_l[7:0]**: the setting is control under SLPIN region. **gip_slpin_gnd_r[7:0]**: the setting is control under SLPIN region. Please refer R58h for detail.									
Restriction	-									

4.27. SET_GIP_SLPIN2: Set GIP Information (Page2-R58h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	0	1	0	1	1	0	0	0	58
Parameter 1	R/W	gip_slpin_hl_l[7:0]								FF
Parameter 2	R/W	gip_slpin_hl_r[7:0]								FF

Description

This command is used to set GIP and 1 time OTP.



gip_slpin_hl_l[7:0]: the setting is control under SLPIN region.
gip_slpin_hl_r[7:0]: the setting is control under SLPIN region.

GIP_CTRL[1:0] = {gip_slpin_gnd, gip_slpin_hl}
- 2'b00: VSP
- 2'b01: GND
- 2'b10: VGHO
- 2'b11: VGLO

Function	Setting	Group definition	Output state
{R57h[7], R58h[7]} gip_slpin_gnd_l/r[7], gip_ slpin_hl_/r[7]	00	ETV1~4	VSP
	01		GND
	10		VGHO
	11		VGLO
{R57h[6], R58h[6]} gip_slpin_gnd_l/r[6], gip_ slpin_hl_/r[6]	00	VGL	VSP
	01		GND
	10		VGHO
	11		VGLO
{R57h[6], R58h[6]} gip_slpin_gnd_l/r[6], gip_ slpin_hl_/r[6]	00	VGL	VSP
	01		GND
	10		VGHO
	11		VGLO
{R57h[5], R58h[5]} gip_slpin_gnd_l/r[5], gip_ slpin_hl_/r[5]	00	VGH	VSP
	01		GND
	10		VGHO
	11		VGLO
{R57h[4], R58h[4]} gip_slpin_gnd_l/r[4], gip_ slpin_hl_/r[4]	00	GCL/ GCH	VSP
	01		GND
	10		VGHO
	11		VGLO
{R57h[3], R58h[3]} gip_slpin_gnd_l/r[3], gip_ slpin_hl_/r[3]	00	STV0~8	VSP
	01		GND
	10		VGHO
	11		VGLO

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	{R57h[2], R58h[2]} gip_slpin_gnd_l/r[2], gip_ slpin_hl_/r[2]	00	CKV1~CKV16	VSP
		01		GND
		10		VGHO
		11		VGLO
	{R57h[1], R58h[1]} gip_slpin_gnd_l/r[1], gip_ slpin_hl_/r[1]	00	RST0~10	VSP
		01		GND
		10		VGHO
		11		VGLO
	{R57h[0], R58h[0]} gip_slpin_gnd_l/r[0], gip_ slpin_hl_/r[0]	00	FLCA/ FLCB/ Sub_FLCA/ Sub_FLCB	VSP
		01		GND
		10		VGHO
		11		VGLO
Restriction	-			

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4.28. SET_IGZO_GIP_INF: Set IGZO GIP Information (Page2-R59h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	1	59
Parameter 1	**R/W**	flc_hole_vgh_dly_sel[1:0]		gip_igzo_intsw_r	gip_igzo_intsw_l	gip_igzo_intdis_r	gip_igzo_intdis_l	gip_igzo_en_r	gip_igzo_en_l	30
Description	This command is used to set IGZO GIP and 1 time OTP. **gip_igzo_en_l**: Enable GIP left side IGZO discharge. //0: disable. //1: enable. **gip_igzo_en_r**: Enable GIP right side IGZO discharge. //0: disable. //1: enable. **gip_igzo_intdis_l**: Enable GIP left side IGZO internal discharge. //0: disable. //1: enable. **gip_igzo_intdis_r**: Enable GIP right side IGZO internal discharge. //0: disable. //1: enable. **gip_igzo_intsw_l**: Enable GIP left side IGZO internal switch. //0: disable. //1: enable. **gip_igzo_intsw_r**: Enable GIP right side IGZO internal switch. //0: disable. //1: enable. **flc_hole_vgh_dly_sel[1:0]**: Flc pull to VGH delay timing select. //2'b00: non-delay. //2'b01: delay 84T. 1T= 25ns //2'b10: delay 171T. 1T= 25ns //2'b11: delay 255T. 1T= 25ns									
Restriction	-									

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4.29. SET_GS=0_GCOUTL2: Set GIP Information (Page2-R5Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	0	5A
Parameter 1	**R/W**	-	-	forward_goutl1_sel[5:0]						23
Parameter 2	**R/W**	-	-	forward_goutl2_sel[5:0]						23
Parameter 3	**R/W**	-	-	forward_goutl3_sel[5:0]						23
Parameter 4	**R/W**	-	-	forward_goutl4_sel[5:0]						23
Parameter 5	**R/W**	-	-	forward_goutl5_sel[5:0]						23
Parameter 6	**R/W**	-	-	forward_goutl6_sel[5:0]						23
Parameter 7	**R/W**	-	-	forward_goutl7_sel[5:0]						23
Parameter 8	**R/W**	-	-	forward_goutl8_sel[5:0]						23
Parameter 9	**R/W**	-	-	forward_goutl9_sel[5:0]						23
Parameter 10	**R/W**	-	-	forward_goutl10_sel[5:0]						23
Parameter 11	**R/W**	-	-	forward_goutl11_sel[5:0]						23
Description	This command is used to set GIP and 1 time OTP. **forward_goutln_sel[5:0]**: forward left side flexible GIP output selection. n=1~11 The mapping table can refer R5Bh.									
Restriction	-									

4.30. SET_GS=0_GCOUTL2: Set GIP Information (Page2-R5Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	1	5B
Parameter 1	**R/W**	-	-	forward_goutl12_sel[5:0]						23
Parameter 2	**R/W**	-	-	forward_goutl13_sel[5:0]						23
Parameter 3	**R/W**	-	-	forward_goutl14_sel[5:0]						23
Parameter 4	**R/W**	-	-	forward_goutl15_sel[5:0]						23
Parameter 5	**R/W**	-	-	forward_goutl16_sel[5:0]						23
Parameter 6	**R/W**	-	-	forward_goutl17_sel[5:0]						23
Parameter 7	**R/W**	-	-	forward_goutl18_sel[5:0]						23
Parameter 8	**R/W**	-	-	forward_goutl19_sel[5:0]						23
Parameter 9	**R/W**	-	-	forward_goutl20_sel[5:0]						23
Parameter 10	**R/W**	-	-	forward_goutl21_sel[5:0]						23
Parameter 11	**R/W**	-	-	forward_goutl22_sel[5:0]						23

Description

This command is used to set GIP and 1 time OTP.

forward_goutln_sel[5:0]: forward left side flexible GIP output selection.
n=12~22.

GOUT_SEL	output	GOUT_SEL	output	GOUT_SEL	output
0x00	STV1	0x14	CKV13	0x28	RST9
0x01	STV2	0x15	CKV14	0x29	RST10
0x02	STV3	0x16	CKV15	0x2A	scan_hole_vgh
0x03	STV4	0x17	CKV16	0x2B	scan_hole_vgh_inv
0x04	STV5	0x18	RST1	0x2C	ETV1
0x05	STV6	0x19	RST2	0x2D	ETV2
0x06	STV7	0x1A	RST3	0x2E	ETV3
0x07	STV8	0x1B	RST4	0x2F	ETV4

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	0x08	CKV1	0x1C	RST5	0x30	STV0	
	0x09	CKV2	0x1D	RST6	0x31	scan_hole	
	0x0A	CKV3	0x1E	RST7	0x32	scan_hole_inv	
	0x0B	CKV4	0x1F	RST8	0x33	RST0	
	0x0C	CKV5	0x20	FLC1	0x34	RST0_INV	
	0x0D	CKV6	0x21	FLC2	0x35	VGH_INV	
	0x0E	CKV7	0x22	VGH	0x36	VGL_INV	
	0x0F	CKV8	0x23	VGL			
	0x10	CKV9	0x24	GCH			
	0x11	CKV10	0x25	GCL			
	0x12	CKV11	0x26	SUB_FLC1			
	0x13	CKV12	0x27	SUB_FLC2			
Restriction	-						

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4.31. SET_GS=0_GCOUTR1: Set GIP Information (Page2-R5Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	0	5C
Parameter 1	**R/W**	-	-	fordward_goutr1_sel[5:0]						23
Parameter 2	**R/W**	-	-	fordward _goutr2_sel[5:0]						23
Parameter 3	**R/W**	-	-	fordward _goutr3_sel[5:0]						23
Parameter 4	**R/W**	-	-	fordward _goutr4_sel[5:0]						23
Parameter 5	**R/W**	-	-	fordward _goutr5_sel[5:0]						23
Parameter 6	**R/W**	-	-	fordward _goutr6_sel[5:0]						23
Parameter 7	**R/W**	-	-	fordward _goutr7_sel[5:0]						23
Parameter 8	**R/W**	-	-	fordward _goutr8_sel[5:0]						23
Parameter 9	**R/W**	-	-	fordward _goutr9_sel[5:0]						23
Parameter 10	**R/W**	-	-	fordward _goutr10_sel[5:0]						23
Parameter 11	**R/W**	-	-	fordward _goutr11_sel[5:0]						23
Description	This command is used to set GIP and 1 time OTP. **forward_goutrn_sel[5:0]**: forward right side flexible GIP output selection. n=1~11 The mapping table can refer R5Dh.									
Restriction	-									

4.32. SET_GS=0_GCOUTR2: Set GIP Information (Page2-R5Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	1	5D
Parameter 1	**R/W**	-	-	fordward _goutr12_sel[5:0]						23
Parameter 2	**R/W**	-	-	fordward _goutr13_sel[5:0]						23
Parameter 3	**R/W**	-	-	fordward _goutr14_sel[5:0]						23
Parameter 4	**R/W**	-	-	fordward _goutr15_sel[5:0]						23
Parameter 5	**R/W**	-	-	fordward _goutr16_sel[5:0]						23
Parameter 6	**R/W**	-	-	fordward _goutr17_sel[5:0]						23
Parameter 7	**R/W**	-	-	fordward _goutr18_sel[5:0]						23
Parameter 8	**R/W**	-	-	fordward _goutr19_sel[5:0]						23
Parameter 9	**R/W**	-	-	fordward _goutr20_sel[5:0]						23
Parameter 10	**R/W**	-	-	fordward _goutr21_sel[5:0]						23
Parameter 11	**R/W**	-	-	fordward _goutr22_sel[5:0]						23

Description

This command is used to set GIP and 1 time OTP.

forward_goutrn_sel[5:0]: forward right side flexible GIP output selection.
n=12~22.

GOUT_SEL	output	GOUT_SEL	output	GOUT_SEL	output
0x00	STV1	0x14	CKV13	0x28	RST9
0x01	STV2	0x15	CKV14	0x29	RST10
0x02	STV3	0x16	CKV15	0x2A	scan_hole_vgh
0x03	STV4	0x17	CKV16	0x2B	scan_hole_vgh_inv
0x04	STV5	0x18	RST1	0x2C	ETV1
0x05	STV6	0x19	RST2	0x2D	ETV2
0x06	STV7	0x1A	RST3	0x2E	ETV3
0x07	STV8	0x1B	RST4	0x2F	ETV4
0x08	CKV1	0x1C	RST5	0x30	STV0

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	0x09	CKV2	0x1D	RST6	0x31	scan_hole	
	0x0A	CKV3	0x1E	RST7	0x32	scan_hole_inv	
	0x0B	CKV4	0x1F	RST8	0x33	RST0	
	0x0C	CKV5	0x20	FLC1	0x34	RST0_INV	
	0x0D	CKV6	0x21	FLC2	0x35	VGH_INV	
	0x0E	CKV7	0x22	VGH	0x36	VGL_INV	
	0x0F	CKV8	0x23	VGL			
	0x10	CKV9	0x24	GCH			
	0x11	CKV10	0x25	GCL			
	0x12	CKV11	0x26	SUB_FLC1			
	0x13	CKV12	0x27	SUB_FLC2			
Restriction	-						

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4.33. SET_GS=1_GCOUTL1: Set GIP Information (Page2-R5Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	0	5E
Parameter 1	**R/W**	-	-	backward_goutl1_sel[5:0]						23
Parameter 2	**R/W**	-	-	backward_goutl2_sel[5:0]						23
Parameter 3	**R/W**	-	-	backward_goutl3_sel[5:0]						23
Parameter 4	**R/W**	-	-	backward_goutl4_sel[5:0]						23
Parameter 5	**R/W**	-	-	backward_goutl5_sel[5:0]						23
Parameter 6	**R/W**	-	-	backward_goutl6_sel[5:0]						23
Parameter 7	**R/W**	-	-	backward_goutl7_sel[5:0]						23
Parameter 8	**R/W**	-	-	backward_goutl8_sel[5:0]						23
Parameter 9	**R/W**	-	-	backward_goutl9_sel[5:0]						23
Parameter 10	**R/W**	-	-	backward_goutl10_sel[5:0]						23
Parameter 11	**R/W**	-	-	backward_goutl11_sel[5:0]						23
Description	This command is used to set GIP and 1 time OTP. **backward_goutln_sel[5:0]**: backward left side flexible GIP output selection. n=1~11. The mapping table can refer R5Fh									
Restriction	-									

4.34. SET_GS=1_GCOUTL2: Set GIP Information (Page2-R5Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	1	5F
Parameter 1	**R/W**	-	-	backward_goutl12_sel[5:0]						23
Parameter 2	**R/W**	-	-	backward_goutl13_sel[5:0]						23
Parameter 3	**R/W**	-	-	backward_goutl14_sel[5:0]						23
Parameter 4	**R/W**	-	-	backward_goutl15_sel[5:0]						23
Parameter 5	**R/W**	-	-	backward_goutl16_sel[5:0]						23
Parameter 6	**R/W**	-	-	backward_goutl17_sel[5:0]						23
Parameter 7	**R/W**	-	-	backward_goutl18_sel[5:0]						23
Parameter 8	**R/W**	-	-	backward_goutl19_sel[5:0]						23
Parameter 9	**R/W**	-	-	backward_goutl20_sel[5:0]						23
Parameter 10	**R/W**	-	-	backward_goutl21_sel[5:0]						23
Parameter 11	**R/W**	-	-	backward_goutl22_sel[5:0]						23

Description

This command is used to set GIP and 1 time OTP.

backward_goutln_sel[5:0]: backward left side flexible GIP output selection.
n=12~22.

GOUT_SEL	output	GOUT_SEL	output	GOUT_SEL	output
0x00	STV1	0x14	CKV13	0x28	RST9
0x01	STV2	0x15	CKV14	0x29	RST10
0x02	STV3	0x16	CKV15	0x2A	scan_hole_vgh
0x03	STV4	0x17	CKV16	0x2B	scan_hole_vgh_inv
0x04	STV5	0x18	RST1	0x2C	ETV1
0x05	STV6	0x19	RST2	0x2D	ETV2
0x06	STV7	0x1A	RST3	0x2E	ETV3
0x07	STV8	0x1B	RST4	0x2F	ETV4
0x08	CKV1	0x1C	RST5	0x30	STV0

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	0x09	CKV2	0x1D	RST6	0x31	scan_hole
	0x0A	CKV3	0x1E	RST7	0x32	scan_hole_inv
	0x0B	CKV4	0x1F	RST8	0x33	RST0
	0x0C	CKV5	0x20	FLC1	0x34	RST0_INV
	0x0D	CKV6	0x21	FLC2	0x35	VGH_INV
	0x0E	CKV7	0x22	VGH	0x36	VGL_INV
	0x0F	CKV8	0x23	VGL		
	0x10	CKV9	0x24	GCH		
	0x11	CKV10	0x25	GCL		
	0x12	CKV11	0x26	SUB_FLC1		
	0x13	CKV12	0x27	SUB_FLC2		
Restriction	-					

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4.35. SET_GS=1_GCOUTR1: Set GIP Information (Page2-R60h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	0	60
Parameter 1	**R/W**	-	-	backward_goutr1_sel[5:0]						23
Parameter 2	**R/W**	-	-	backward_goutr2_sel[5:0]						23
Parameter 3	**R/W**	-	-	backward_goutr3_sel[5:0]						23
Parameter 4	**R/W**	-	-	backward_goutr4_sel[5:0]						23
Parameter 5	**R/W**	-	-	backward_goutr5_sel[5:0]						23
Parameter 6	**R/W**	-	-	backward_goutr6_sel[5:0]						23
Parameter 7	**R/W**	-	-	backward_goutr7_sel[5:0]						23
Parameter 8	**R/W**	-	-	backward_goutr8_sel[5:0]						23
Parameter 9	**R/W**	-	-	backward_goutr9_sel[5:0]						23
Parameter 10	**R/W**	-	-	backward_goutr10_sel[5:0]						23
Parameter 11	**R/W**	-	-	backward_goutr11_sel[5:0]						23
Description	This command is used to set GIP and 1 time OTP. **backward_goutrn_sel[5:0]**: backward right side flexible GIP output selection. n=1~11. The mapping table can refer R61h									
Restriction	-									

4.36. SET_GS=1_GCOUTR2: Set GIP Information (Page2-R61h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	1	61
Parameter 1	**R/W**	-	-	backward_goutr12_sel[5:0]						23
Parameter 2	**R/W**	-	-	backward_goutr13_sel[5:0]						23
Parameter 3	**R/W**	-	-	backward_goutr14_sel[5:0]						23
Parameter 4	**R/W**	-	-	backward_goutr15_sel[5:0]						23
Parameter 5	**R/W**	-	-	backward_goutr16_sel[5:0]						23
Parameter 6	**R/W**	-	-	backward_goutr17_sel[5:0]						23
Parameter 7	**R/W**	-	-	backward_goutr18_sel[5:0]						23
Parameter 8	**R/W**	-	-	backward_goutr19_sel[5:0]						23
Parameter 9	**R/W**	-	-	backward_goutr20_sel[5:0]						23
Parameter 10	**R/W**	-	-	backward_goutr21_sel[5:0]						23
Parameter 11	**R/W**	-	-	backward_goutr22_sel[5:0]						23

Description

This command is used to set GIP and 1 time OTP.

backward_goutrn_sel[5:0]: backward right side flexible GIP output selection. n=12~22.

GOUT_SEL	output	GOUT_SEL	output	GOUT_SEL	output
0x00	STV1	0x14	CKV13	0x28	RST9
0x01	STV2	0x15	CKV14	0x29	RST10
0x02	STV3	0x16	CKV15	0x2A	scan_hole_vgh
0x03	STV4	0x17	CKV16	0x2B	scan_hole_vgh_inv
0x04	STV5	0x18	RST1	0x2C	ETV1
0x05	STV6	0x19	RST2	0x2D	ETV2
0x06	STV7	0x1A	RST3	0x2E	ETV3
0x07	STV8	0x1B	RST4	0x2F	ETV4
0x08	CKV1	0x1C	RST5	0x30	STV0

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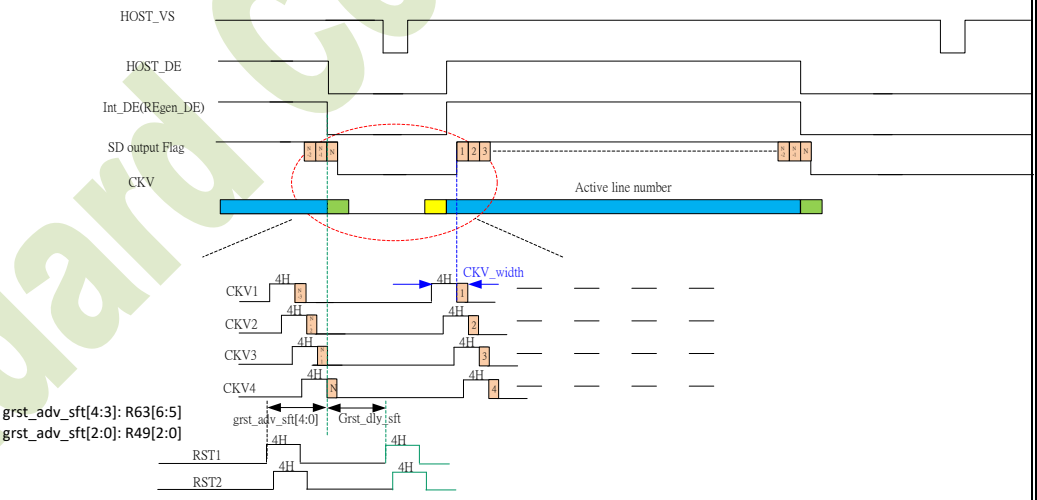
	0x09	CKV2	0x1D	RST6	0x31	scan_hole	
	0x0A	CKV3	0x1E	RST7	0x32	scan_hole_inv	
	0x0B	CKV4	0x1F	RST8	0x33	RST0	
	0x0C	CKV5	0x20	FLC1	0x34	RST0_INV	
	0x0D	CKV6	0x21	FLC2	0x35	VGH_INV	
	0x0E	CKV7	0x22	VGH	0x36	VGL_INV	
	0x0F	CKV8	0x23	VGL			
	0x10	CKV9	0x24	GCH			
	0x11	CKV10	0x25	GCL			
	0x12	CKV11	0x26	SUB_FLC1			
	0x13	CKV12	0x27	SUB_FLC2			
Restriction	-						

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4.37. SET_GCL/CKV_predmy: Set GIP Information (Page2-R62h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	0	62
Parameter 1	**R/W**	gcl_mask_num[3:0]				hole_ckv_front_dmy[3:0]				00
Description	This command is used to set GIP and 1 time OTP. **hole_ckv_front_dmy[3:0]**: select dummy gCKV numbers before scan hole. //2'b0000: no dummy ckv //2'b0001: 1 dummy ckv //2'b0010: 2 dummy ckv //2'b0011: 3 dummy ckv ... //2'b0011: 15 dummy ckv **gcl_mask_num[3:0]**: gGCL high pulse keep N line. //2'b0000: no use //2'b0001: gGCL 1 line pulse. //2'b0010: gGCL 2 line pulse. //2'b0011: gGCL 3 line pulse.									
Restriction	-									

4.38. SET_RST_ADV_SFT: Set GIP Information (Page2-R63h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	1	63
Parameter 1	**R/W**	-	grst_adv_sft[4:3]		ecoab_fix_ckv_mask	hole_ckv_back_dmy[3:0]				00
Description	This command is used to set GIP and 1 time OTP. **grst_adv_sft[4:3]**: gRST advance shift line follow last data.  **Ecoab_fix_ckv_mask**:. **hole_ckv_back_dmy[3:0]**: select dummy gCKV numbers after scan hole. //2'b0000: no dummy ckv //2'b0001: 1 dummy ckv //2'b0010: 2 dummy ckv //2'b0011: 3 dummy ckv ... //2'b0011: 15 dummy ckv									
Restriction	-									

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4.39. GIP_GAS_L_OUT: Set GCOUTL output during abnormal power off (Page2-R64h)

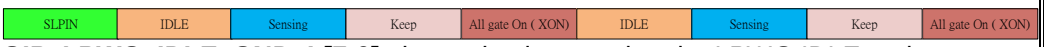
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	0	64
Parameter 1	**R/W**	GAS_L_SEL[7:0]								FF
Parameter 2	**R/W**	GAS_L_SEL[15:8]								FF
Parameter 3	**R/W**	-	-	GAS_L_SEL[21:16]						3F
Description	This command is about GCOUT output level during abnormla power off. **GAS_L_SEL[21:0]**: GCOUT_L[22:1] GAS_L_SEL[x] =1, tie to VGH GAS_L_SEL[x] =0, tie to VGL									
Restriction	-									

4.40. GIP_GAS_R_OUT: Set GCOUTL output during abnormal power off (Page2-R65h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	1	65
Parameter 1	**R/W**	GAS_R_SEL[7:0]								FF
Parameter 2	**R/W**	GAS_R_SEL[15:8]								FF
Parameter 3	**R/W**	-	-	GAS_R_SEL[21:16]						3F
Description	This command is about GCOUT output level during abnormla power off. **GAS_R_SEL[21:0]**: GCOUT_R[22:1] GAS_R_SEL[x] =1, tie to VGH GAS_R_SEL[x] =0, tie to VGL									
Restriction	-									

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4.41. SET_GIP_LPWG_IDLE1: Set GIP Information (Page2-R66h)

CMD/Pas	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	0	66
Parameter 1	**R/W**	GIP_LPWG_IDLE_GND_L[7:0]								FF
Parameter 2	**R/W**	GIP_LPWG_IDLE_GND_R[7:0]								FF
Description	This command is used to set GIP and 1 time OTP.  **GIP_LPWG_IDLE_GND_L[7:0]**: the setting is control under LPWG IDLE region. **GIP_LPWG_IDLE_GND_R [7:0]**: the setting is control under LPWG IDLE region. Please refer R67h for detail.									
Restriction	-									

4.42. SET_GIP_LPWG_IDLE2: Set GIP Information (Page2-R67h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	67
Parameter 1	**R/W**	GIP_LPWG_IDLE_HL_L[7:0]								FF
Parameter 2	**R/W**	GIP_LPWG_IDLE_HL_R[7:0]								FF

Description

This command is used to set GIP and 1 time OTP.



GIP_LPWG_IDLE_HL_L[7:0]: the setting is control under LPWG IDLE region.
GIP_LPWG_IDLE_HL_R [7:0]: the setting is control under LPWG IDLE region.

GIP_CTRL[1:0] = {GIP_LPWG_IDLE_GND, GIP_LPWG_IDLE_HL }
- 2'b00: VSP
- 2'b01: GND
- 2'b10: VGHO
- 2'b11: VGLO

Function	Setting	Group definition	Output state
{R66h[7], R67h[7]} gip_lpwg_idle_gnd_l/r[7] gip_lpwg_idle_hl_l/r[7]	00	ETV1~4	VSP
	01		GND
	10		VGHO
	11		VGLO
{R66h[6], R67h[6]} gip_lpwg_idle_gnd_l/r[6] gip_lpwg_idle_hl_l/r[6]	00	VGL	VSP
	01		GND
	10		VGHO
	11		VGLO
{R66h[5], R67h[5]} gip_lpwg_idle_gnd_l/r[5] gip_lpwg_idle_hl_l/r[5]	00	VGH	VSP
	01		GND
	10		VGHO
	11		VGLO
{R66h[4], R67h[4]} gip_lpwg_idle_gnd_l/r[4] gip_lpwg_idle_hl_l/r[4]	00	GCL/ GCH	VSP
	01		GND
	10		VGHO
	11		VGLO
{R66h[3], R67h[3]} gip_lpwg_idle_gnd_l/r[3] gip_lpwg_idle_hl_l/r[3]	00	STV0~8	VSP
	01		GND
	10		VGHO
	11		VGLO
{R66h[2], R67h[2]} gip_lpwg_idle_gnd_l/r[2] gip_lpwg_idle_hl_l/r[2]	00	CKV1~CKV16	VSP
	01		GND
	10		VGHO
	11		VGLO

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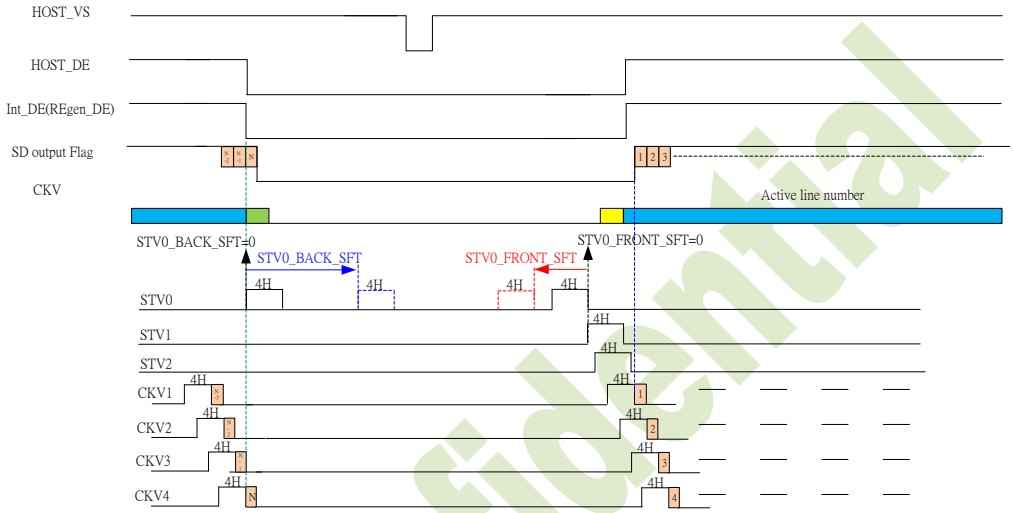
	{R66h[1], R67h[1]} gip_lpwg_idle_gnd_l/r[1] gip_lpwg_idle_hl_l/r[1]	00	RST1~10	VSP
		01		GND
		10		VGHO
		11		VGLO
	{R66h[0], R67h[0]} gip_lpwg_idle_gnd_l/r[0] gip_lpwg_idle_hl_l/r[0]	00	FLCA/ FLCB/ Sub_FLCA/ Sub_FLCB	VSP
		01		GND
		10		VGHO
		11		VGLO
Restriction	-			

4.43. SET_GIP_WIDTH_1H: Set GIP Information (Page2-R68h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	0	68
Parameter 1	**R/W**	flc_tp_keep_en	vgl_inv_tp_swp_en	vgh_inv_tp_swp_en	inx_10p95_en	rst_1line_en	ckv_1line_en	etv_1line_en	stv_1line_en	00
Description	This command is used to set GIP and 1 time OTP. **FLC_TP_KEEP_EN**: Enable FLC keep function during TP term at SLPOUT + Display ON region. **VGL_INV_TP_SWP_EN**: Switch VGL group to VGH group for VGL signal. 0: VGL signal refer VGL group at TP term 1: VGL signal refer VGH group at TP term **VGH_INV_TP_SWP_EN**: Switch VGL group to VGH group for VGL signal. 0: VGH signal refer VGH group at TP term 1: VGH signal refer VGL group at TP term **INX_10p95_EN**: **RST_1LINE_EN**: Enable RST wdith be 1H. 0: Disable 1: Enable **CKV_1LINE_EN**: Enable CKV wdith be 1H. 0: Disable 1: Enable **ETV_1LINE_EN**: Enable ETV wdith be 1H. 0: Disable 1: Enable **STV_1LINE_EN**: Enable STV wdith be 1H. 0: Disable 1: Enable									
Restriction	-									

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4.44. SET_GIP_STV0 timing: Set GIP Information (Page2-R69h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	1	69
Parameter 1	**R/W**	STV0_Front_SFT[2:0]			STV0_Back_SFT[4:0]					00
Description	This command is used to set GIP and 1 time OTP. **STV0_Front_SFT[2:0]**: Set shift distance between STV0 falling and STV1 risiing. **STV0_Back_SFT[2:0]**: Set shift distance between STV0 rising and last source data.  									
Restriction	-									

4.45. SET_GIP_STV0 timing: Set GIP Information (Page2-R6Ah)

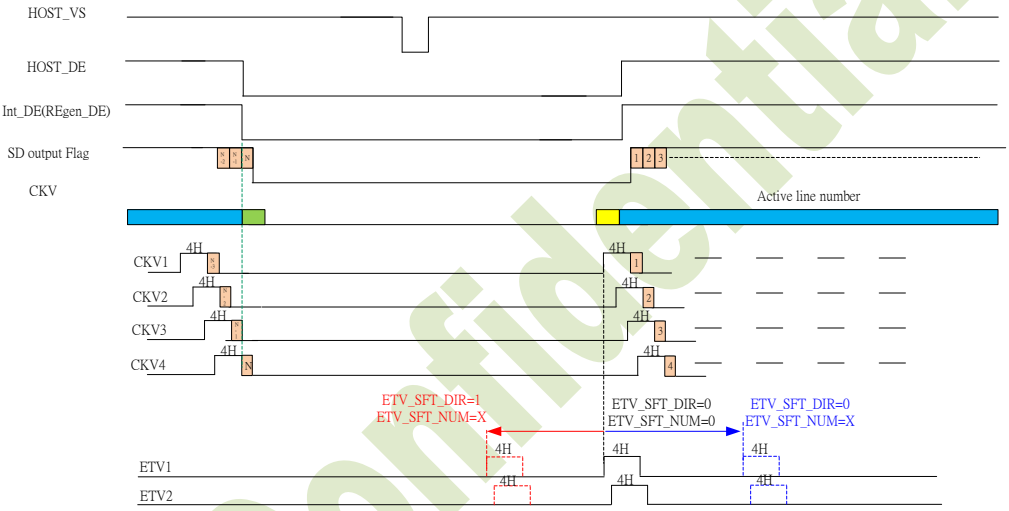
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	1	0	6A
Parameter 1	**R/W**	ETV_WIDTH[3:0]				STV0_WIDTH[3:0]				00
Description	This command is used to set GIP and 1 time OTP. **ETV_WIDTH[3:0]**: Set ETV Width **STV0_WDITH[3:0]**: Set STV0 width									

ETV_WIDTH[3:0]/ STV0_WDITH[3:0]	Width (H)
0000	Inhibit
0001	2H
0010	3H
0011	4H
0100	5H
0101	6H
●●●	+1H/ step
1111	16H

Restriction	-

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4.46. SET_GIP_STV0 timing: Set GIP Information (Page2-R6Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	1	1	6B
Parameter 1	**R/W**	ETV_EN	ETV_ABN_CLR	ETV_SFT_DIR	ETV_SFT_NUM[4:0]					00
Description	This command is used to set GIP and 1 time OTP. **ETV_EN**: Enable ETV singal. **ETV_ABN_CLR**: Clear ETV output at abnormal ETV. **ETV_SFT_DIR**: Set ETV shift direction. 0: Advance direction 1: Backward direction **ETV_SFT_NUM[4:0]**: Set shift line number.  									
Restriction	-									

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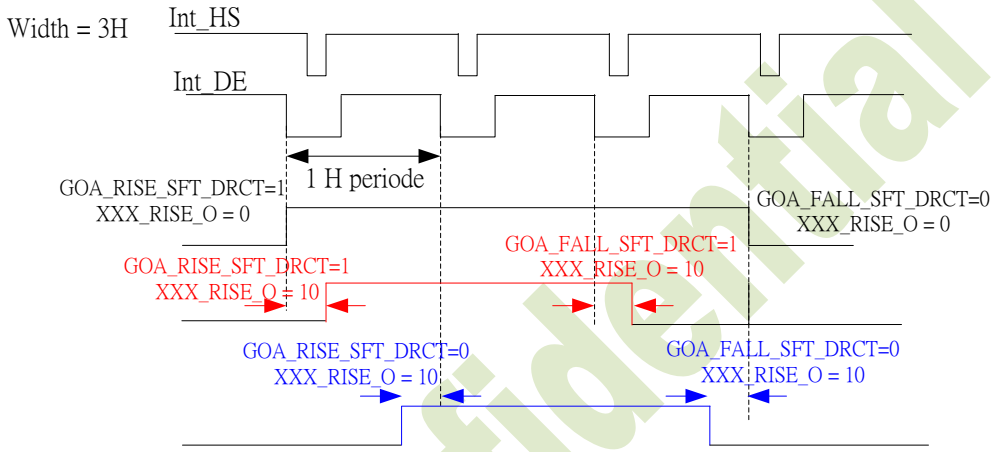
4.47. SET_GIP_RISE_FALL_TIME: Set GIP Information (Page2-R6Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	1	0	0	6C
Parameter 1	**R/W**	-	getv_rise_o[6:0]							**00**
Parameter 2	**R/W**	-	getv_rise_e[6:0]							**00**
Parameter 3	**R/W**	-	getv_fall_o[6:0]							**20**
Parameter 4	**R/W**	-	getv_fall_e[6:0]							**20**

Description

This command is used to set GIP and 1 time OTP.

The rise/ fall time need refer R42h[2:1](goa_fall_sft_drct/ goa_rise_sft_drct) to judge shift direction.



getv_rise_o[6:0]: Odd gETV rising edge fine tune.
//2'b000000: 0T
//2'b000001: 1T : 25ns
//2'b000010: 2T : 50ns
//2'b000011: 3T : 75ns
...
//2'b111111: 127T : 3.2us
getv_rise_e[6:0]: Even gETV rising edge fine tune.
//2'b000000: 0T
//2'b000001: 1T : 25ns
//2'b000010: 2T : 50ns
//2'b000011: 3T : 75ns
...
//2'b111111: 127T : 3.2us
getv_fall_o[6:0]: Odd gETV falling edge fine tune.
//2'b000000: 0T
//2'b000001: 1T : 25ns
//2'b000010: 2T : 50ns
//2'b000011: 3T : 75ns
...
//2'b111111: 127T : 3.2us
getv_fall_e[6:0]: Even gETV falling edge fine tune.
//2'b000000: 0T
//2'b000001: 1T : 25ns
//2'b000010: 2T : 50ns
//2'b000011: 3T : 75ns
...
//2'b111111: 127T : 3.2us

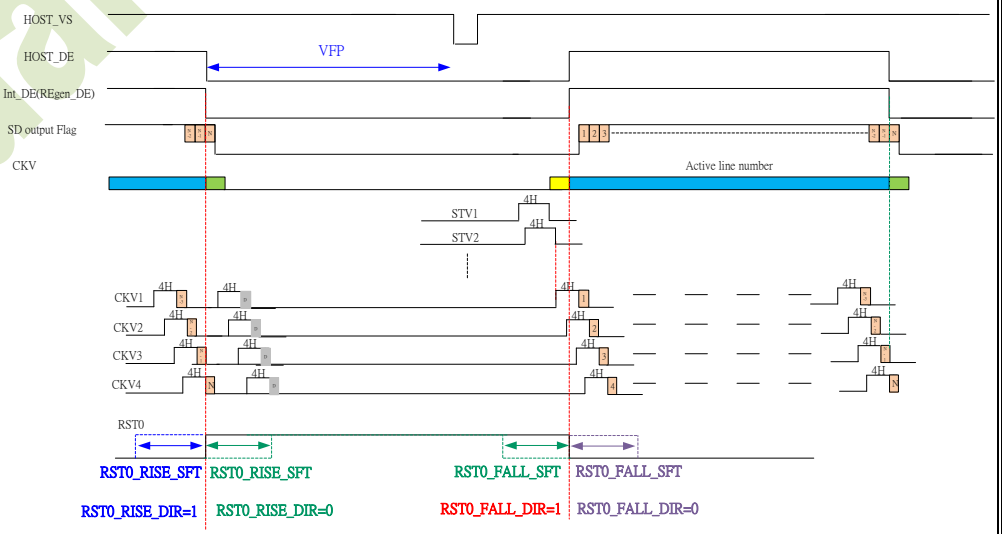
Restriction	-

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4.48. SET_GIP_RST0_TIMING1: Set GIP Information (Page2-R6Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	1	0	1	6D
Parameter 1	**R/W**	-		RST0_RISE_SFT[4:0]					RST0_RISE_DIR	**00**
Description	This command is used to set GIP and 1 time OTP. **RST0_RISE_SFT[4:0]**: Control RST shifter line number at rising edge. //2'b000000: 0 Line //2'b000001: 1 Line ... : 1 line / step //2'b111111: 31 lines **RST0_RISE_DIR**: Control RST0 rising edge shift direction. 0: RST0 rising edge is shift forard to V-porch region. 1: RST0 rising edge is shift forard to display region.. Pease refer R6Eh for detail.									
Restriction	-									

4.49. SET_ GIP_RST0_TIMING2: Set GIP Information (Page2-R6Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	1	1	0	6E
Parameter 1	**R/W**	-		RST0_FALL_SFT[4:0]					RST0_FALL_DIR	**00**
Description	This command is used to set GIP and 1 time OTP. **RST0_FALL_SFT[4:0]**: Control RST shifter line number at falling edge. //2'b000000: 0 Line //2'b000001: 1 Line ... : 1 line / step //2'b111111: 31 lines **RST0_FALL_DIR**: Control RST0 falling edge shift direction. 0: RST0 falling edge is shift forard to V-porch region. 1: RST0 falling edge is shift forard to display region.. Note: R41h_PA9(RST_RISE_OCS) & PA12(RST_FALL_OCS) for ocsd base fine tune. Generate GIP timing (STV0) - R6Dh/ R6Eh (line base) - R41h_PA9(rising)/ R41h_PA12h(falling) (OSD base)  									
Restriction	-									

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4.50. GIP_POWERON_L_OUT: Set GCOUTL output during power on (Page2-R6Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	6F
Parameter 1	**R/W**	POWERON_L_SEL[7:0]								00
Parameter 2	**R/W**	POWERON _L_SEL[15:8]								00
Parameter 3	**R/W**	PWRON_HL_REG_DIS	-	POWERON _L_SEL[21:16]						00
Description	This command is about GCOUT output level during power on sequnce. **POWERON_HL_REG_DIS**: Set GCOUT output level reference source. 0: From regsiter (R6Fh & R70h) 1: Internal control **POWERON_L_SEL[21:0]**: GCOUT_L[22:1] POWERON _L_SEL[x] =1, tie to VGH POWERON _L_SEL[x] =0, tie to VGL									
Restriction	-									

4.51. GIP_POWERON_R_OUT: Set GCOUTR output during power on (Page2-R70h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	0	70
Parameter 1	**R/W**	POWERON_R_SEL[7:0]								00
Parameter 2	**R/W**	POWERON _R_SEL[15:8]								00
Parameter 3	**R/W**	-	-	POWERON _R_SEL[21:16]						00
Description	This command is about GCOUT output level during power on sequnce. **POWERON_R_SEL[21:0]**: GCOUT_R[22:1] POWERON _R_SEL[x] =1, tie to VGH POWERON _R_SEL[x] =0, tie to VGL									
Restriction	-									

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4.52. GIP_POWEROFF_L_OUT: Set GCOUTL output during power off (Page2-R71h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	1	71
Parameter 1	**R/W**	POWEROFF_L_SEL[7:0]								FF
Parameter 2	**R/W**	POWEROFF _L_SEL[15:8]								FF
Parameter 3	**R/W**	PWROFF_HL_REG_DIS	-	POWEROFF _L_SEL[21:16]						3F
Description	This command is about GCOUT output level during power off sequnce. **POWEROFF_HL_REG_DIS**: Set GCOUT output level reference source. 0: From regsiter (R71h & R72h) 1: Internal control **POWERON_L_SEL[21:0]**: GCOUT_L[22:1] POWERON _L_SEL[x] =1, tie to VGH POWERON _L_SEL[x] =0, tie to VGL									
Restriction	-									

4.53. GIP_POWEROFF_R_OUT: Set GCOUTR output during power off (Page2-R72h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	2	0	72
Parameter 1	**R/W**	POWEROFF _R_SEL[7:0]								FF
Parameter 2	**R/W**	POWEROFF _R_SEL[15:8]								FF
Parameter 3	**R/W**	-	3F	POWEROFF _R_SEL[21:16]						3F
Description	This command is about GCOUT output level during power off sequnce. **POWERON_R_SEL[21:0]**: GCOUT_R[22:1] POWERON _R_SEL[x] =1, tie to VGH POWERON _R_SEL[x] =0, tie to VGL									
Restriction	-									

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4.54. GIP_GAS_L_OUT: Set GCOUTL output during abnormal power off (Page2-R73h)

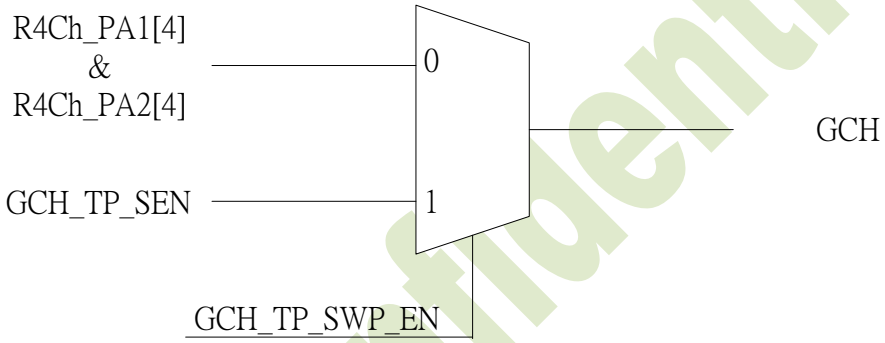
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	2	0	72
Parameter 1	**R/W**	-	-	-	-	GCH_TP_SEN	GCH_TP_SWP EN	RST0_INV_TP_SEN	RST0_INV_TP_SWP_EN	0A

Description

This command is about GCOUT output level during power off sequnce.

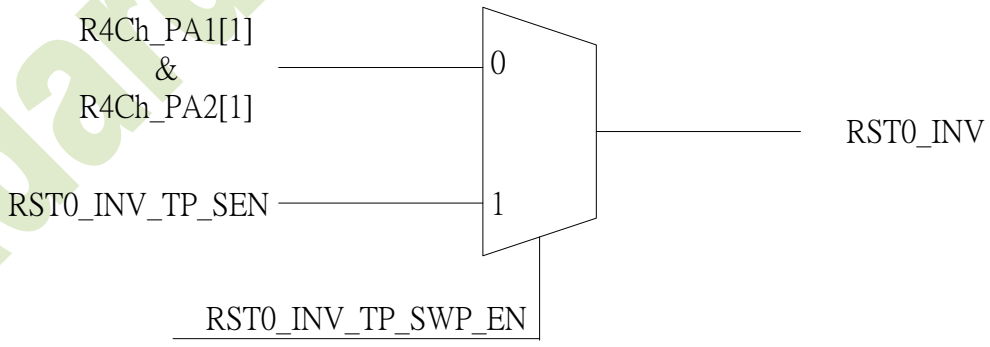
GCH_TP_SEN: Control GCH output level during TP term.
0: VGL
1: VGH

GCH_TP_SWP_EN: Control GCH reference group during TP term.
0: Refer GCL group
1: Refer GCH_TP_SEN control



RST0_INV_TP_SEN: Control RST0 INV singal output level during TP term.
0: VGL
1: VGH

RST0_INV_TP_SWP_EN: Control RST0 INV reference group during TP term.
0: Refer RST group
1: Refer RST0_INV_TP_SEN control



Restriction: -

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5. Page 3 Command Description

5.1. SET_DB_PB: Set Debug Probe (Page3-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	probe_sel[7:0]								00
Description	This command is used to set debug probe signal. **probe_sel[7:0]**: APR debug probe select.									
Restriction	-									

5.2. RD_IC_ST: Read IC Status (Page3-R3Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	0	3A
Parameter 1	**R**	core_state[7:0]								00
Description	This command is used to read IC status. **core_state[7:0]**: read IC corestate. //01h: initdly state. //02h: start state. //04h: free-run state. //08h: waiteof state. //10h: normal state. //20h: bist state. //40h: end state. //80h: idle state.									
Restriction	-									

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5.3. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	r_lb_mbist_w_setup_sel	r_lb_mbist_r_setup_sel	r_lb_mbist_r_hold_sel[1:0]		r_lb_debug_sub_ram_sel[3:0]				D0
Description	This command is used to set line buffer information. **r_lb_debug_sub_ram_sel[3:0]**: line-buffer mbist debug sub-ram select. //01h: for sub-ramA. //02h: for sub-ramB. //04h: for sub-ramC. //08h: for sub-ramD. **r_lb_mbist_r_hold_sel[1:0]**: line-buffer mbist read hold time select. //2'b0000: 18.75ns. //2'b0001: 31.25ns. //2'b0010: 43.72ns. //2'b0011: 56.25ns. **r_lb_mbist_r_setup_sel**: line-buffer mbist read setup time select. //0: 0.625ns. //1: 18.75ns. **r_lb_mbist_w_setup_sel**: line-buffer mbist write setup time select. //0: 0.625ns. //1: 18.75ns.									
Restriction	-									

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5.4. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	treg[7:0]								00
Description	This command is used enter testmode. **treg[7:0]**: testmode register key and select.									
Restriction	-									

5.5. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	-	-	-	-	-	-	-	r_lb_mbist_auto_en	00
Description	This command is used to set line buffer information. **r_lb_mbist_auto_en**: line-buffer mbist auto run enable. //0: disable //1: enable									
Restriction	-									

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5.6. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	-	-	-	r_lb_mbist_pattern_sel[12:8]					00
Parameter 2	**R/W**	r_lb_mbist_pattern_sel[7:0]								00
Description	This command is used to set line buffer information. **r_lb_mbist_pattern_sel[12:8]**: line-buffer mbist pattern select[12:8]. **r_lb_mbist_pattern_sel[7:0]**: line-buffer mbist pattern select[7:0].									
Restriction	-									

5.7. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	r_lb_mbist_ext_pat[47:40]								00
Parameter 2	**R/W**	r_lb_mbist_ext_pat[39:32]								00
Parameter 3	**R/W**	r_lb_mbist_ext_pat[31:24]								00
Parameter 4	**R/W**	r_lb_mbist_ext_pat[23:16]								00
Parameter 5	**R/W**	r_lb_mbist_ext_pat[15:8]								00
Parameter 6	**R/W**	r_lb_mbist_ext_pat[7:0]								00
Description	This command is used to set line buffer information. **r_lb_mbist_ext_pat[47:0]**: line-buffer mbist external flexible pattern [47:0]									
Restriction	-									

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5.8. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	-	-	-	r_lb_debug_adx_sel[4:0]					00
Description	This command is used to set line buffer information. **r_lb_debug_adx_sel[4:0]**: line-buffer mbist debug address-x select[4:0].									
Restriction	-									

5.9. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	-	-	r_lb_debug_ady_sel[7:0]						00
Description	This command is used to set line buffer information. **r_lb_debug_ady_sel[7:0]**: line-buffer mbist debug address-y select[4:0].									
Restriction	-									

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5.10. SET_LINE_BUF_INF: Set Line Buffer Information (Page3-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	-	-	-	-	-	-	r_lb_debug_read_trig	r_lb_debug_write_trig	00
Description	This command is used to set line buffer information. **r_lb_debug_write_trig**: line-buffer mbist debug mode write trigger. **r_lb_debug_read_trig**: line-buffer mbist debug mode read trigger.									
Restriction	-									

5.11. RD_LINE_BUF_INF: Read Line Buffer Information (Page3-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R**	repaire_wline_num_0[7:0]								00
Description	This command is used to read line buffer information. **repaire_wline_num_0[7:0]**: read only, line-buffer fail address_0									
Restriction	-									

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5.12. RD_LINE_BUF_INF: Read Line Buffer Information (Page3-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R**	repaire_wline_num_1[7:0]								00
Description	This command is used to read line buffer information. **repaire_wline_num_0[7:0]**: read only, line-buffer fail address_1									
Restriction	-									

5.13. RD_SRAMA_INF: Read SramA Information (Page3-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R**	debug_sram_rd_data_a[47:40]								00
Parameter 2	**R**	debug_sram_rd_data_a[39:32]								00
Parameter 3	**R**	debug_sram_rd_data_a[31:24]								00
Parameter 4	**R**	debug_sram_rd_data_a[23:16]								00
Parameter 5	**R**	debug_sram_rd_data_a[15:8]								00
Parameter 6	**R**	debug_sram_rd_data_a[7:0]								00
Description	This command is used to read sramA information. **debug_sram_rd_data_a[47:0]**: read line-buffer sub-ramA debug data[47:0].									
Restriction	-									

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5.14. RD_SRAMB_INF: Read SramB Information (Page3-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R**	debug_sram_rd_data_b[47:40]								00
Parameter 2	**R**	debug_sram_rd_data_b[39:32]								00
Parameter 3	**R**	debug_sram_rd_data_b[31:24]								00
Parameter 4	**R**	debug_sram_rd_data_b[23:16]								00
Parameter 5	**R**	debug_sram_rd_data_b[15:8]								00
Parameter 6	**R**	debug_sram_rd_data_b[7:0]								00
Description	This command is used to read sramB information. **debug_sram_rd_data_b[47:0]**: read line-buffer sub-ramB debug data[47:0].									
Restriction	-									

5.15. RD_SRAMC_INF: Read SramC Information. (Page3-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R**	debug_sram_rd_data_c[47:40]								00
Parameter 2	**R**	debug_sram_rd_data_c[39:32]								00
Parameter 3	**R**	debug_sram_rd_data_c[31:24]								00
Parameter 4	**R**	debug_sram_rd_data_c[23:16]								00
Parameter 5	**R**	debug_sram_rd_data_c[15:8]								00
Parameter 6	**R**	debug_sram_rd_data_c[7:0]								00
Description	This command is used to read sramC information. **debug_sram_rd_data_c[47:0]**: read line-buffer sub-ramC debug data[47:0].									
Restriction	-									

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5.16. RD_SRAMC_INF: Read SramD Information. (Page3-R48h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R**	debug_sram_rd_data_d[47:40]								00
Parameter 2	**R**	debug_sram_rd_data_d[39:32]								00
Parameter 3	**R**	debug_sram_rd_data_d[31:24]								00
Parameter 4	**R**	debug_sram_rd_data_d[23:16]								00
Parameter 5	**R**	debug_sram_rd_data_d[15:8]								00
Parameter 6	**R**	debug_sram_rd_data_d[7:0]								00
Description	This command is used to read sramD information. **debug_sram_rd_data_d[47:0]**: read line-buffer sub-ramD debug data[47:0].									
Restriction	-									

5.17. SET_GPIO_INF: Set GPIO Information. (Page3-R49h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	-	-	-	r_replace_probe_en	r_gpio_probe_sel_1[1:0]		r_gpio_probe_sel_0[1:0]		04
Description	This command is used to set GPIO information. **r_gpio_probe_sel_0[1:0]**: testmode debug probe to gpio[7:4] select. **r_gpio_probe_sel_1[1:0]**: testmode deubg probe to gpio[3:0] select. **r_replace_probe_en**: testmode replace debug probe for AFE enable.									
Restriction	-									

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5.18. SET_TE/TE1_INF: Set TE/TE1 Information. (Page3-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	r_te1_probe_sel[3:0]				r_te_probe_sel[3:0]				00
Description	This command is used to set TE/TE1 information. **r_te_probe_sel[3:0]**: testmode debug probe to TE select. **r_te1_probe_sel[3:0]**: testmode debug probe to TE1 select.									
Restriction	-									

5.19. SET_PWM_INF: Set LEDPWM Information. (Page3-R4Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	1	4B
Parameter 1	**R/W**	-	-	-	-	r_pwm_probe[3:0]				00
Description	This command is used to set LEDPWM information. **r_pwm_probe[3:0]**: testmode debug probe to LEDPWM select.									
Restriction	-									

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5.20. SET_SPI_INF: Set SPI Information. (Page3-R55h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	0	1	55
Parameter 1	**R/W**	-	-	-	-	-	cmd_wr_s_en	cmd_wr_m_en	spi_rd_s	00
Description	This command is used to set SPI information. **spi_rd_s**: spi read from slave. 0: disable 1: enable **cmd_wr_s_en**: command write for slave IC. 0: Disable 1: Enable **cmd_wr_m_en**: command write for master IC. 0: Disable 1: Enable									
Restriction	-									

5.21. SET_OTP_INF: Set OTP Information. (Page3-R56h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	0	56
Parameter 1	**R/W**	otp_mask[7:0]								00
Description	This command is used to set OTP information. **otp_mask[7:0]**: otp write data mask.									
Restriction	-									

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5.22. SET_OTP_INF: Set OTP Information. (Page3-R57h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	1	57
Parameter 1	**R/W**	-	-	-	-	-	otp_pa[10:8]			00
Parameter 2	**R/W**	otp_pa[7:0]								00
Description	This command is used to set OTP information. **otp_pa[10:0]**: otp address [10:0] for testmode									
Restriction	-									

5.23. SET_OTP_INF: Set OTP Information. (Page3-R58h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	0	58
Parameter 1	**R/W**	otp_all_wr	otp_pgm_bit[1:0]		otp_single_rd	otp_single_wr	otp_load_dis	otp_int_vpp	0	00
Description	This command is used to set OTP information. **otp_int_vpp**: otp internal vpp select. //0: external power. //1: internal power. **otp_load_dis**: disable otp reload. //0: reload. //1: no reload. **otp_single_wr**: otp single write command. //0: disable. //1: enable. **otp_single_rd**: otp single read command. //0: disable. //1: enable. **otp_pgm_bit[1:0]**: otp pgm mode select. **otp_all_wr**: otp all write enable. //0: disable. //1: enable.									
Restriction	-									

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5.24. RD_OTP_INF: Read OTP Information. (Page3-R59h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	1	59
Parameter 1	**R**	otp_single_rd_data[7:0]								00
Description	This command is used to read OTP information. **otp_single_rd_data[7:0]**: read data from OTP for single read operation.									
Restriction	-									

5.25. SET_OTP_INF: Set OTP Information. (Page3-R5Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	0	5A
Parameter 1	**R/W**	otp_single_wr_data[7:0]								00
Description	This command is used to set OTP information. **otp_single_wr_data[7:0]**: write data for OTP write data.									
Restriction	-									

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5.26. SET_OTP_INF: Set OTP Information. (Page3-R5Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	1	5B
Parameter 1	**R/W**	reg_ptr	reg_ptc	reg_pwe	reg_pclk	reg_pprog	reg_pce	reg_pdstb	reg_pdin	00
Description	This command is used to set OTP information. **reg_pdin**: PDIN for OTP. **reg_pdstb**: PDSTB for OTP. **reg_pce**: PCE for OTP. **reg_pprog**: PPROG for OTP. **reg_pclk**: PCLK for OTP. **reg_pwe**: PWE for OTP. **reg_ptc**: PTC for OTP. **reg_ptr**: PTR for OTP.									
Restriction	-									

5.27. SET_OTP_INF: Set OTP Information. (Page3-R5Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	0	5C
Parameter 1	**R/W**	otp_sw_sel	0	sw_pprog	reg_paio[2:0]			reg_pas[1:0]		00
Description	This command is used to set OTP information. **reg_pas[1:0]**: PAS for OTP. **reg_paio[2:0]**: PAIO for OTP. **sw_pprog**: software pprog setting. **otp_sw_sel**: otp switch select.									
Restriction	-									

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5.28. SET_OTP_INF: Set OTP Information. (Page3-R5Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	0	1	5D
Parameter 1	**R/W**	otp_pwr_rdy_bypass	otp_tvps_sel[1:0]		otp_pg_speedup	reg_ptm[3:0]				00
Description	This command is used to set OTP information. **reg_ptm[3:0]**: PTM for OTP. **otp_pg_speedup**: otp program speed up (only for simulation). **otp_tvps_sel[1:0]**: VPP select. **otp_pwr_rdy_bypass**: otp power ready bypass.									
Restriction	-									

5.29. SET_OTP_INF: Set OTP Information. (Page3-R5Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	1	0	5E
Parameter 1	**R/W**	pg_width_sel[1:0]		rd_clk_width[2:0]			otp_sw	-	-	58
Description	This command is used to set OTP information. **standby_reload_en**: standby reload enable. //0: no reload. //1: reload. **otp_sw**: VPP switch control. **rd_clk_width[2:0]**: read clock width , 100 ns per step. **pg_width_sel[1:0]**: program width select. //2'b00: 4us. //2'b01: 4.8us. //2'b10: 6.4us. //2'b11: 8us.									
Restriction	-									

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6. Page 4 Command Description

6.1. SET_CE_INF: Set Color Enhance Information (Page4-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	-	rg_edg_sp_prot	rg_sat_sp_prot	rg_hue_en	ce_sat_mode[1:0]		ce_edg_mode_en_pre	ce_sat_mode_en_pre	7C
Description	This command is used to set color enhance information. **ce_sat_mode_en_pre**: saturation enable of color enhancement. //0: disable. //1: enable **ce_edg_mode_en_pre**: edge enable of color enhancement. //0: disable. //1: enable **ce_sat_mode[1:0]**: saturation strength mode selection of color enhance. //2'b00: normal //2'b01: low strength //2'b10: mid strength //2'b11: high strength **rg_hue_en**: hue enable of color enhancement. //0: disable. //1: enable **rg_sat_sp_prot**: special pattern protection at saturation enable. //0: disable. //1: enable **rg_edg_sp_prot**: special pattern protection at edge enable. //0: disable. //1: enable									
Restriction	-									

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6.2. SET_CE_INF: Set Color Enhance Information (Page4-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	rg_ce_SG016[3:0]				rg_ce_SG000[3:0]				A0
Parameter 2	**R/W**	rg_ce_SG048[3:0]				rg_ce_SG032[3:0]				FE
Parameter 3	**R/W**	rg_ce_SG080[3:0]				rg_ce_SG064[3:0]				EE
Parameter 4	**R/W**	rg_ce_SG112[3:0]				rg_ce_SG096[3:0]				CD
Parameter 5	**R/W**	rg_ce_SG144[3:0]				rg_ce_SG128[3:0]				BC
Parameter 6	**R/W**	rg_ce_SG176[3:0]				rg_ce_SG160[3:0]				9A
Parameter 7	**R/W**	re_ce_SG208[3:0]				rg_ce_SG192[3:0]				67
Parameter 8	**R/W**	rg_ce_SG240[3:0]				rg_ce_SG224[3:0]				34
Parameter 9	**R/W**	-	-	-	-	rg_ce_SG255[3:0]				00
Description	This command is used to set color enhance information. gain = 1+(rg_ce_SGxxx/16) **rg_ce_SG000[3:0]**: global gain of avg. global gain of avg. Dmax = 000 **rg_ce_SG016[3:0]**: global gain of avg. global gain of avg. Dmax = 016 **rg_ce_SG032[3:0]**: global gain of avg. global gain of avg. Dmax = 032 **rg_ce_SG048[3:0]**: global gain of avg. global gain of avg. Dmax = 048 **rg_ce_SG064[3:0]**: global gain of avg. global gain of avg. Dmax = 064 **rg_ce_SG080[3:0]**: global gain of avg. global gain of avg. Dmax = 080 **rg_ce_SG096[3:0]**: global gain of avg. global gain of avg. Dmax = 096 **rg_ce_SG112[3:0]**: global gain of avg. global gain of avg. Dmax = 112 **rg_ce_SG128[3:0]**: global gain of avg. global gain of avg. Dmax = 128 **rg_ce_SG144[3:0]**: global gain of avg. global gain of avg. Dmax = 144 **rg_ce_SG160[3:0]**: global gain of avg. global gain of avg. Dmax = 160 **rg_ce_SG176[3:0]**: global gain of avg. global gain of avg. Dmax = 176 **rg_ce_SG192[3:0]**: global gain of avg. global gain of avg. Dmax = 192 **rg_ce_SG208[3:0]**: global gain of avg. global gain of avg. Dmax = 208 **rg_ce_SG224[3:0]**: global gain of avg. global gain of avg. Dmax = 224 **rg_ce_SG240[3:0]**: global gain of avg. global gain of avg. Dmax = 240 **rg_ce_SG255[3:0]**: global gain of avg. global gain of avg. Dmax = 255									
Restriction	-									

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6.3. SET_CE_INF: Set Color Enhance Information (Page4-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	-	-	-	rg_ce_THCOLOR2[2:0]			rg_col_en	rg_ce_vv_en	01
Parameter 2	**R/W**	-	-	rg_ce_THREF[2:0]			rg_ce_THSAT[2:0]			25
Parameter 3	**R/W**	-	-	rg_ce_THGREY[2:0]			rg_ce_THCOLOR[2:0]			04
Parameter 4	**R/W**	rg_ce_SAT_OFFS[7:0]								08
Parameter 5	**R/W**	rg_ce_REF_OFFS[7:0]								0A
Parameter 6	**R/W**	rg_ce_COL_OFFs[7:0]								00
Description	This command is used to set color enhance information. **rg_ce_vv_en**: reflection and saturation pixel protection (local gain) enable. //0: disable. //1: enable **rg_col_en**: 2nd colorful pixel protection enable. //0: disable. //1: enable **rg_ce_THCOLOR2[2:0]**: threshold of colorful pixel protection. **rg_ce_THSAT[2:0]**: threshold of saturation pixel protection. **rg_ce_THREF[2:0]**: threshold of reflection pixel protection. **rg_ce_THCOLOR[2:0]**: threshold of colorful pixel protection. **rg_ce_THGREY[2:0]**: threshold of gray pixel protection. **rg_ce_SAT_OFFS[7:0]**: offset of saturation pixel protection. **rg_ce_REF_OFFS[7:0]**: offset of reflection pixel protection. **rg_ce_COL_OFFs[7:0]**: offset of colorful pixel protection.									
Restriction	-									

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6.4. SET_CE_INF: Set Color Enhance Information (Page4-R3B)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	-	-	rg_ce_HUE_TBL00[5:0]						1D
Parameter 2	**R/W**	-	-	rg_ce_HUE_TBL01[5:0]						1F
Parameter 3	**R/W**	-	-	rg_ce_HUE_TBL02[5:0]						1F
Parameter 4	**R/W**	-	-	rg_ce_HUE_TBL03[5:0]						20
Parameter 5	**R/W**	-	-	rg_ce_HUE_TBL04[5:0]						21
Parameter 6	**R/W**	-	-	rg_ce_HUE_TBL05[5:0]						22
Parameter 7	**R/W**	-	-	rg_ce_HUE_TBL06[5:0]						22
Parameter 8	**R/W**	-	-	rg_ce_HUE_TBL07[5:0]						22
Parameter 9	**R/W**	-	-	rg_ce_HUE_TBL08[5:0]						22
Parameter 10	**R/W**	-	-	rg_ce_HUE_TBL09[5:0]						22
Parameter 11	**R/W**	-	-	rg_ce_HUE_TBL10[5:0]						22
Parameter 12	**R/W**	-	-	rg_ce_HUE_TBL11[5:0]						22
Description	This command is used to set color enhance information. gain = (1~63)/32 24 band to 48 band 1: HUE_TBL00 2: (HUE_TBL00+HUE_TBL01)/2 3: HUE_TBL01 … 48: (HUE_TBL23+HUE_TBL00)/2 **rg_ce_HUE_TBL00[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL01[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL02[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL03[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL04[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL05[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL06[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL07[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL08[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL09[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL10[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL11[5:0]**: Hue factor of saturation gain.									
Restriction	-									

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6.5. SET_CE_INF: Set Color Enhance Information (Page4-R3C)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	-	-	rg_ce_HUE_TBL12[5:0]						22
Parameter 2	**R/W**	-	-	rg_ce_HUE_TBL13[5:0]						22
Parameter 3	**R/W**	-	-	rg_ce_HUE_TBL14[5:0]						22
Parameter 4	**R/W**	-	-	rg_ce_HUE_TBL15[5:0]						22
Parameter 5	**R/W**	-	-	rg_ce_HUE_TBL16[5:0]						22
Parameter 6	**R/W**	-	-	rg_ce_HUE_TBL17[5:0]						22
Parameter 7	**R/W**	-	-	rg_ce_HUE_TBL18[5:0]						22
Parameter 8	**R/W**	-	-	rg_ce_HUE_TBL19[5:0]						20
Parameter 9	**R/W**	-	-	rg_ce_HUE_TBL20[5:0]						20
Parameter 10	**R/W**	-	-	rg_ce_HUE_TBL21[5:0]						20
Parameter 11	**R/W**	-	-	rg_ce_HUE_TBL22[5:0]						1F
Parameter 12	**R/W**	-	-	rg_ce_HUE_TBL23[5:0]						1E
Description	This command is used to set color enhance information. **rg_ce_HUE_TBL12[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL13[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL14[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL15[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL16[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL17[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL18[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL19[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL20[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL21[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL22[5:0]**: Hue factor of saturation gain. **rg_ce_HUE_TBL23[5:0]**: Hue factor of saturation gain.									
Restriction	-									

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6.6. SET_CE_INF: Set Color Enhance Information (Page4-R3D)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	rg_ce_EDG_FAC2[3:0]				rg_ce_EDG_FAC1[3:0]				85
Description	This command is used to set color enhance information. **rg_ce_EDG_FAC1[3:0]**: Edge enhance gain k1. Edge enhance gain k1 Ysharp = Yn + k1*(2*Yn-Yn+1-Yn-1) + k2*(2*Yn-Yn+2-Yn-2) **rg_ce_EDG_FAC2[3:0]**: Edge enhance gain k2. Edge enhance gain k2 Ysharp = Yn + k1*(2*Yn-Yn+1-Yn-1) + k2*(2*Yn-Yn+2-Yn-2)									
Restriction	-									

6.7. SET_CE_INF: Set Color Enhance Information (Page4-R3E)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	-	-	-	rg_xy_fac_mode	rg_roi_auto	rg_roi_en	rg_ce_dim_reset	rg_ce_sat_dim_en	09
Description	This command is used to set color enhance information. **rg_ce_sat_dim_en**: saturation global gain dimming enable. //0: disable. //1: enable **rg_ce_dim_reset**: dimming function reset enable, same as software reset. //0: disable. //1: enable **rg_roi_en**: ROI(region of interest) enable. //0: disable. //1: enable **rg_roi_auto**: ROI auto mode enable. //0: disable. //1: enable **g_xy_fac_mode**: AMS down sample xy factor manual select mode enable. //0: disable. //1: enable									
Restriction	-									

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6.8. SET_CE_INF: Set Color Enhance Information (Page4-R3F)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	rg_ce_sat_DLT_FD[3:0]				rg_ce_sat_FPS_FD[3:0]				22
Description	This command is used to set color enhance information. **rg_ce_sat_FPS_FD[3:0]**: every FPS to decrease/increase global gain. **rg_ce_sat_DLT_FD[3:0]**: increase/decrease step for dimming function.									
Restriction	-									

6.9. SET_CE_INF: Set Color Enhance Information (Page4-R40)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	rg_bk_color[7:0]								00
Parameter 2	**R/W**	rg_bk_color[15:8]								00
Parameter 3	**R/W**	rg_bk_color[23:16]								00
Description	This command is used to set color enhance information. **rg_bk_color[23:0]**: background color select for ROI, R=[23:16], G=[15:8], B=[7:0].									
Restriction	-									

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6.10. SET_CE_INF: Set Color Enhance Information (Page4-R41)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	rg_EN_TC_AMS_DET	rg_EN_LE_AMS_DE	rg_EN_CE_AMS_DET	0	rg_PAT_MODE_SEL[3:0]				EF
Parameter 2	**R/W**	rg_SP_SATDIFF_TH[7:0]								05
Parameter 3	**R/W**	rg_SP_WHITE_TH[7:0]								FA
Parameter 4	**R/W**	rg_SP_COLOR_TH[7:0]								05
Parameter 5	**R/W**	rg_SP_GREY_TH[7:0]								05
Parameter 6	**R/W**	rg_SP_SWCGP_NUM_TH[7:0]								FB
Description	This command is used to set color enhance information. **rg_PAT_MODE_SEL[3:0]**: special pattern detection enable. **rg_EN_CE_AMS_DET**: AMS detection enable for color enhancement. //0: disable. //1: enable **rg_EN_LE_AMS_DE**: AMS detection enable for light enhancement. //0: disable. //1: enable **rg_EN_TC_AMS_DET**: AMS detection enable for TCON. //0: disable. //1: enable **rg_SP_SATDIFF_TH[7:0]**: saturation difference threshold for special pattern protection. **rg_SP_WHITE_TH[7:0]**: white difference threshold for special pattern protection. **rg_SP_COLOR_TH[7:0]**: color difference threshold for special pattern protection. **rg_SP_GREY_TH[7:0]**: grey difference threshold for special pattern protection. **rg_SP_SWCGP_NUM_TH[7:0]**: Pixel number threshold of saturation difference, White difference, color difference, and grey difference.									
Restriction	-									

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6.11. SET_CE_INF: Set Color Enhance Information (Page4-R42)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	-	-	-	-	-	rg_AMMD_SC_CNT_RESET	rg_AMMD_SMP_FNUM_RESET	rg_AMS_PREF_VMAX_RESET	00
Parameter 2	**R/W**	rg_AMMD_SMP_FNUM[7:0]								05
Parameter 3	**R/W**	-	-	rg_AMMD_SC_CNT_UBTH[5:0]						08
Parameter 4	**R/W**	-	-	rg_AMMD_SC_CNT_LBTH[5:0]						05
Parameter 5	**R/W**	rg_AMMD_SC_DEC_STEP[3:0]				rg_AMMD_SC_INC_STEP[3:0]				46
Parameter 6	**R/W**	rg_AMMD_VMAX_SUM_TH[7:0]								01
Parameter 7	**R/W**	rg_AMMD_VMAX_SUM_TH[15:8]								00
Description	This command is used to set color enhance information. **rg_AMS_PREF_VMAX_RESET**: reset Pre Vmax enable. //0: disable. //1: enable **rg_AMMD_SMP_FNUM_RESET**: reset AMMD sample frame counter enable. //0: disable. //1: enable **rg_AMMD_SC_CNT_RESET**: reset AMMD sc_cnt enable. //0: disable. //1: enable **rg_AMMD_SMP_FNUM[7:0]**: period to detect moving mode. **rg_AMMD_SC_CNT_UBTH[5:0]**: upper boundary for AMMD sc_cnt. **rg_AMMD_SC_CNT_LBTH[5:0]**: lower boundary for AMMD sc_cnt. **rg_AMMD_SC_INC_STEP[3:0]**: increase step of AMMD sc_cnt. **rg_AMMD_SC_DEC_STEP[3:0]**: decrease step of AMMD sc_cnt. **rg_AMMD_VMAX_SUM_TH[15:0]**: VMAX sum threshold for AMS Previous frame and current frame difference.									
Restriction	-									

6.12. SET_CE_INF: Set Color Enhance Information (Page4-R43)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	rg_x_fac_m1[7:0]								00
Parameter 2	**R/W**	rg_y_fac_m1[7:0]								00

Description	This command is used to set color enhance information.
	
	rg_x_fac_m1[7:0]: AMS down sample x factor for manual select mode.
	force given x_factor -1
	x_factor = (size*256)/width);
	if(width!=size) x_factor++;
	rg_y_fac_m1[7:0]: AMS down sample y factor for manual select mode.
	force given y_factor -1
	y_factor = (size*256)/height);
	if(height!=size) y_factor++;									

Restriction	-									

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6.13. SET_CE_INF: Set Color Enhance Information (Page4-R44)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	rg_ce_gating_en[7:0]								FF
Description	This command is used to set color enhance information. **rg_ce_gating_en[7:0]**: AMS down sample y factor for manual select mode. //01h: enable clock at every VSYNC or ce_en rising edge. //02h: enable clock gating at every V porch. //others: reserved.									
Restriction	-									

6.14. SET_CE_INF: Set Color Enhance Information (Page4-R45)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	rg_ce_gate_selA[7:0]								FF
Description	This command is used to set color enhance information. **rg_ce_gate_selA[7:0]**: clock gating select.									
Restriction	-									

6.15. SET_CE_INF: Set Color Enhance Information (Page4-R46)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	rg_ce_gate_selB[7:0]								FF
Description	This command is used to set color enhance information. **rg_ce_gate_selA[7:0]**: clock gating select.									
Restriction	-									

6.16. SET_CE_INF: Set Color Enhance Information (Page4-R47)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	rg_ce_gate_selC[7:0]								FF
Description	This command is used to set color enhance information. **rg_ce_gate_selA[7:0]**: clock gating select.									
Restriction	-									

6.17. SET_CE_INF: Set Color Enhance Information (Page4-R48)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	rg_ce_gate_selD[7:0]								FF
Description	This command is used to set color enhance information. **rg_ce_gate_selA[7:0]**: clock gating select.									
Restriction	-									

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6.18. SET_CE_INF: Set Color Enhance Information (Page4-R49)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	rg_ce_ctrl_sel[7:0]								1B
Description	This command is used to set color enhance information. **rg_ce_ctrl_sel[7:0]**: //01h: down sampling timing enable. //02h: checksum clock enable. //04h: internal last line enable. //08h: special pattern protection for hue enhancement. //10h: special pattern protection for edge enhancement. //20h: vsync rising edge select. //40h: saturation mode(H/M/L) update at every frame. //80h: VBLK clock gating with pwr_slpout.									
Restriction	-									

6.19. SET_CE_INF: Set Color Enhance Information (Page4-R4A)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	rg_BinF_IP_BYPASS	-	rg_dbg_ce_en	rg_dbg_mux_sel[4:0]					00
Description	This command is used to set color enhance information. **rg_dbg_mux_sel[4:0]**: debug signals output select. **rg_dbg_ce_en**: debug signals output enable. //0: disable. //1: enable **rg_BinF_IP_BYPASS**: hsync output bypass enable. //0: disable. //1: enable									
Restriction	-									

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6.20. SET_CE_INF: Set Color Enhance Information (Page4-R4B)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	1	4B
Parameter 1	**R/W**	-	-	-	-	rg_LE_DIM_DBIT[1:0]		rg_LE_MAP_FUN[1:0]		01
Parameter 2	**R/W**	rg_LE_L_BOUND[7:0]								00
Parameter 3	**R/W**	rg_LE_U_BOUND[7:0]								FF
Description	This command is used to set color enhance information. **rg_LE_MAP_FUN[1:0]**: statistics method for LE module. **rg_LE_DIM_DBIT[1:0]**: judge AMMD function. **rg_LE_L_BOUND[7:0]**: statistics lower bound for LE module. **rg_LE_U_BOUND[7:0]**: statistics upper bound for LE module.									
Restriction	-									

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6.21. RD_CE_CHK: Read Color Enhance Checksum (Page4-R50)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	50
Parameter 1	**R**	ce_top_in_chksum1[7:0]								00
Parameter 2	**R**	ce_top_out_chksum1[7:0]								00
Parameter 3	**R**	ColEng2LE_TP_type[1:0]		ro_ColEng_AMS_DET_idx[5:0]						00
Parameter 4	**R**	ColEng2LE_mean[7:0]								00
Parameter 5	**R**	ce_top_in_chksum2[7:0]								00
Parameter 6	**R**	ce_top_out_chksum2[7:0]								00
Description	This command is read color enhance checksum. **ce_top_in_chksum1[7:0]**: read only **ce_top_out_chksum1[7:0]**: read only **ro_ColEng_AMS_DET_idx[5:0]**: read only **ColEng2LE_TP_type[1:0]**: read only **ColEng2LE_mean[7:0]**: read only **ce_top_in_chksum2[7:0]**: read only **ce_top_out_chksum2[7:0]**: read only									
Restriction	-									

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7. Page 5 Command Description

7.1. SET_LE_INF: Set LE Information (Page5-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	-	-	RG_BL	RG_DD	RG_BCTRL	RG_2FRM_DIMO_ENB	RG_USER_PWM_WT	RG_SLR_AMB_CTL	39
Description	This command is used to set LE information. **RG_SLR_AMB_CTL**: sun light readable consider Ambient light enable. //0: disable. //1: enable. **RG_USER_PWM_WT**: user decide PWM_WT value enable (r_46_00[5:0] ~ r_46_09[5:0]. //0: disable. //1: enable. **RG_2FRM_DIMO_ENB**: //0: only display delay 2frame signal for pwm pulse. //1: always on for pwm pulse. **RG_BCTRL**: brightness control. //0: disable. //1: enable **RG_DD**: display dimming enable. //0: disable. //1: enable. **RG_BL**: back light enable. //0: disable. //1: enable.									
Restriction	-									

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7.2. SET_LE_INF: Set LE Information (Page5-R32)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	RG_WRCABCMB[7:0]								00
Description	This command is used to set LE information. **Back Light PWM =** **((PWM_VAL * RG_WRDISBV) + RG_PWM_FRAC) /256) + RG_PWM_OFFSET** **RG_WRCABCMB[7:0]**: Brightness control minpwm value.									
Restriction	-									

7.3. SET_LE_INF: Set LE Information (Page5-R33)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	RG_WRDISBV[7:0]								FF
Description	This command is used to set LE information. **RG_WRDISBV[7:0]**: Wright display brightness value.									
Restriction	-									

7.4. SET_LE_INF: Set LE Information (Page5-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	RG_AMB_IN[31:24]								00
Parameter 2	**R/W**	RG_AMB_IN[23:16]								00
Parameter 3	**R/W**	RG_AMB_IN[15:8]								00
Parameter 4	**R/W**	RG_AMB_IN[7:0]								00
Description	This command is used to set LE information. **RG_AMB_IN[31:0]**: Ambient light value.									
Restriction	-									

7.5. SET_LE_INF: Set LE Information (Page5-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	-	-	-	RG_BE_STR000[4:0]					00
Parameter 2	**R/W**	-	-	-	RG_BE_STR064[4:0]					00
Parameter 3	**R/W**	-	-	-	RG_BE_STR128[4:0]					00
Parameter 4	**R/W**	-	-	-	RG_BE_STR192[4:0]					00
Parameter 5	**R/W**	-	-	-	RG_BE_STR255[4:0]					00
Description	This command is used to set LE information. **RG_BE_STRn[4:0]**: Global gain for brightness enhance. n=000,064,128,192,255.									
Restriction	-									

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7.6. SET_LE_INF: Set LE Information (Page5-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	-	RG_CABC_InHouse_en	RG_CABC_TP_BLK_EN	RG_SLRH_UI_REF	RG_SLRH_REF[1:0]		RG_SLRH_EN	RG_BE_EN	3C
Description	This command is used to set LE information. **RG_BE_EN**: brightness enhance enable. //0: disable. //1: enable. **RG_SLRH_EN**: Enable SLR. //0: disable. //1: enable. **RG_SLRH_REF[0]**: Prevent sun light readable Over Enhance enable. //0: disable. //1: enable. **RG_SLRH_REF[1]**: Prevnet sun light readable Color Un-Continuum enable. //0: disable. //1: enable. **RG_SLRH_UI_REF**: Enable UI referance.(his = total_num/8) //0: disable. //1: enable. **RG_CABC_TP_BLK_EN**: CABC test pattern black detection enable use RG_CABC_TP_BLK_PWM. //0: disable. //1: enable. **RG_CABC_InHouse_en**: Enable In House. //0: disable. //1: enable.									
Restriction	-									

7.7. SET_LE_INF: Set LE Information (Page5-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	-	RG_SLRH_STR[6:0]							00
Description	This command is used to set LE information. **RG_SLRH_STR[6:0]**: Fixed SLR strength.									
Restriction	-									

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7.8. SET_LE_INF: Set LE Information (Page5-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3F
Parameter 1	**R/W**	RG_CABC_CRV_REF	RG_CABC_BLK_CUT_SEL	RG_CABC_PWM_SYNC_FD	RG_CABC_CUT_TH[4:0]					E0
Description	This command is used to set LE information. **RG_CABC_CUT_TH[4:0]**: CABC strengh parameter. 平均亮度大於128的圖不做CABC，但是可藉由RG_CABC_CUT_TH的設定，強制讓所有的輸入圖做CABC **RG_CABC_PWM_SYNC_FD**: PWM SYNC. //0: always. //1: follow dimming sync. **RG_CABC_BLK_CUT_SEL**: cut_pt_T3 =0 enable. (y average < RG_LE_L_BOUND) **RG_CABC_CRV_REF**: CABC Histogram parameter.									
Restriction	-									

7.9. SET_LE_INF: Set LE Information (Page5-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	RG_CABC_PWM_WT_UI[7:0]								80
Parameter 2	**R/W**	RG_CABC_PWM_WT_PIC[7:0]								80
Parameter 3	**R/W**	RG_CABC_PWM_WT_MOV[7:0]								80
Description	This command is used to set LE information. **RG_CABC_PWM_WT_UI[7:0]**: CABC PWM_WT UI mode value. **RG_CABC_PWM_WT_PIC[7:0]**: CABC PWM_WT PICTURE mode value. **RG_CABC_PWM_WT_MOV[7:0]**: CABC PWM_WT MOVE mode value.									
Restriction	-									

7.10. SET_LE_INF: Set LE Information (Page5-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	RG_CABC_MIN_PWM[7:0]								00
Description	This command is used to set LE information. **RG_CABC_MIN_PWM[7:0]**: CABC PWM MIN value.									
Restriction	-									

7.11. SET_LE_INF: Set LE Information (Page5-R42)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	RG_CABC_MAX_PWM[7:0]								80
Description	This command is used to set LE information. **RG_CABC_MAX_PWM[7:0]**: CABC PWM MIN value.									
Restriction	-									

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7.12. SET_LE_INF: Set LE Information (Page5-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	RG_CABC_TP_BLK_PWM[7:0]								00
Description	This command is used to set LE information. **RG_CABC_TP_BLK_PWM[7:0]**: CABC test pattern black detection enable value.									
Restriction	-									

7.13. SET_LE_INF: Set LE Information (Page5-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	-	RG_CABC_CRV_TURN_TH[6:0]							46
Description	This command is used to set LE information. **RG_CABC_CRV_TURN_TH[6:0]**: CABC Histogram parameter.									
Restriction	-									

7.14. SET_LE_INF: Set LE Information (Page5-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	-	-	rg_LE_IP_BYPASS	RG_LE_TP_EN[4:0]					1F
Description	This command is used to set LE information. **RG_LE_TP_EN[4:0]**: Test Pattern Detection Enable. **rg_LE_IP_BYPASS**: LE IP bypass enable. //0: disable. //1: enable.									
Restriction	-									

7.15. SET_LE_INF: Set LE Information (Page5-R46)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	RG_LE_CUT2PWM_INP0[7:0]								00
Parameter 2	**R/W**	RG_LE_CUT2PWM_INP1[7:0]								01
Parameter 3	**R/W**	RG_LE_CUT2PWM_INP2[7:0]								06
Parameter 4	**R/W**	RG_LE_CUT2PWM_INP3[7:0]								0E
Parameter 5	**R/W**	RG_LE_CUT2PWM_INP4[7:0]								1C
Parameter 6	**R/W**	RG_LE_CUT2PWM_INP5[7:0]								2D
Parameter 7	**R/W**	RG_LE_CUT2PWM_INP6[7:0]								44
Parameter 8	**R/W**	RG_LE_CUT2PWM_INP7[7:0]								60
Description	This command is used to set LE information. **RG_LE_CUT2PWM_INPn[7:0]**: Gamma transfer. n=0~7.									
Restriction	-									

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7.16. SET_LE_INF: Set LE Information (Page5-R47)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	RG_LE_PWM2CUT_INP0[7:0]								00
Parameter 2	**R/W**	RG_LE_PWM2CUT_INP1[7:0]								87
Parameter 3	**R/W**	RG_LE_PWM2CUT_INP2[7:0]								BA
Parameter 4	**R/W**	RG_LE_PWM2CUT_INP3[7:0]								DF
Description	This command is used to set LE information. **RG_LE_PWM2CUT_INPn[7:0]**: Gamma transfer. n=0~3.									
Restriction	-									

7.17. SET_LE_INF: Set LE Information (Page5-R48)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	RG_LE_DIM_DLT[3:0]				RG_LE_DIM_FPS[3:0]				11
Description	This command is used to set LE information. **RG_LE_DIM_FPS[3:0]**: Fix-Delta dimming frames for each step. **RG_LE_DIM_DLT[3:0]**: Fixed-delta dimming for each step.									
Restriction	-									

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7.18. SET_LE_INF: Set LE Information (Page5-R49)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	-	-	RG_LE_CHKSUM_EN	RG_LE_DIM_MAD_EN	RG_LE_DIM_EN	RG_LE_DIM_MAD[2:0]			09
Description	This command is used to set LE information. **RG_LE_DIM_MAD[2:0]**: MAD: minimum allow difference. **RG_LE_DIM_EN**: LE dimming enable. //0: disable. //1: enable. **RG_LE_DIM_MAD_EN**: MAD function enable / disable. //0: disable. //1: enable. **RG_LE_CHKSUM_EN**: check sum enable. //0: disable. //1: enable.									
Restriction	-									

7.19. SET_LE_INF: Set LE Information (Page5-R4A)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	-	ECO_AE_GATING_SOT123	ECO_AE_GATING_HS_CMD	-	-	-	-	-	00
Parameter 2	**R/W**	RG_LE_L_BOUND[7:0]								00
Description	This command is used to set LE information. **ECO_AE_GATING_HS_CMD**: **ECO_AE_GATING_SOT123**: **RG_LE_L_BOUND[7:0]**: LE Low Bound.(cut_pt_T3)									
Restriction	-									

7.20. SET_LE_INF: Set LE Information (Page5-R4B)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	1	4B
Parameter 1	**R/W**	-	-	-	-	-	-	RG_LE_MAP_FUN[1:0]		01
Description	This command is used to set LE information. **RG_LE_MAP_FUN[1:0]**: Select 4 type for RGB transfer to Y share with CE.									
Restriction	-									

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7.21. SET_LE_INF: Set LE Information (Page5-R4E)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	1	0	4E
Parameter 1	**R/W**	-	-	RG_AMB_CABC_PWM0[5:0]						00
Parameter 2	**R/W**	-	-	RG_AMB_CABC_PWM1[5:0]						00
Parameter 3	**R/W**	-	-	RG_AMB_CABC_PWM2[5:0]						00
Parameter 4	**R/W**	-	-	RG_AMB_CABC_PWM3[5:0]						00
Parameter 5	**R/W**	-	-	RG_AMB_CABC_PWM4[5:0]						00
Parameter 6	**R/W**	-	-	RG_AMB_CABC_PWM5[5:0]						00
Parameter 7	**R/W**	-	-	RG_AMB_CABC_PWM6[5:0]						00
Parameter 8	**R/W**	-	-	RG_AMB_CABC_PWM7[5:0]						00
Parameter 9	**R/W**	-	-	RG_AMB_CABC_PWM8[5:0]						00
Parameter 10	**R/W**	-	-	RG_AMB_CABC_PWM9[5:0]						00
Description	This command is used to set LE information. **RG_AMB_CABC_PWMn[5:0]**: CABC strength for AMB. n=0~9.									
Restriction	-									

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7.22. SET_LE_INF: Set LE Information (Page5-R50)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	0	0	50
Parameter 1	**R/W**	-	-	-	-	RG_AMB_SLRH_STR0[3:0]				00
Parameter 2	**R/W**	-	-	-	-	RG_AMB_SLRH_STR1[3:0]				00
Parameter 3	**R/W**	-	-	-	-	RG_AMB_SLRH_STR2[3:0]				00
Parameter 4	**R/W**	-	-	-	-	RG_AMB_SLRH_STR3[3:0]				00
Parameter 5	**R/W**	-	-	-	-	RG_AMB_SLRH_STR4[3:0]				04
Parameter 6	**R/W**	-	-	-	-	RG_AMB_SLRH_STR5[3:0]				08
Parameter 7	**R/W**	-	-	-	-	RG_AMB_SLRH_STR6[3:0]				0A
Parameter 8	**R/W**	-	-	-	-	RG_AMB_SLRH_STR7[3:0]				0C
Parameter 9	**R/W**	-	-	-	-	RG_AMB_SLRH_STR8[3:0]				0E
Parameter 10	**R/W**	-	-	-	-	RG_AMB_SLRH_STR9[3:0]				0F
Description	This command is used to set LE information. **RG_AMB_SLRH_STR0[3:0]**: SLR strength for AMB. n=0~9.									
Restriction	-									

7.23. SET_LE_INF: Set LE Information (Page5-R55)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	0	1	55
Parameter 1	**R/W**	RG_LEDPWM_POL	RG_LEDON_POL	RG_LEDON_EN	RG_PWM_OFFSET[4:0]					C0
Description	This command is used to set LE information. **RG_PWM_OFFSET[4:0]**: Offset CABC_PWM output value. **RG_LEDON_EN**: Enable LED_flag. //0: disable. //1: enable. **RG_LEDON_POL**: LED_flag polar. **RG_LEDPWM_POL**: PWM_pulse polar.									
Restriction	-									

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7.24. SET_LE_INF: Set LE Information (Page5-R56)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	0	56
Parameter 1	**R/W**	RG_SEL_CLK_DIV[7:0]								00
Description	This command is used to set LE information. **RG_SEL_CLK_DIV[7:0]**: Oscillator divider config.									
Restriction	-									

7.25. SET_LE_INF: Set LE Information (Page5-R57)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	1	57
Parameter 1	**R/W**	RG_N_VAL[7:0]								00
Description	This command is used to set LE information. **RG_N_VAL[7:0]**: PWM_pulse scale. ((N+1)* osc_clk)									
Restriction	-									

7.26. SET_LE_INF: Set LE Information (Page5-R58)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	0	58
Parameter 1	**R/W**	RG_PWM_SYNC2VS	RG_DIM_FD_EN	RG_INC_STP_FT[2:0]			RG_DEC_STP_FT[2:0]			49
Description	This command is used to set LE information. **RG_DEC_STP_FT[2:0]**: Decrease Fixed step for BL Dimming. (Frame period) **RG_INC_STP_FT[2:0]**: Increase Fixed step for BL Dimming. (Frame period) **RG_DIM_FD_EN**: Select Fixed Delta/step for BL Dimming. **RG_PWM_SYNC2VS**: PWM value sync to Vsync.									
Restriction	-									

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7.27. SET_LE_INF: Set LE Information (Page5-R59)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	1	59
Parameter 1	**R/W**	RG_INC_FPS_FT[3:0]				RG_DEC_FPS_FT[3:0]				11
Description	This command is used to set LE information. **RG_DEC_FPS_FT[3:0]**: Decrease Fixed step for BL Dimming. (Frame period) **RG_INC_FPS_FT[3:0]**: Increase Fixed step for BL Dimming. (Frame period)									
Restriction	-									

7.28. SET_LE_INF: Set LE Information (Page5-R5A)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	1	5A
Parameter 1	**R/W**	RG_DEC_DLT_FD[7:0]								40
Parameter 2	**R/W**	RG_INC_DLT_FD[7:0]								40
Description	This command is used to set LE information. **RG_DEC_DLT_FD[7:0]**: Increase Fixed delta for BL Dimming. (Delta) **RG_INC_DLT_FD[7:0]**: Decrease Fixed delta for BL Dimming. (Delta)									
Restriction	-									

7.29. SET_LE_INF: Set LE Information (Page5-R5B)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	1	5B
Parameter 1	**R/W**	RG_INC_FPS_FD[3:0]				RG_DEC_FPS_FD[3:0]				11
Description	This command is used to set LE information. **RG_DEC_FPS_FD[3:0]**: Decrease Fixed delta for BL Dimming. (Frame period) **RG_INC_FPS_FD[3:0]**: Increase Fixed delta for BL Dimming. (Frame period)									
Restriction	-									

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7.30. SET_LE_INF: Set LE Information (Page5-R5C)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	0	5C
Parameter 1	**R/W**	RG_DIM_RESET	RG_DIM_RESET_MODE	RG_CABC_PWM_SEL	RG_CABC_PWM_BUND	RG_PWM_MIN_MODE	RG_DIM_CTRL_MODE	RG_DIM_CTRL_IMD	RG_DIM_DEC_CTRL	20
Description	This command is used to set LE information. **RG_DIM_DEC_CTRL**: For BL DIM debug. **RG_DIM_CTRL_IMD**: For BL DIM debug. **RG_DIM_CTRL_MODE**: For BL DIM debug. **RG_PWM_MIN_MODE**: For BL DIM debug. **RG_CABC_PWM_BUND**: For BL DIM debug. **RG_CABC_PWM_SEL**: For BL DIM debug Select Calculate/Force CABC PWM. **RG_DIM_RESET_MODE**: For BL DIM debug. **RG_DIM_RESET**: For BL DIM debug.									
Restriction	-									

7.31. SET_LE_INF: Set LE Information (Page5-R5D)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	1	5D
Parameter 1	**R/W**	RG_DIM_RESET_VAL[7:0]								00
Description	This command is used to set LE information. **RG_DIM_RESET_VAL[7:0]**: For BL DIM debug.									
Restriction	-									

7.32. SET_LE_INF: Set LE Information (Page5-R5E)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	0	5E
Parameter 1	**R/W**	RG_PWM_FRAC[7:0]								00
Description	This command is used to set LE information. **RG_PWM_FRAC[7:0]**: For BL DIM debug PWM value offset.									
Restriction	-									

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7.33. SET_LE_INF: Set LE Information (Page5-R5F)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	1	5F
Parameter 1	**R/W**	RG_CABC_PWM[7:0]								00
Description	This command is used to set LE information. **RG_CABC_PWM[7:0]**: For BL DIM debug CABC PWM value.									
Restriction	-									

7.34. SET_LE_INF: Set LE Information (Page5-R60)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	0	60
Parameter 1	**R/W**	RG_MANU_RESET_VAL[7:0]								00
Description	This command is used to set LE information. **RG_MANU_RESET_VAL[7:0]**: For BL DIM debug MANU PWM value.									
Restriction	-									

7.35. SET_LE_INF: Set LE Information (Page5-R61)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	1	61
Parameter 1	**R/W**	RG_LE_GATING_EN[7:0]								FE
Description	This command is used to set LE information. **RG_LE_GATING_EN[7:0]**: For LE debug gating enable.									
Restriction	-									

7.36. SET_LE_INF: Set LE Information (Page5-R62)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	0	62
Parameter 1	**R/W**	RG_BC_GATING_EN[7:0]								FE
Description	This command is used to set LE information. **RG_BC_GATING_EN[7:0]**: For BC debug gating enable.									
Restriction	-									

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7.37. SET_LE_INF: Set LE Information (Page5-R63)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	1	63
Parameter 1	**R/W**	-	RG_EXT_PWMI_SEL	RG_PWM8B14B_SEL	RG_BE_AMB_CTRL	RG_BE_MODE[1:0]		RG_SEL_DBG_PWM[1:0]		00
Description	This command is used to set LE information. **RG_SEL_DBG_PWM[1:0]**: BL debug select. **RG_BE_MODE[1:0]**: brightness enhance mode select. **RG_BE_AMB_CTRL**: 8bit for 14bit resolution. **RG_PWM8B14B_SEL**: 8bit-14bit selection **RG_EXT_PWMI_SEL**:									
Restriction	-									

7.38. SET_LE_INF: Set LE Information (Page5-R64)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	0	64
Parameter 1	**R/W**	RG_dbg_mux_sel[3:0]				dbg_mux_en	RG_LE_DBG_SEL[2:0]			08
Description	This command is used to set LE information. **RG_LE_DBG_SEL[2:0]**: histrogram select. **RG_BE_AMB_CTRL**: debug select. **RG_dbg_mux_sel[3:0]**: debug select.									
Restriction	-									

7.39. SET_LE_INF: Set LE Information (Page5-R65)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	1	65
Parameter 1	**R/W**	RG_SEL_PWMI_DET_CLK_DIV[7:0]								00
Description	This command is used to set LE information. **RG_SEL_PWMI_DET_CLK_DIV[7:0]**: pwmi_det clock select.									
Restriction	-									

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7.40. RD_ACC_INF: Read ACC Information (Page5-R66)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	0	66
Parameter 1	**R**	ACC0_FINAL[7:0]								00
Parameter 2	**R**	ACC1_FINAL[7:0]								00
Parameter 3	**R**	ACC2_FINAL[7:0]								00
Parameter 4	**R**	ACC3_FINAL[7:0]								00
Parameter 5	**R**	ACC4_FINAL[7:0]								00
Parameter 6	**R**	ACC5_FINAL[7:0]								00
Parameter 7	**R**	ACC6_FINAL[7:0]								00
Description	This command is used to read ACC information. **ACCn_FINAL[7:0]**: read only (final_his0) n=0~6.									
Restriction	-									

7.41. RD_LE_INF: Read LE Information (Page5-R67)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	67
Parameter 1	**R**	GAIN0_M32[7:0]								00
Parameter 2	**R**	GAIN1_M32[7:0]								00
Parameter 3	**R**	GAIN2_M32[7:0]								00
Parameter 4	**R**	GAIN3_M32[7:0]								00
Parameter 5	**R**	GAIN4_M32[7:0]								00
Parameter 6	**R**	GAIN5_M32[7:0]								00
Parameter 7	**R**	GAIN6_M32[7:0]								00
Description	This command is used to read LE information. **GAINn_M32[7:0]**: read only (LE_gainn[10:3]) n=0~6.									
Restriction	-									

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7.42. RD_LE_INF: Read LE Information (Page5-R68)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	0	68
Parameter 1	**R**	Ygry_avg[7:0]								00
Description	This command is used to read LE information. **Ygry_avg[7:0]**: read only. (y gray num avgerge)									
Restriction	-									

7.43. RD_LE_INF: Read LE Information (Page5-R69)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	1	69
Parameter 1	**R**	-	slope_cond0	hv_cond1	hv_cond0	u2u4_cond1	u2u4_cond0	u4[14:13]		00
Description	This command is used to read **u4[14:13]**: Read only **u2u4_cond0**: Read only **u2u4_cond1**: Read only **hv_cond0**: Read only **hv_cond1**: Read only **slope_cond0**: Read only									
Restriction	-									

7.44. RD_LE_INF: Read LE Information (Page5-R6A)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	1	0	6A
Parameter 1	**R**	target_cutpt[7:0]								00
Parameter 2	**R**	output_cutpt[7:0]								00
Parameter 3	**R**	target_PWM[7:0]								00
Parameter 4	**R**	output_PWM[7:0]								00
Description	This command is used to read **target_cutpt[7:0]**: read only **output_cutpt[7:0]**: read only **target_PWM[7:0]**: read only **output_PWM[7:0]**: read only									
Restriction	-									

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8. Page 6 Command Description

8.1. SET_R_CONT_INF: Set Red Contrast Information (Page6-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	r_contrast[7:0]								80
Description	This command is used to set red gain of contrast information. **r_contrast[7:0]**: $D_{contrast} = D_{in} * Gain$. //00h: 0.500. //80h: 1.000. //FFh: 1.496.									
Restriction	-									

8.2. SET_G_CONT_INF: Set Green Contrast Information (Page6-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	g_contrast[7:0]								80
Description	This command is used to set green gain of contrast information. **g_contrast[7:0]**: $D_{contrast} = D_{in} * Gain$. //00h: 0.500. //80h: 1.000. //FFh: 1.496.									
Restriction	-									

8.3. SET_B_CONT_INF: Set Blue Contrast Information (Page6-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	b_contrast[7:0]								80
Description	This command is used to set blue gain of contrast information. **b_contrast[7:0]**: $D_{contrast} = D_{in} * Gain$. //00h: 0.500. //80h: 1.000. //FFh: 1.496.									
Restriction	-									

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8.4. SET_DG_INF: Set Digital Gamma Information (Page6-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	-	-	dither_en	d_gma_en	dgma_ext_pol_en	dgma_ext_pol_enb	contrast_en	bright_en	08
Description	This command is used to set digital gamma information. **bright_en**: bright enable. //0: disable. //1: enable. **contrast_en**: contrast enable. //0: disable. //1: enable. **dgma_ext_pol_enb**: //0: disable. //1: enable. **dgma_ext_pol_en**: //0: disable. //1: enable. **d_gma_en**: digital gamma enable. //0: disable. //1: enable. **dither_en**: dither enable. //0: disable. //1: enable.									
Restriction	-									

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8.5. SET_R_BRT_INF: Set Red Brightness Information (Page6-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	r_bright[7:0]								40
Description	This command is used to set red offset of brightness information. **r_bright[7:0]**: $D_{brightness} = D_{contrast}$ + Offset. //00h: -64. //40h: 0. //FFh: 191.									
Restriction	-									

8.6. SET_G_BRT_INF: Set Green Brightness Information (Page6-R3Dh)

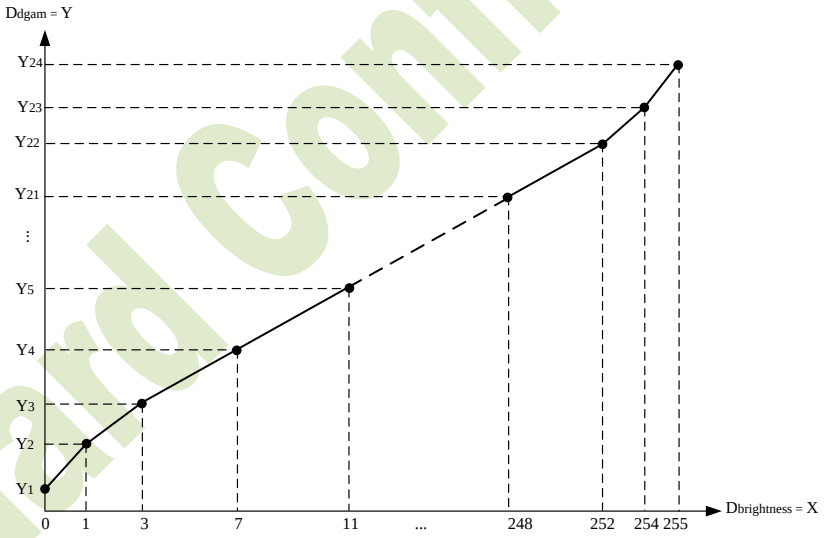
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	g_bright[7:0]								
Description	This command is used to set green offset of brightness information. **g_bright[7:0]**: $D_{brightness} = D_{contrast}$ + Offset. //00h: -64. //40h: 0. //FFh: 191.									
Restriction	-									

8.7. SET_B_BRT_INF: Set Blue Brightness Information (Page6-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	b_bright[7:0]								
Description	This command is used to set blue offset of brightness information. **b_bright[7:0]**: $D_{brightness} = D_{contrast}$ + Offset. //00h: -64. //40h: 0. //FFh: 191.									
Restriction	-									

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8.8. SET_R_DG_INF: Set Red Digital Gamma Information (Page6-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	r_point_1[7:0]								00
Parameter 2	**R/W**	r_point_2[7:0]								04
Parameter 3	**R/W**	r_point_3[7:0]								0C
Parameter 4	**R/W**	r_point_4[7:0]								1C
Parameter 5	**R/W**	r_point_5[7:0]								2C
Parameter 6	**R/W**	r_point_6[7:0]								3C
Parameter 7	**R/W**	r_point_7[7:0]								5C
Parameter 8	**R/W**	r_point_8[7:0]								7C
Parameter 9	**R/W**	r_point_4[9:8]		r_point_3[9:8]		r_point_2[9:8]		r_point_1[9:8]		00
Parameter 10	**R/W**	r_point_8[9:8]		r_point_7[9:8]		r_point_6[9:8]		r_point_5[9:8]		00
Description	This command is used to set red digital gamma information. **r_point_n[9:0]**: Red digital gamma with 10-bit correction. n=1~8. Red digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Yn) * (X – Xn) / (X_{n+1} – Xn)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering. 								
Restriction	-									

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8.9. SET_R_DG_INF: Set Red Digital Gamma Information (Page6-R40h)

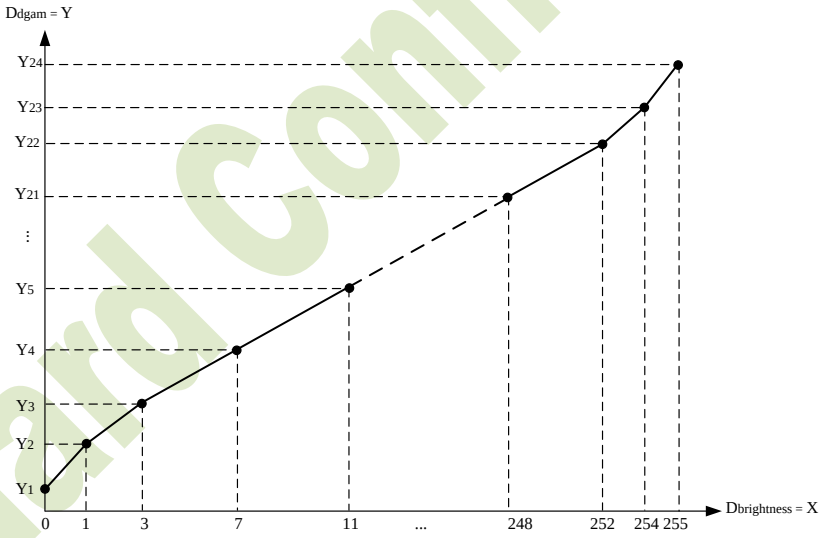
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	r_point_9[7:0]								BC
Parameter 2	**R/W**	r_point_10[7:0]								FC
Parameter 3	**R/W**	r_point_11[7:0]								7C
Parameter 4	**R/W**	r_point_12[7:0]								FC
Parameter 5	**R/W**	r_point_13[7:0]								00
Parameter 6	**R/W**	r_point_14[7:0]								80
Parameter 7	**R/W**	r_point_15[7:0]								00
Parameter 8	**R/W**	r_point_16[7:0]								40
Parameter 9	**R/W**	r_point_12[9:8]		r_point_11[9:8]		r_point_10[9:8]		r_point_9[9:8]		50
Parameter 10	**R/W**	r_point_16[9:8]		r_point_15[9:8]		r_point_14[9:8]		r_point_13[9:8]		FA
Description	This command is used to set red digital gamma information. **r_point_n[9:0]**: Red digital gamma with 10-bit correction. n=9~16. Red digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.									
Restriction	-									

8.10. SET_R_DG_INF: Set Red Digital Gamma Information (Page6-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	r_point_17[7:0]								80
Parameter 2	**R/W**	r_point_18[7:0]								A0
Parameter 3	**R/W**	r_point_19[7:0]								C0
Parameter 4	**R/W**	r_point_20[7:0]								D0
Parameter 5	**R/W**	r_point_21[7:0]								E0
Parameter 6	**R/W**	r_point_22[7:0]								F0
Parameter 7	**R/W**	r_point_23[7:0]								F8
Parameter 8	**R/W**	r_point_24[7:0]								FC
Parameter 9	**R/W**	r_point_20[9:8]		r_point_19[9:8]		r_point_18[9:8]		r_point_17[9:8]		FF
Parameter 10	**R/W**	r_point_24[9:8]		r_point_23[9:8]		r_point_22[9:8]		r_point_21[9:8]		FF
Description	This command is used to set red digital gamma information. **r_point_n[9:0]**: Red digital gamma with 10-bit correction. n=17~24. Red digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.									
Restriction	-									

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8.11. SET_G_DG_INF: Set Green Digital Gamma Information (Page6-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	g_point_1[7:0]								00
Parameter 2	**R/W**	g_point_2[7:0]								04
Parameter 3	**R/W**	g_point_3[7:0]								0C
Parameter 4	**R/W**	g_point_4[7:0]								1C
Parameter 5	**R/W**	g_point_5[7:0]								2C
Parameter 6	**R/W**	g_point_6[7:0]								3C
Parameter 7	**R/W**	g_point_7[7:0]								5C
Parameter 8	**R/W**	g_point_8[7:0]								7C
Parameter 9	**R/W**	g_point_4[9:8]		g_point_3[9:8]		g_point_2[9:8]		g_point_1[9:8]		00
Parameter 10	**R/W**	g_point_8[9:8]		g_point_7[9:8]		g_point_6[9:8]		g_point_5[9:8]		00
Description	This command is used to set green digital gamma information. **g_point_n[9:0]**: Green digital gamma with 10-bit correction. n=1~8. Green digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.  								
Restriction	-									

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8.12. SET_G_DG_INF: Set Green Digital Gamma Information (Page6-R43h)

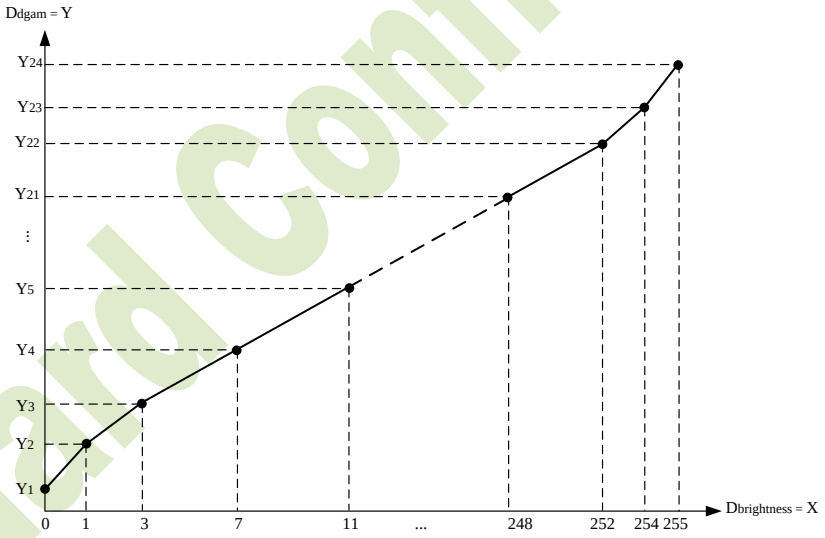
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	g_point_9[7:0]								BC
Parameter 2	**R/W**	g_point_10[7:0]								FC
Parameter 3	**R/W**	g_point_11[7:0]								7C
Parameter 4	**R/W**	g_point_12[7:0]								FC
Parameter 5	**R/W**	g_point_13[7:0]								00
Parameter 6	**R/W**	g_point_14[7:0]								80
Parameter 7	**R/W**	g_point_15[7:0]								00
Parameter 8	**R/W**	g_point_16[7:0]								40
Parameter 9	**R/W**	g_point_12[9:8]		g_point_11[9:8]		g_point_10[9:8]		g_point_9[9:8]		50
Parameter 10	**R/W**	g_point_16[9:8]		g_point_15[9:8]		g_point_14[9:8]		g_point_13[9:8]		FA
Description	This command is used to set green digital gamma information. **g_point_n[9:0]**: Green digital gamma with 10-bit correction. n=9~16. Green digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.									
Restriction	-									

8.13. SET_G_DG_INF: Set Green Digital Gamma Information (Page6-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	g_point_17[7:0]								80
Parameter 2	**R/W**	g_point_18[7:0]								A0
Parameter 3	**R/W**	g_point_19[7:0]								C0
Parameter 4	**R/W**	g_point_20[7:0]								D0
Parameter 5	**R/W**	g_point_21[7:0]								E0
Parameter 6	**R/W**	g_point_22[7:0]								F0
Parameter 7	**R/W**	g_point_23[7:0]								F8
Parameter 8	**R/W**	g_point_24[7:0]								FC
Parameter 9	**R/W**	g_point_20[9:8]		g_point_19[9:8]		g_point_18[9:8]		g_point_17[9:8]		FF
Parameter 10	**R/W**	g_point_24[9:8]		g_point_23[9:8]		g_point_22[9:8]		g_point_21[9:8]		FF
Description	This command is used to set green digital gamma information. **g_point_n[9:0]**: Green digital gamma with 10-bit correction. n=17~24. Green digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.									
Restriction	-									

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8.14. SET_B_DG_INF: Set Blue Digital Gamma Information (Page6-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	b_point_1[7:0]								00
Parameter 2	**R/W**	b_point_2[7:0]								04
Parameter 3	**R/W**	b_point_3[7:0]								0C
Parameter 4	**R/W**	b_point_4[7:0]								1C
Parameter 5	**R/W**	b_point_5[7:0]								2C
Parameter 6	**R/W**	b_point_6[7:0]								3C
Parameter 7	**R/W**	b_point_7[7:0]								5C
Parameter 8	**R/W**	b_point_8[7:0]								7C
Parameter 9	**R/W**	b_point_4[9:8]		b_point_3[9:8]		b_point_2[9:8]		b_point_1[9:8]		00
Parameter 10	**R/W**	b_point_8[9:8]		b_point_7[9:8]		b_point_6[9:8]		b_point_5[9:8]		00
Description	This command is used to set blue digital gamma information. **b_point_n[9:0]**: Blue digital gamma with 10-bit correction. n=1~8. Blue digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Yn) * (X – Xn) / (X_{n+1} – Xn)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.  								
Restriction	-									

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8.15. SET_B_DG_INF: Set Blue Digital Gamma Information (Page6-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	b_point_9[7:0]								BC
Parameter 2	**R/W**	b_point_10[7:0]								FC
Parameter 3	**R/W**	b_point_11[7:0]								7C
Parameter 4	**R/W**	b_point_12[7:0]								FC
Parameter 5	**R/W**	b_point_13[7:0]								00
Parameter 6	**R/W**	b_point_14[7:0]								80
Parameter 7	**R/W**	b_point_15[7:0]								00
Parameter 8	**R/W**	b_point_16[7:0]								40
Parameter 9	**R/W**	b_point_12[9:8]		b_point_11[9:8]		b_point_10[9:8]		b_point_9[9:8]		50
Parameter 10	**R/W**	b_point_16[9:8]		b_point_15[9:8]		b_point_14[9:8]		b_point_13[9:8]		FA
Description	This command is used to set blue digital gamma information. **b_point_n[9:0]**: Blue digital gamma with 10-bit correction. n=9~16. Blue digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.									
Restriction	-									

8.16. SET_B_DG_INF: Set Blue Digital Gamma Information (Page6-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	b_point_17[7:0]								80
Parameter 2	**R/W**	b_point_18[7:0]								A0
Parameter 3	**R/W**	b_point_19[7:0]								C0
Parameter 4	**R/W**	b_point_20[7:0]								D0
Parameter 5	**R/W**	b_point_21[7:0]								E0
Parameter 6	**R/W**	b_point_22[7:0]								F0
Parameter 7	**R/W**	b_point_23[7:0]								F8
Parameter 8	**R/W**	b_point_24[7:0]								FC
Parameter 9	**R/W**	b_point_20[9:8]		b_point_19[9:8]		b_point_18[9:8]		b_point_17[9:8]		FF
Parameter 10	**R/W**	b_point_24[9:8]		b_point_23[9:8]		b_point_22[9:8]		b_point_21[9:8]		FF
Description	This command is used to set blue digital gamma information. **b_point_n[9:0]**: Blue digital gamma with 10-bit correction. n=17~24. Blue digital gamma are defined with 24 register values Y1~Y24 for level X1~X24= 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. Y on X between Xn and Xn+1 is interpolated with the following equations. $Y = Yn + (Y_{n+1} – Y_n) * (X – X_n) / (X_{n+1} – X_n)$. The gamma correction output 10-bit data Ddgam is then fed to 8-bit DAC and OP to drive the source lines on the panel with dithering.									
Restriction	-									

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8.17. SET_DG_INF: Set Digital Gamma Information (Page6-R48h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	esd_bright_en	esd_bright_step[1:0]			esd_bright[4:0]				00
Description	This command is used to set **esd_bright[4:0]**: **esd_bright_step[1:0]**: **esd_bright_en**: //0: disable. //1: enable.									
Restriction	-									

8.18. REV_REG: Reserved Register (Page6-R49h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	0	0	0	0	0	0	0	0	00
Description	This command is a reserved register.									
Restriction	-									

8.19. REV_REG: Reserved Register (Page6-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	1	1	1	1	1	1	1	1	FF
Description	This command is a reserved register.									
Restriction	-									

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8.20. SET_DITH_INF: Set Dither Table Information (Page6-R56h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	0	56
Parameter 1	**R/W**	-	-	-	r_detect_127	h_table_en	dith_tabset_sel	hfrc_sel[1:0]		0A
Parameter 2	**R/W**	-	-	-	-	-	ext_dith_tab_en	h_table_sel[1:0]		00
Parameter 3	**R/W**	ext_dith_tab1_0[7:0]								00
Parameter 4	**R/W**	ext_dith_tab1_1[7:0]								00
Parameter 5	**R/W**	ext_dith_tab1_2[7:0]								00
Parameter 6	**R/W**	ext_dith_tab1_3[7:0]								00
Parameter 7	**R/W**	ext_dith_tab1_4[7:0]								00
Parameter 8	**R/W**	ext_dith_tab1_5[7:0]								00
Parameter 9	**R/W**	ext_dith_tab1_6[7:0]								00
Parameter 10	**R/W**	ext_dith_tab1_7[7:0]								00
Description	This command is used to set dither table information. **hfrc_sel[1:0]**: hfrc table select. **dith_tabset_sel**: dither table select. **h_table_en**: h_table enable. //0: disable. //1: enable. **r_detect_127**: detect 127 enable. //0: disable. //1: enable. **h_table_sel[1:0]**: h_table select. **ext_dith_tab_en**: external dither table enable. //0: disable. //1: enable.									
Restriction	-									

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9. Page 7 Command Description

9.1. SET_MIPI_INF: Set MIPI Information (Page7-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	r_lp11_check_enb	r_din_eq_z_sel[1:0]		r_hs_settle[4:0]					40
Description	This command is used to set MIPI information. **r_hs_settle[4:0]**: high speed detetct sync after settle time. //00h: dphy settle ready + 4~7.9UI(default) //01h: dphy settle ready + 8~11.9UI //10h: dphy settle ready + 68~71.9UI //11h: dphy settle ready + 72~75.9UI //1Eh: dphy settle ready + 124~127.9UI //1Fh: dphy settle ready + 128~131.9UI **r_din_eq_z_sel[1:0]**: data input zero count select. //2'b00: check 0 bit zero input //2'b01: check 4 bit zero input //2'b10: check 8 bit zero input(default) //2'b11: check 16 bit zero input **r_lp11_check_enb**: low power mode check enb. //0: disable. //1: enable.									
Restriction	-									

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9.2. SET_MIPI_INF: Set MIPI Information (Page7-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	TA_SURE[2:0]			r_esd_prot	r_wc_esd_det_en	r_en_cmd_queue	r_hster_delay[1:0]		06
Description	This command is used to set MIPI information. **r_hster_delay[1:0]**: data lane termial enable delay select. //2'b00: 0UI //2'b01: 4~7.9UI //2'b10: 8~11.9UI(default) //2'b11: 12~15.9UI **r_en_cmd_queue**: command queue enable. //0: disable. //1: enable. **r_wc_esd_det_en**: Word count esd detect. //0: disable. //1: enable. **r_esd_prot**: esd protect. //0: when esd detect, command signal clear //1: bypass esd protect. **TA_SURE[2:0]**: Tta-sure time select. //00h: 1T //01h: 2T //02h: 4T //03h: 6T //04h: 8T //05h: 10T //06h: 12T //07h: 14T T:80MHz OSC period									
Restriction	-									

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9.3. SET_MIPI_INF: Set MIPI Information (Page7-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	r_crc_esd_clr	r_cdcnt[1:0]		cd_case1_3_disable	r_ta_get[3:0]				44
Description	This command is used to set MIPI information. **r_ta_get[3:0]**: Tta-get time select. //00h: 1T //01h: 2T //10h: 32T //11h: 34T //1Eh: 60T //1Fh: 62T T:80MHz OSC period **Cd_case1_3_disable**: **r_cdcnt[1:0]**: contention detect counter. //2'b00: 3T //2'b01: 7T //2'b10: 11T //2'b11: 23T T:80MHz OSC period **r_crc_esd_clr**: crc esd detect clear. //0: mipi esd detect function that detect crc error, function enable. //1: mipi esd detect function that detect crc error, function disable.									
Restriction	-									

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9.4. SET_MIPI_INF: Set MIPI Information (Page7-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**		r_tx_pnsw	r_lptx_bpn[1:0]		r_t_lpx_rl[3:0]				D8
Description	This command is used to set MIPI information. **r_t_lpx_rl[3:0]**: low power TX mode rising/falling width. //00h: 16T //01h: 2T //08h: 9T(default) //09h: 10T //0Eh: 15T //0Fh: 16T T:80MHz OSC period **r_lptx_bpn[1:0]**: low power TX mode BPN function //2'b00: force 0, not enhance LPTXP/LPTXN //2'b01 or 2'b10: enhance LPTXP/LPTXN by data change //2'b11: force 1, always enhance LPTXP/LPTXN **r_tx_pnsw**: low power TX mode PN swap //0: lptx bpn swap //1: lptx bpn not swap(default)									
Restriction	-									

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9.5. SET_MIPI_INF: Set MIPI Information (Page7-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	cd_compare_sel[1:0]		VCID[1:0]		mipi_clk_stop_sel	pwr_mipi_en_sel	ulpslp11_check_enb	tx_eotp	08
Description	This command is used to set MIPI information. **tx_eotp**: low power TX mod include eotp packet. //0: LPTX packet end not include eotp packet //1: LPTX packet end include eotp packet(default) **ulpslp11_check_enb**: **pwr_mipi_en_sel**: **mipi_clk_stop_sel**: **VCID[1:0]**: **cd_compare_sel [1:0]**:									
Restriction	-									

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9.6. SET_MIPI_INF: Set MIPI Information (Page7-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	-	-	r_non_gated_dclk	r_non_gated_mipiclk	r_dynamic_lane0_en	r_en_dynamic_terr	r_dynamic_de_adj_sel	r_back_chip_non_gated	08
Description	This command is used to set MIPI information. **r_back_chip_non_gated**: cascade mode back chip non gating. //0: cascade mode bcak chip dynamic clock gating(dfault) //1: cascade mode back chip clock not gating **r_dynamic_de_adj_sel**: dynamic DE adjust select. //0: cascade mode front chip after DE falling 25T clock gating(default) //1: cascade mode front chip after DE falling 50T clock gating **r_en_dynamic_terr**: dynamic terminal enable. //0: cascade mode DPHY terminal enable by enter high mode(default) //1: cascade mode DPHY terminal dynamic disabe by internal control **r_dynamic_lane0_en**: lane0 terminal dynamic enable. //0: cascade mode DPHY lane0 terminal enable by enter high mode //1: cascade mode DPHY lane0 terminal dynamic disabe by internal control(default) **r_non_gated_mipiclk**: mipi clk non gating. //0: mipi clock non gating(default) //1: mipi clock dynamic gating **r_non_gated_dclk**: dclk non gating. //0: mipi video 2 pixel clock non gating(default) //1: mipi video 2 pixel clock dynamic gating									
Restriction	-									

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9.7. SET_MIPI_INF: Set MIPI Information (Page7-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	-	-	r_mipi_dclk_inv	r_non_gated_ret_osccck	r_en_glb_dynamic	r_mipi_overlap[1:0]		r_extra_dummy	00
Description	This command is used to set MIPI information. **r_extra_dummy**: cascade mode extra dummy. //0: r_mipi_overlap=2'b11, 4T dummy(default) //1: r_mipi_overlap=2'b11, 8T dummy **r_mipi_overlap[1:0]**: RL type mode overlap select. //2'b00: 0T(default) //2'b01: 1T //2'b10: 2T //2'b11: r_extra_dummy=0, 4T dummy r_extra_dummy=1, 8T dummy T: mipi video 2pixel clock **r_en_glb_dynamic**: global dynamic enable. //0: disable.global dynamic mipi clock & video 2pixel clock gating(default) //1: enable.global dynamic mipi clock & video 2pixel clock gating **r_non_gated_ret_osccck**: low power TX mode osc non gating. //0: low power TX mode osc clock gating(default) //1: low power TX mode osc clock non gating **r_mipi_dclk_inv**: dclk inverse. //0: mipi video 2pixel clock not inverse //1: mipi video 2pixel clock inverse									
Restriction	-									

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9.8. SET_MIPI_INF: Set MIPI Information (Page7-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	-	r_wr_parainit[6:0]							00

Description	This command is used to set MIPI information.
	
	r_wr_parainit[6:0]: long packet initial start read/ write parameter.
	//00h: prepare to write/read address_x=00h
	//01h: prepare to write/read address_x=01h
	//80h: prepare to write/read address_x=80h
	//81h: prepare to write/read address_x=81h
	//FEh: prepare to write/read address_x=FEh
	//FFh: prepare to write/read address_x=FFh									

Restriction	-									

9.9. SET_MIPI_INF: Set MIPI Information (Page7-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	-	-	r_tm_hster_switch	r_tm_hster_en[4:0]					00

Description	This command is used to set MIPI information.
	
	r_tm_hster_en[4:0]: Test mode for terminal switch select.
	//r_tm_hster_switch =1
	//bit0: mipi data lane0 terminal switch
	//bit1: mipi data lane1 terminal switch
	//bit2: mipi data lane2 terminal switch
	//bit3: mipi data lane3 terminal switch
	//bit4: mipi ck lane terminal switch
	r_tm_hster_switch: Test mode for terminal switch.
	//0: terminal test mode disable
	//1: terminal test mode enable									

Restriction	-									

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9.10. SET_MIPI_INF: Set MIPI Information (Page7-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	-	-	-	r_en_prbs	r_prbs_bit_err_lansel[1:0]		r_prbs_err_clr	r_err_sts_cnt_clr	00
Description	This command is used to set MIPI information. **r_err_sts_cnt_clr**: error status counter clear. //0: error report counter not clear //1: error report counter clear **r_prbs_err_clr**: prbs error clear. //0: prbs error status not clear //1: prbs errort status clear **r_prbs_bit_err_lansel[1:0]**: prbs lane select. //2b‘00: watch data lane0 prbs bit error count //2b‘01: watch data lane1 prbs bit error count //2b‘10: watch data lane2 prbs bit error count //2b‘11: watch data lane3 prbs bit error count **r_en_prbs**: prbs test mode enable. //0: disable. //1: enable.									
Restriction	-									

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9.11. SET_MIPI_INF: Set MIPI Information (Page7-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	r_cd_en	r_false_ctrl_err_rpt	r_periph_tout_err_rpt	r_lp_sync_err_rpt	r_esc_entry_err_rpt	r_eot_sync_err_rpt	r_sot_sync_err_rpt	r_sot_err_rpt	00
Description	This command is used to set MIPI information. **r_sot_err_rpt**: error report item. //0: disable. //1: enable. **r_sot_sync_err_rpt**: error report item. //0: disable. //1: enable. **r_eot_sync_err_rpt**: error report item. //0: disable. //1: enable. **r_esc_entry_err_rpt**: error report item. //0: disable. //1: enable. **r_lp_sync_err_rpt**: error report item. //0: disable. //1: enable. **r_periph_tout_err_rpt**: error report item. //0: disable. //1: enable. **r_false_ctrl_err_rpt**: error report item. //0: disable. //1: enable. **r_cd_en**: error report item. //0: disable. //1: enable.									
Restriction	-									

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9.12. SET_MIPI_INF: Set MIPI Information (Page7-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	r_mipi_err_rpt_en	r_protol_violation	r_invld_transmit_length	r_invld_vcid_err_rpt	r_invld_dtype_err_rpt	r_en_crc_det	r_ecc_on	r_ecc_1bit_err_rpt	80
Description	This command is used to set MIPI information. **r_ecc_1bit_err_rpt**: error report item. //0: disable. //1: enable. **r_ecc_on**: error report item. //0: disable. //1: enable. **r_en_crc_det**: error report item. //0: disable. //1: enable. **r_invld_dtype_err_rpt**: error report item. //0: disable. //1: enable. **r_invld_vcid_err_rpt**: error report item. //0: disable. //1: enable. **r_invld_transmit_length**: error report item. //0: disable. //1: enable. **r_protol_violation**: error report item. //0: disable. //1: enable. **r_mipi_err_rpt_en**: main error report enable. //0: disable. //1: enable.									
Restriction	-									

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9.13. SET_MIPI_INF: Set MIPI Information (Page7-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	index_pg7_dummy			pwr_lp11_det_enb	r_hsrx_ff_ter_cnt[1:0]		r_hsrx_ff_en_ck_sel	r_dcs_idle_mode_sel	02
Description	This command is used to set MIPI information. **r_dcs_idle_mode_sel**: //0: //1: **r_hsrx_ff_en_ck_sel**: //0: //1: **r_hsrx_ff_ter_cnt[1:0]**: **pwr_lp11_det_en**: **index_pg7_dummy**: reserved register.									
Restriction	-									

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10. Page 8 Command Description

10.1. SET_PWR_SEQ: Set Power On/Off Sequence (Page8-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	r_off_num[3:0]				start_num[3:0]				68
Description	This command is used to set power on/off sequence. **start_num[3:0]**: power-on start-state frame number, 0.5 frame per step. **r_off_num[3:0]**: power-off end-state frame number, 0.5 frame per step.									
Restriction	-									

10.2. SET_PWR_SEQ: Set Power On/Off Sequence (Page8-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	r_regen_delay_num[3:0]				r_waiteof_num[1:0]		freerun_num[1:0]		4A
Description	This command is used to set power on/off sequence. **freerun_num[1:0]**: freerun-state frame number, 1-frame per step. **r_waiteof_num[1:0]**: waiteof-state frame number, 1-frame per step. **r_regen_delay_num[3:0]**: delay regen start for corestate cascade sync, 0.2us per step.									
Restriction	-									

10.3. SET_PWR_SEQ: Set Power On/Off Sequence (Page8-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	-	-	-	r_nokia_flow_enb	r_all_gate_on_sel	r_tx_check_en	r_slp_in_skip_vs_f	r_skip_str2fr_en	07
Description	This command is used to set power on/off sequence. **r_skip_str2fr_en**: dummy, no function. **r_slp_in_skip_vs_f**: exit normal-state with/without tx vs. //0: wait vs. //1: no-need vs. **r_tx_check_en**: check DE at free-run. //0: disable. //1: enable. **r_all_gate_on_sel**: gip all-gate-on trigger selection. //0: sleep-in. //1: display-off. **r_nokia_flow_enb**: nokia flow, need sleep-out/display-on command. //0: enable. //1: disable.									
Restriction	-									

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10.4. SET_PWR_SEQ: Set Power On/Off Sequence (Page8-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	-	-	-	r_oneFtenL	-	tm_skip_all	-	for_sim	00
Description	This command is used to set power on/off sequence. **for_sim**: for simulation. **tm_skip_all**: skip power-on. //0: disable. //1: enable. **r_oneFtenL**: ten-line mode, for simulation.									
Restriction	-									

10.5. SET_BIST_PAT: Set BIST Pattern (Page8-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	-	r_flicker_tgl_enb	r_pixel_inv	r_chessboard_sel	border_sel[2:0]			r_pattern_keep	00
Description	This command is used to set BIST information. **r_pattern_keep**: pattern keep. //0: disable. //1: enable. **border_sel[1:0]**: border pattern, border color select, W/R/G/B. //2'b00: W //2'b01: R //2'b10: G //2'b11: B **border_sel[2]**: border pattern, 1 or 2 pixel-width select. //0: 1pixel //1: 2pixel **r_chessboard_sel**: pattern keep. //0: 1x1 pixel chang. //1: total 8x8 grid. **r_pixel_inv**: invert output pixel data. **r_flicker_tgl_enb**: flicker pattern frame toggle enable bar.									
Restriction	-									

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10.6. SET_BIST_PAT: Set BIST Pattern (Page8-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	r_gray_pat_sel_r[7:0]								80
Parameter 2	**R/W**	r_gray_pat_sel_g[7:0]								80
Parameter 3	**R/W**	r_gray_pat_sel_b[7:0]								80
Description	This command is used to set BIST information. **r_gray_pat_sel_r[7:0]**: gray pattern R data depth select. //00h: red gray 0. //01h: red gray 1. //80h: red gray 128. //09h: red gray 129. //0Eh: red gray 254. //0Fh: red gray 255. **r_gray_pat_sel_g[7:0]**: gray pattern G data depth select. //00h: green gray 0. //01h: green gray 1. //80h: green gray 128. //09h: green gray 129. //0Eh: green gray 254. //0Fh: green gray 255. **r_gray_pat_sel_b[7:0]**: gray pattern B data depth select. //00h: blue gray 0. //01h: blue gray 1. //80h: blue gray 128. //09h: blue gray 129. //0Eh: blue gray 254. //0Fh: blue gray 255.									
Restriction	-									

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10.7. SET_BIST_PAT: Set BIST Pattern (Page8-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	r_pattern_sel[7:0]								FF
Parameter 2	**R/W**	r_pattern_sel[15:8]								FF
Description	This command is used to set BIST output pattern. **r_pattern_sel[0]=1**: white pattern output. **r_pattern_sel[1]=1**: balck pattern output. **r_pattern_sel[2]=1**: red pattern output. **r_pattern_sel[3]=1**: green pattern output. **r_pattern_sel[4]=1**: blue pattern output. **r_pattern_sel[5]=1**: gray 128 level pattern output. **r_pattern_sel[6]=1**: color bar pattern output. **r_pattern_sel[7]=1**: horizontal 256 gray scale pattern output. **r_pattern_sel[8]=1**: vertical 256 gray scale pattern output. **r_pattern_sel[9]=1**: crosstalk pattern output. **r_pattern_sel[10]=1**: 1x1 chess board pattern output. **r_pattern_sel[11]=1**: white border pattern output. **r_pattern_sel[12]=1**: 1x1 sub-pixel pattern output. **r_pattern_sel[13]=1**: column flicker pattern output. **r_pattern_sel[14]=1**: v-line pattern output. **r_pattern_sel[15]=1**: h-line pattern output.									
Restriction	-									

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10.8. SET_BIST_PAT: Set BIST Pattern (Page8-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	r_pattern_frm_sel[7:0]								78
Parameter 2	**R/W**	-	-	-	-	-	-	-	r_pattern_frm_sel[8]	00
Description	This command is used to set BIST pattern to BIST pattern time. **r_pattern_frm_sel[8:0]**: bist pattern frame number select, 1-frame per step.									
Restriction	-									

10.9. SET_H_BLK: Set Horizontal Blanking (Page8-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	r_hbp[7:0]								50
Description	This command is used to set display horizontal back porch. **r_hbp[7:0]**: set display horizontal back porch. r_hbp[7:0] follow TX input timing setting.									
Restriction	-									

10.10. SET_H_BLK: Set Horizontal Blanking (Page8-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	r_hfp[7:0]								DA
Description	This command is used to set display horizontal front porch. **r_hfp[7:0]**: set display horizontal front porch. r_hfp[7:0] follow TX input timing setting.									
Restriction	-									

10.11. SET_V_BLK: Set Vertical Blanking (Page8-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	r_vbp[7:0]								0A
Description	This command is used to set display vertical back porch. **r_vbp[7:0]**: set display vertical back porch. r_vbp[7:0] follow TX input timing setting.									
Restriction	-									

10.12. SET_V_BLK: Set Vertical Blanking (Page8-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	r_vfp[7:0]								14
Description	This command is used to set display vertical front porch. **r_vfp[7:0]**: set display vertical front porch. r_vfp[7:0] follow TX input timing setting.									
Restriction	-									

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10.13. SET_H_BLK: Set Horizontal Blanking (Page8-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	r_hfp_least[7:0]								0A
Description	This command is used to set display minimum horizontal front porch. **r_hfp_least[7:0]**: set display minimum horizontal front porch. r_hfp_least[7:0]: follow TX input timing setting.									
Restriction	-									

10.14. SET_FREE_H_BLK: Set Freerun Horizontal Blanking (Page8-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	r_hfp_free[7:0]								DA
Description	This command is used to set freerun horizontal front porch. **r_hfp_free[7:0]**: freerun horizontal front porch setting.									
Restriction	-									

10.15. SET_FREE_V_BLK: Set Freerun Vertical Blanking (Page8-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	r_vfp_free[7:0]								75
Description	This command is used to set freerun vertical front porch. **r_vfp_free[7:0]**: freerun vertical front porch setting.									
Restriction	-									

10.16. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	-	r_hole_time_sel_doze[2:0]			r_hole_time_sel[3:0]				03
Description	This command is used to set GIP hole information. **r_hole_time_sel[3:0]**: pulse-width for each hole, 200us~50us.(10us/step) **r_hole_time_sel_doze[2:0]**: for doze_long_v mode enable in long-h , 80us~150us. (10us/step) hole select									
Restriction	-									

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10.17. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R48h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	r_vs_clr_hole_qsync_en	r_scan_hole_in_vbp_en	r_vs_clr_hole_cnt_en	r_hole_num[4:0]					0C
Description	This command is used to set GIP hole information. **r_hole_num[4:0]**: hole number select. //00h: without hole //01h: 1hole //10h: 2hole //11h: 3hole //1Eh: 31hole //1Fh: 32hole **r_vs_clr_hole_cnt_en**: vs clear hole_cnt enable. //0h:disable //1h:enable **r_scan_hole_in_vbp_en**: scan hole open in vbp enable. //0h:disable //1h:enable **r_vs_clr_hole_qsync_en**: vs clear hole enable. //0h:disable //1h:enable									
Restriction	-									

10.18. REV_REG: Reserved Register (Page8-R49h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	r_hole_sft_front[7:0]								**00**
Description	This command is no function. **r_hole_sft_front[7:0]**: dummy, no function.									
Restriction	-									

10.19. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	r_hole_sft_back[7:0]								00

Description	This command is used to set GIP hole information.
	
	r_hole_sft_back[7:0]: shift scan hole back line number.
	//00h: no shift
	//01h: 1line
	//80h: 128line
	//81h: 129line
	//FEh: 254line
	//FFh: 255line									

Restriction	-									

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10.20. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R4Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	1	4B
Parameter 1	**R/W**	r_hole_dly_4_source[1:0]		r_rescan_num_gip[2:0]			r_rescan_num_sd[2:0]			18
Description	This command is used to set GIP hole information. **r_rescan_num_sd[2:0]**: long_h only, rescan line num after hole finished, for source. **r_rescan_num_gip[2:0]**: long_h only, rescan line num after hole finished for gip. (qeual to ckv-width -1) **r_hole_dly_4_source[1:0]**: long_h only, delay hole to inform SD to pull GND. //2b'00: 0us. //2b'01: 2us. //2b'10: 4us. //2b'11: 8us.									
Restriction	-									

10.21. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R4Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4C
Parameter 1	**R/W**	r_hole_dis_1st_longh[7:0]								96
Parameter 2	**R/W**	-	-	-	-	-	-	-	r_hole_dis_1st_longh[8]	00
Description	This command is used to set GIP hole information. **r_hole_dis_1st_longh[8:0]**: long_h only, manual mode, line count.									
Restriction	-									

10.22. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R4Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	1	0	4E
Parameter 1	**R/W**	r_hole_dis_2nd_longh[7:0]								96
Parameter 2	**R/W**	-	-	-	-	-	-	-	r_hole_dis_2nd_longh[8]	00
Description	This command is used to set GIP hole information. **r_hole_dis_2nd_longh[8:0]**: long_h only, line count between each hole.									
Restriction	-									

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10.23. SET_GIP_HOL_INF: Set GIP Hole Information (Page8-R50h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	0	0	50
Parameter 1	**R/W**	r_hole_dis_1st_longv[7:0]								04
Description	This command is used to set GIP hole information. **r_hole_dis_1st_longv[7:0]**: long_v only, line num of 1st hole after last DE.									
Restriction	-									

10.24. SET_GIP_LINE_SHT:Set GIP Line Shift (Page8-R51h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	0	1	51
Parameter 1	**R/W**	r_line_shift_step[3:0]				r_line_shift_range[3:0]				00
Parameter 2	**R/W**	r_te_hblk_range_inv	r_line_shift_en	r_line_shift_direction	r_line_shift_mode	r_line_shift_rate[3:0]				00
Description	This command is used to set GIP line shift information. **r_line_shift_en**: line-shift function enable. //0:disable //1:enable **r_line_shift_step[3:0]**: line shift step per next changed, 1~16. //00: 1 line //01: 2 line … //0E: 15 line //0F: 16 line **r_line_shift_range[3:0]**: total line shift number, 1~16. //00h: total shift 1 line //01h: total shift 2 line … //0Eh: total shift 15 line //0Fh: total shift 16 line **r_line_shift_rate[3:0]**: frame number of next shift changed, 1~16 //00: 1 frame //01: 2 frames … //0E: 15 frames //0F: 16 frames Ex: line_shift_step=1, line_shift_range=4, line_shift_rate=1 Range/step=4, total shift 4 times, each time chnaged by 1 frame. **r_line_shift_mode**: //0: restart at original line number. //1: go back from the end of line number. **r_line_shift_direction**: line-shift direction. //0: line number decrease. //1: line number increase. **r_te_hblk_range_inv**: invert TE mode1 hblk range.									
Restriction	-									

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10.25. SET_TE1_INF: Set TE1 Information (Page8-R52h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	1	0	52
Parameter 1	**R/W**	r_te1_active_high_en	r_te1_enable	r_hole_dly_4_gip[3:0]				r_pump_clk_en	r_set_by_vs_en	83
Description	This command is used to set TE1 information. **r_set_by_vs_en**: //0:disable //1:enable **r_pump_clk_en**: //0:disable //1:enable **r_hole_dly_4_gip[3:0]**: long_h_only, delay hole for 0~60us to inform SD to pull GND. **r_te1_enable**: TE1 enable. //0:disable //1:enable **r_te1_active_high_en**: TE1 active-high/low selection. //0:High //1:Low									
Restriction	-									

10.26. SET_TE1_INF: Set TE1 Information (Page8-R53h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	1	1	53
Parameter 1	**R/W**	r_te1_position[7:0]								00
Parameter 2	**R/W**	-	-	-	-	r_te1_position[11:8]				00
Description	This command is used to set TE1 information. **r_te1_position[11:0]**: TE1 start position. (line)									
Restriction	-									

10.27. SET_TE1_INF: Set TE1 Information (Page8-R54h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	0	0	54
Parameter 1	**R/W**	r_te1_pulse_width[7:0]								00
Parameter 2	**R/W**	-	-	-	-	r_te1_pulse_width[11:8]				00
Description	This command is used to set TE1 information. **r_te1_pulse_width[11:0]**: TE1 pulse-width. (line)									
Restriction	-									

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10.28. SET_Q_SYN_INF: Set Q-Sync Information (Page8-R55h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	0	1	55
Parameter 1	**R/W**	r_hole_dis_1st_qsync[7:0]								A6
Parameter 2	**R/W**	-	-	-	r_hole_dis_1st_qsync[12:8]					04
Description	This command is used to set Q-sync information. **r_hole_dis_1st_qsync[12:0]**: q-sync only, distance for 1st hole from vs if sync_to_disp_vs_en.									
Restriction	-									

10.29. SET_Q_SYN_INF: Set Q-Sync Information (Page8-R56h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	0	56
Parameter 1	**R/W**	r_hole_dis_2nd_qsync[7:0]								A6
Parameter 2	**R/W**	-	-	-	r_hole_dis_2nd_qsync[12:8]					04
Description	This command is used to set Q-sync information. **r_hole_dis_1st_qsync[12:0]**: q-sync only, distance select between each hole.									
Restriction	-									

10.30. SET_TCON_INF: Set TCON Information (Page8-R57)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	1	57
Parameter 1	**R/W**	-	-	-	-	regen_hs_width [1:0]		slpin_range_sel [1:0]		05
Description	This command is used to set TCON information. **slpin_range_sel[1:0]**: **regen_hs_width[1:0]**: regen hsync pulse width, N*4T. (4T/step)									
Restriction	-									

10.31. SET_TCON_INF: Set TCON Information (Page8-R58)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	0	58
Parameter 1	**R/W**	r_vs_width[3:0]				r_vs_r_adj[3:0]				25
Description	This command is used to set TCON information. **r_vs_r_adj[3:0]**: IP vs position select for color engine. (line) **r_vs_width[3:0]**: IP vs pulse width select for color engine. (line)									
Restriction	-									

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10.32. SET_TCON_INF: Set TCON Information (Page8-R59)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	1	59
Parameter 1	**R/W**	r_ll_width[3:0]				r_ll_r_adj[3:0]				71
Description	This command is used to set TCON information. **r_ll_r_adj[3:0]**: IP last-line position select for color engine. (line) **r_ll_width[3:0]**: IP last-line pulse width select for color engine. (line)									
Restriction	-									

10.33. SET_RES_INF: Set Resolution Information (Page8-R5A)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	0	5A
Parameter 1	**R/W**	r_h_res_set[7:0]								00
Description	This command is used to set panel horizontal resolution. **r_h_res_set[7:0]**: horizontal resolution setting. For example: 1200RGBx2000 resolution. R5Ah=B0h. R5Bh=70h. R5Ch=74h.									
Restriction	-									

10.34. SET_RES_INF: Set Resolution Information (Page8-R5B)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	1	5B
Parameter 1	**R/W**	r_v_res_set[7:0]								00
Description	This command is used to set panel vertical resolution. **r_v_res_set[7:0]**: vertical resolution setting. For example: 1200RGBx2000 resolution. R5Ah=B0h. R5Bh=70h. R5Ch=74h.									
Restriction	-									

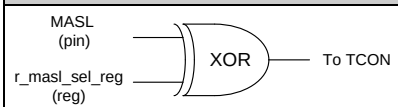
10.35. SET_RES_INF: Set Resolution Information (Page8-R5C)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	0	5C
Parameter 1	**R/W**	r_v_res_set[11:8]				r_h_res_set[11:8]				00
Description	This command is used to set panel resolution. **r_h_res_set[11:8]**: horizontal resolution setting. **r_v_res_set[11:8]**: vertical resolution setting. For example: 1200RGBx2000 resolution. R5Ah=B0h. R5Bh=70h. R5Ch=74h.									
Restriction	-									

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10.36. SET_TCON_INF: Set TCON Information (Page8-R5D)

<table>
<tr><th>CMD/PAs</th><th>R/W</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>R/W</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>5D</td></tr>
<tr><td>Parameter 1</td><td>R/W</td><td>r_masl_sel_reg</td><td>-</td><td>r_mode_sel_reg</td><td>r_bist_en_sel_reg</td><td colspan="2">r_lansel_sel_reg [1:0]</td><td>r_pnsw_sel_reg</td><td>r_masloc</td><td>01</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to set TCON information, and some H/W pin and Register do XOR function.

For example:
<table>
<tr><th colspan="2">Combination Logic</th><th colspan="3">Truth table</th><th>To Tcon</th></tr>
<tr><th colspan="2"></th><th>PIN</th><th>REG</th><th>To Tcon</th><th></th></tr>
<tr><td colspan="2" rowspan="4"> 
</td><td>0</td><td>0</td><td>0</td><td rowspan="4">Master/Slave IC selection.
MASL = 0, Slave IC.
MASL = 1, Master IC.</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td></tr>
</table>

r_masloc: MASLOC register setting.
//0:Slave
//1:Master
r_pnsw_sel_reg: PNSW use register enable. (XOR)

r_lansel_sel_reg[1:0]: LANSEL[1:0] use register enable. (XOR)

r_bist_en_sel_reg: BIST_EN use register enable. (XOR)

r_mode_sel_reg: MODE use register enable. (XOR)

r_masl_sel_reg: MASL use register enable. (XOR)</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
</table>

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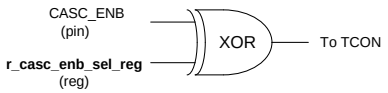
10.37. SET_TCON_INF: Set TCON Information (Page8-R5E)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	0	5E
Parameter 1	**R/W**	r_bist_en	r_pnsw	r_casc_enb_sel_reg	r_dswap_sel_reg[1:0]		r_dual_sel_reg	r_dswap[1:0]		00

Description

This command is used to set TCON information, and some H/W pin and Register do XOR function.

For example:

Combination Logic	Truth table			To Tcon
	PIN	REG	To Tcon	
	0	0	0	One/Two chip IC selection. CASC_ENB = 0, Two chip. CASC_ENB = 1, One chip..
	0	1	1	
	1	0	1	
	1	1	0	

r_dswap[1:0]: DSWAP[1:0] register setting.

r_dswap[1:0]	CKP	CKN	D0P	D0N	D1P	D1N	D2P	D2N	D3P	D3N
MIPI lanes mapping table										
00	CKP	CKN	D1P	D1N	D2P	D2N	D3P	D3N	D0P	D0N
01	CKP	CKN	D1P	D1N	D0P	D0N	D3P	D3N	D2P	D2N
10	CKP	CKN	D2P	D2N	D1P	D1N	D0P	D0N	D3P	D3N
11	CKP	CKN	D0P	D0N	D1P	D1N	D2P	D2N	D3P	D3N

r_dual_sel_reg: DUAL use register enable. (XOR)

r_dswap_sel_reg[1:0]: DSWAP_SEL[1:0] use register enable. (XOR)

r_casc_enb_sel_reg: CASC_ENB use register enable. (XOR)

r_pnsw: PNSW register setting.

r_bist_en: BIST_EN register setting.
//0: Exit BIST mode.
//1: BIST mode.

Restriction	-

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10.38. SET_TCON_INF: Set TCON Information (Page8-R5F)

<table>
<tr><th>CMD/PAs</th><th>R/W</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>R/W</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>5F</td></tr>
<tr><td>Parameter 1</td><td>R/W</td><td>slp_out</td><td>disp_on</td><td>r_dual</td><td>r_masl</td><td>-</td><td>r_mode</td><td colspan="2">r_lansel[1:0]</td><td>00</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to set TCON information.

Slp_out:

disp_on:

r_lansel[1:0]: LANSEL[1:0] register setting.
<table>
<tr><th>LANSEL[1:0]</th><th>MIPI lane number</th></tr>
<tr><td>00</td><td>T.B.D.</td></tr>
<tr><td>01</td><td>2 lanes</td></tr>
<tr><td>10</td><td>3 lanes</td></tr>
<tr><td>11</td><td>4 lanes</td></tr>
</table>
r_mode: MODE register setting.
//0: Multi-drop mode.
//1: LR mode.
r_masl: MASL register setting.
//0: Slave IC.
//1: Master IC.
r_dual: DUAL register setting.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
</table>

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10.39. SET_TCON_INF: Set TCON Information (Page8-R60)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	0	60
Parameter 1	**R/W**	r_led_en_reg	reg_wr_protect2	reg_wr_protect1	-	r_nbw	-	r_all_sel_pin	r_casc_enb	E0
Description	This command is used to set TCON information. **r_casc_enb**: CASC_ENB register setting. //0: two chip application. //1: one chip application. **r_all_sel_pin**: all select pin enable. **r_nbw**: NBW register setting. **reg_wr_protect1**: //0: no protect //1: write protect **reg_wr_protect2**: //0: no protect //1: write protect **r_led_en_reg**: register setting									
Restriction	-									

10.40. SET_TCON_INF: Set TCON Information (Page8-R61)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	1	61
Parameter 1	**R/W**	-	-	-	-	-	-	vs_inv_sel	hs_inv_sel	00
Description	This command is used to set TCON information. **hs_inv_sel**: invert hs for dat_framing. **vs_inv_sel**: invert vs for dat_framing.									
Restriction	-									

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10.41. SET_TCON_INF: Set TCON Information (Page8-R62)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	0	62
Parameter 1	**R/W**	-	r_rescan_in_blk_en	r_doze_long_v_en	-	r_sync_to_disp_vs_en	r_cas_slave_sync_en	r_qsync_en	r_scan_long_h_en	05
Description	This command is used to set TCON information. **r_scan_long_h_en**: touch scan long-h mode enable. //0: disable //1: enable **r_qsync_en**: touch scan q-sync mode enable. //0: disable //1: enable **r_cas_slave_sync_en**: cascade sync de_regen enable. //0: disable //1: enable **r_sync_to_disp_vs_en**: q-sync only, hole count synchronize to display vs. //0: disable //1: enable **r_doze_long_v_en**: doze long-v mode for firmware to enable. //0: disable //1: enable **r_rescan_in_blk_en**: v-blanking rescan enable for q-sync. //0: disable //1: enable									
Restriction	-									

10.42. SET_HW RST_DEG_INF: Set H/W Reset De-glitch Information (Page8-R63)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	1	63
Parameter 1	**R/W**	-	-	-	-	-	rst_deglitch_sel	-	-	04
Description	This command is used to set TCON information. **r_rescan_in_blk_en**: hw-rst de-glitch select. //0: 5us. //1: 10us.									
Restriction	-									

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10.43. SET_TCON_INF: Set TCON Information (Page8-R64)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	0	64
Parameter 1	**R/W**	slpout_reload_en	standby_reload_en	r_bist_gating_en	r_clk_pol	r_non_gated_en	r_lvds_skew_sel[2:0]			A0
Description	This command is used to set TCON information. **r_lvds_skew_sel[2:0]**: dat_framing clock skew select. **r_non_gated_en**: clkgen no glock gating enable. //0: disable //1: enable **r_clk_pol**: dat_framing clock pol select. **r_bist_gating_en**: bist gating enable. //0: disable //1: enable **slpout_reload_en**: //0: reload again //1: no reload again									
Restriction	-									

10.44. SET_TCON_INF: Set TCON Information (Page8-R65)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	1	65
Parameter 1	**R/W**	r_disp_pll_clk_sel	clkdet_en	r_oscd_sel[1:0]		r_dat_gating_en	r_gip_gating_en	1	r_datapath_gating_en	7F
Description	This command is used to set TCON information. **r_datapath_gating_en**: datapath clock gating enable. //0: disable //1: enable **r_gip_gating_en**: gip clock gating enable. //0: disable //1: enable **r_dat_gating_en**: dat_framing clock gating enable. //0: disable //1: enable **r_oscd_sel[1:0]**: clkgen datapath osc select. //2b'00: 20Mhz. //2b'01: 40MHz. //2b'11: 80MHz. **clkdet_en**: clock detect enable. //0: disable //1: enable **r_disp_pll_clk_sel**: clkgen normal display clock select PLL clock enable. //0: disable //1: enable									
Restriction	-									

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10.45. SET_TCON_INF: Set TCON Information (Page8-R66)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	0	66
Parameter 1	**R/W**	0	0	r_masl_rd_sel	r_8to10_sel	hfrc	r_ext_pll_clk_en	r_ext_osct_clk_en	r_ext_oscd_clk_en	08
Description	This command is used to set TCON information. **r_ext_oscd_clk_en**: clkgen external OSCD enable. //0: disable //1: enable **r_ext_osct_clk_en**: clkgen external OSCT enable. //0: disable //1: enable **r_ext_pll_clk_en**: clkgen external pll clock enable. //0: disable //1: enable **hfrc**: dither hfrc enable. //0: disable //1: enable **r_8to10_sel**: datapath 8-bit to 10-bit select. //0: insert MSB[7:6]. //1: insert 2b'00. **r_masl_rd_sel**: clock detect enable. //0: disable //1: enable									
Restriction	-									

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10.46. SET_TCON_INF: Set TCON Information (Page8-R67)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	67
Parameter 1	**R/W**	-	-	-	-	-	-	-	pin_dg_enb	00
Description	This command is used to set TCON information. **pin_dg_enb**: pin-deglitch enable bar. //0: disable //1: enable									
Restriction	-									

10.47. SET_TCON_INF: Set TCON Information (Page8-R68)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	0	68
Parameter 1	**R/W**	-	-	-	r_resend_en	r_sync_slp_out_en	r_sync_disp_on_en	r_time_out_sel[1:0]		00
Description	This command is used to set TCON information. **r_time_out_sel[1:0]**: sync_ctrl packet wait time select. **r_sync_disp_on_en**: sync_ctrl sync display-on command enable. //0: disable //1: enable **r_sync_slp_out_en**: sync_ctrl sync sleep-out command enable. //0: disable //1: enable **r_resend_en**: sync_ctrl resend packet enable. //0: disable //1: enable									
Restriction	-									

10.48. SET_SRAM_INF: Set SRAM Information (Page8-R69)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	1	69
Parameter 1	**R/W**	r_repaire_wline_0[7:0]								FF
Description	This command is used to set SRAM information. **r_repaire_wline_0[7:0]**: sram_ctrl line buffer repaire address_0.									
Restriction	-									

10.49. SET_SRAM_INF: Set SRAM Information (Page8-R6A)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	1	0	6A
Parameter 1	**R/W**	r_repaire_wline_1[7:0]								FF
Description	This command is used to set SRAM information. **r_repaire_wline_1[7:0]**: sram_ctrl line buffer repaire address_1.									
Restriction	-									

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10.50. SET_SRAM_INF: Set SRAM Information (Page8-R6D)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	1	0	1	6D
Parameter 1	**R/W**	r_sh2c_sft_sel[3:0]				r_rd2lc_sft_sel[3:0]				00
Description	This command is used to set SRAM information. **r_rd2lc_sft_sel[3:0]**: sram_ctrl rd2lc shift select. **r_sh2c_sft_sel[3:0]**: sram_ctrl sh2c shift select.									
Restriction	-									

10.51. SET_SRAM_INF: Set SRAM Information (Page8-R6E)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	1	1	0	6E
Parameter 1	**R/W**	-	-	-	r_int_read_latch_sel	r_new_de_sft_sel[3:0]				00
Description	This command is used to set SRAM information. **r_new_de_sft_sel[3:0]**: sram_ctrl new de shift select. **r_int_read_latch_sel**: sram_ctrl internal read latch data time select. //0: sh2c rising edge -> 1T. //1: sh2c rising edge -> 2T.									
Restriction	-									

10.52. SET_TCON_INF: Set TCON Information (Page8-R6F)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	1	1	1	6F
Parameter 1	**R/W**	-	-	-	r_pll_refint_sel[1:0]		r_pll_ref_sel[1:0]		r_disp_gating_en	01
Description	This command is used to set TCON information. **r_disp_gating_en**: clkgen sys_4pixel_clk gating enable. //0: disable //1: enable **r_pll_ref_sel[1:0]**: clkgen PLL_REF_CLK_OUT slock select. //2b'00: osct_div64. //2b'01: osct_div32. //2b'10: osct_div16. **r_pll_refint_sel[1:0]**: clkgen AD_OSC_APR clock select. //2b'00: osct_div64. //2b'01: osct_div32. //2b'10: osct_div16.									
Restriction	-									

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10.53. SET_TCON_INF: Set TCON Information (Page8-R70)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	0	70
Parameter 1	**R/W**	-	-	-	-	-	-	-	r_tm_reg_key_en	00
Description	This command is used to set TCON information. **r_tm_reg_key_en**: testmode register key lock, CP need trim this bit after all trim finished. //0: disable //1: enable									
Restriction	-									

10.54. SET_TP_AFE_INF: Set TP_AFE Information (Page8-R72)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	1	0	72
Parameter 1	**R/W**	tp_afe_trim1_left[7:0]								3F
Parameter 2	**R/W**	tp_afe_trim2_left[7:0]								08
Parameter 3	**R/W**	tp_afe_trim3_left[7:0]								00
Parameter 4	**R/W**	tp_afe_trim4_left[7:0]								00
Parameter 5	**R/W**	tp_afe_trim1_right[7:0]								3F
Parameter 6	**R/W**	tp_afe_trim2_right[7:0]								08
Parameter 7	**R/W**	tp_afe_trim3_right[7:0]								00
Parameter 8	**R/W**	tp_afe_trim4_right[7:0]								00
Description	This command is used to set TP AFE trim information. **tp_afe_trimn_left[7:0]**: n=1~4. **tp_afe_trimn_right[7:0]**: n=1~4.									
Restriction	-									

10.55. SET_INT_CRC_INF: Set Internal CRC Information (Page8-R73)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	1	1	73
Parameter 1	**R/W**	-	-	-	-	-	r_crc_mux_sel[1:0]		r_crc_mux_en	00
Description	This command is used to set internal CRC information. **r_crc_mux_en**: r_crc_mux_en: Internal data path crc enable. //0: disable. //1: enable. **r_crc_mux_sel[1:0]**: Internal data path crc mux select. //2b'00: sram_out. //2b'01: coleng_out. //2b'10: dgamma out. //2b'11: NA (sram_out).									
Restriction	-									

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10.56. SET_INT_CRC_INF: Set Internal CRC Information (Page8-R74)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	0	0	74
Parameter 1	**R/W**	crc_internal_o[7:0]								00
Description	This command is used to check internal CRC information. **crc_internal_o[7:0]**: internal data path crc result. Valuable only when r_crc_mux_en=1.									
Restriction	-									

10.57. REV_REG: Reserved Register (Page8-R75)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	0	1	75
Parameter 1	**R/W**	index_pg8_dummy								00
Description	This command is a reserved register.									
Restriction	-									

10.58. REV_REG: Reserved Register (Page8-R76)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	1	0	76
Parameter 1	**R/W**	index_pg8_dummy								00
Description	This command is a reserved register.									
Restriction	-									

10.59. REV_REG: Reserved Register (Page8-R77)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	1	1	77
Parameter 1	**R/W**	index_pg8_dummy								00
Description	This command is a reserved register.									
Restriction	-									

10.60. REV_REG: Reserved Register (Page8-R78)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	1	0	0	0	78
Parameter 1	**R/W**	index_pg8_dummy								00
Description	This command is a reserved register.									
Restriction	-									

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10.61. SET PRAM/DARM repair: PRAM/DRAM repair function (Page8-R7B)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	1	0	1	1	7B
Parameter 1	**R/W**	-	dram_ins2_repair_en	dram_ins1_repair_en	dram_ins0_repair_en	pram_ins3_repair_en	pram_ins2_repair_en	pram_ins1_repair_en	pram_ins0_repair_en	00
Parameter 2	**R/W**	pram_ins0_repair_addr[7:0]								
Parameter 3	**R/W**	pram_ins1_repair_addr[7:0]								
Parameter 4	**R/W**	pram_ins2_repair_addr[7:0]								
Parameter 5	**R/W**	pram_ins3_repair_addr[7:0]								
Parameter 6	**R/W**	dram_ins0_repair_addr[7:0]								
Parameter 7	**R/W**	dram _ins1_repair_addr[7:0]								
Parameter 8	**R/W**	dram _ins2_repair_addr[7:0]								
Description	This command is a reserved register. Pram_ins0~3_repair_en: **pram_ins0~3_repiar_en**: Enable PRAM repair function **pram_ins0~3_repiar_addr[7:0]**: Set repair addres for PRAM. **dram_ins0~2_repiar_en**: Enable DRAM repair function **dram_ins0~2_repiar_addr[7:0]**: Set repair addres for DRAM.									
Restriction	-									

10.62. SET PRAM/DARM failure status: PRAM/DRAM failure states(Page8-R7C)

CMD/Pas	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	1	1	0	0	7C
Parameter 1	**R/W**	-	dram_ins2_bist_fail	dram_ins1_bist_fail	dram_ins0_bist_fail	pram_ins3_bist_fail	pram_ins2_bist_fail	pram_ins1_bist_fail	pram_ins0_bist_fail	00
Parameter 2	**R/W**	pram_ins0_fail_addr[7:0]								
Parameter 3	**R/W**	pram_ins1_ fail _addr[7:0]								
Parameter 4	**R/W**	pram_ins2_ fail _addr[7:0]								
Parameter 5	**R/W**	pram_ins3_ fail _addr[7:0]								
Parameter 6	**R/W**	dram_ins0_ fail _addr[7:0]								
Parameter 7	**R/W**	dram _ins1_ fail _addr[7:0]								
Parameter 8	**R/W**	dram _ins2_ fail _addr[7:0]								
Description	This command is a reserved register. Pram_ins0~3_repair_en: **pram_ins0~3_bist_fail**: Enable PRAM bist function for fail detection **pram_ins0~3_ fail _addr[7:0]**: Record fail addres for PRAM. **dram_ins0~2_ bist_fail**: Enable DRAM bist function for fail detection **dram_ins0~2_ fail _addr[7:0]**: Record fail addres for DRAM.									
Restriction	-									

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11. Page 9 Command Description

11.1. SET_CP_VER_INF: Set CP Version Information (Page9-R5Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	1	5D
Parameter 1	**R/W**	cp_version_msb[7:0]								00
Description	This command is used to set CP version information. **cp_version_msb[7:0]**:									
Restriction	-									

11.2. SET_CP_VER_INF: Set CP Version Information (Page9-R5Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	0	5E
Parameter 1	**R/W**	cp_version_lsb[7:0]								00
Description	This command is used to set CP version information. **cp_version_llsb[7:0]**:									
Restriction	-									

11.3. RD_IC_VER_INF: Read IC Version Information (Page9-R5Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	1	5F
Parameter 1	**R**	version_id[7:0]								11
Description	This command is used to read IC version information. **version_id[7:0]**: 01h: JD9366TC AA version. 02h: JD9366TC AB version. 03h: JD9366TC AB version. 04h: JD9366TC AD version.									
Restriction	-									

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11.4. RD_OTP_IND_INF: Read OTP Index Information (Page9-R60h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	0	60
Parameter 1	**R**	index_gip	index_ce	index_le	index_dgamma_set	index_r_dgamma	index_g_dgamma	index_b_dgamma	index_dither	00
Description	This command is used to read OTP index information. **index_dither** : if any register in dither group has been OTP, then index_dither = '1'. **index_b_dgamma** : if any register in b_dgamma group has been OTP, then index_b_dgamma = '1'. **index_g_dgamma** : if any register in g_dgamma group has been OTP, then index_g_dgamma = '1'. **index_r_dgamma** : if any register in r_dgamma group has been OTP, then index_r_dgamma = '1'. **index_dgamma_set** : if any register in dgamma_set group has been OTP, then index_dgamma_set = '1'. **index_le** : if any register in le group has been OTP, then index_le = '1'. **index_ce** : if any register in ce group has been OTP, then index_ce = '1'. **index_gip** : if any register in gip group has been OTP, then index_gip = '1'.									
Restriction	-									

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11.5. RD_OTP_IND_INF: Read OTP Index Information (Page9-R61h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	1	61
Parameter 1	**R**	index_mipi	index_pg7_dummy	index_corestate	index_regen	index_glue	index_framing	index_group	index_esd	00
Description	This command is used to read OTP index information. **index_esd** : if any register in esd group has been OTP, then index_esd = '1'. **index_group** : if any register in group has been OTP, then index_group = '1'. **index_framing** : if any register in framing group has been OTP, then index_framing = '1'. **index_glue** : if any register in glue group has been OTP, then index_glue = '1'. **index_regen** : if any register in regen group has been OTP, then index_regen = '1'. **index_corestate** : if any register in corestate group has been OTP, then index_corestate = '1'. **index_pg7_dummy** : if any register in pg7_dummy group has been OTP, then index_pg7_dummy = '1'. **index_mipi** : if any register in mipi group has been OTP, then index_mipi = '1'.									
Restriction	-									

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11.6. RD_OTP_IND_INF: Read OTP Index Information (Page9-R62h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	0	62
Parameter 1	**R**	index_sync_ctrl	index_repaire	index_sram_ctrl	index_clk_gen	index_tm	index_pw	index_gas	index_tsen	00
Description	This command is used to read OTP index information. **index_tsen** : if any register in tsen group has been OTP, then index_tsen = '1'. **index_gas** : if any register in gas group has been OTP, then index_gas = '1'. **index_pw** : if any register in pw group has been OTP, then index_pw = '1'. **index_tm** : if any register in tm group has been OTP, then index_tm = '1'. **index_ clk_gen** : if any register in clk_gen group has been OTP, then index_clk_gen = '1'. **index_ sram_ctrl** : if any register in sram_ctrl group has been OTP , then index_sram_ctrl = '1'. **index_ repaire** : if any register in repaire group has been OTP, then index_repaire = '1'. **index_ sync_ctrl** : if any register in sync_ctrl group has been OTP, then index_sync_ctrl = '1'.									
Restriction	-									

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11.7. RD_OTP_IND_INF: Read OTP Index Information (Page9-R63h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	1	63
Parameter 1	**R**	index_pw_tp	index_dphy_ana	index_misc1	index_misc2	index_misc3	index_misc4	index_vddd	index_vcipr	**00**
Description	This command is used to read OTP index information. **index_vcipr** : if any register in vcipr group has been OTP, then index_ vcipr = '1'. **index_vddd** : if any register in vddd group has been OTP, then index_vddd = '1'. **index_misc4** : if any register in misc4 group has been OTP, then index_misc4 = '1'. **index_misc3** : if any register in misc3 group has been OTP, then index_misc3 = '1'. **index_misc2** : if any register in misc2 group has been OTP, then index_misc2 = '1'. **index_misc1** : if any register in misc1 group has been OTP, then index_misc1 = '1'. **index_dphy_ana** : if any register in dphy_ana group has been OTP, then index_dphy_ana = '1'. **index_pw_tp** : if any register in pw_tp group has been OTP, then index_pw_tp = '1'.									
Restriction	-									

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11.8. RD_OTP_IND_INF: Read OTP Index Information (Page9-R64h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	0	64
Parameter 1	**R**	index_pw_ana1	index_dcdc_ana1	index_sd_ana	index_gam	index_sram	index_afe	index_trimming1	index_trimming2	00
Description	This command is used to read OTP index information. **index_trimming2** : if any register in trimming2 group has been OTP, then index_trimming2 = '1'. **index_trimming1** : if any register in trimming1 group has been OTP, then index_trimming1 = '1'. **index_afe** : if any register in afe group has been OTP, then index_afe = '1'. **index_sram** : if any register in sram group has been OTP, then index_sram = '1'. **index_gam** : if any register in gam group has been OTP, then index_gam = '1'. **index_sd_ana** : if any register in sd_ana group has been OTP, then index_sd_ana = '1'. **index_dcdc_ana1** : if any register in dcdc_ana1 group has been OTP, then index_dcdc_ana1 = '1'. **index_pw_ana1** : if any register in pw_ana1 group has been OTP, then index_ pw_ana1 = '1'.									
Restriction	-									

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11.9. RD_OTP_IND_INF: Read OTP Index Information (Page9-R65h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	0	1	65
Parameter 1	**R**	index_sacode	index_icinfo	index_gamma	index_sd_dmy	index_ana_dmy	index_trimming3	index_trimming4	index_trimmingtp	00
Description	This command is used to read OTP index information. **index_trimmingtp** : if any register in trimmingtp group has been OTP, then index_trimmingtp = '1'. **index_trimming4** : if any register in trimming4 group has been OTP, then index_trimming4 = '1'. **index_trimming3** : if any register in trimming3 group has been OTP, then index_trimming3 = '1'. **index_ana_dmy** : if any register in ana_dmy group has been OTP, then index_ana_dmy = '1'. **index_sd_dmy** : if any register in sd_dmy group has been OTP, then index_sd_dmy = '1'. **index_gamma** : if any register in gamma group has been OTP, then index_gamma = '1'. **index_icinfo** : if any register in icinfo group has been OTP, then index_icinfo = '1'. **index_sacode** : if any register in sacode group has been OTP, then index_sacode = '1'.									
Restriction	-									

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11.10. RD_OTP_IND_INF: Read OTP Index Information (Page9-R66h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	0	66
Parameter 1	**R**	-	-	-	index_vcom_5	index_vcom_4	index_vcom_3	index_vcom_2	index_vcom_1	00
Description	This command is used to read OTP index information. **index_vcom_1** : if any register in vcom_1 group has been OTP, then index_ vcom_1 = '1'. **index_vcom_2** : if any register in vcom_2 group has been OTP, then index_ vcom_1 = '1'. **index_vcom_3** : if any register in vcom_3 group has been OTP, then index_ vcom_3 = '1'. **index_vcom_4** : if any register in vcom_4 group has been OTP, then index_ vcom_4 = '1'. **index_vcom_5** : if any register in vcom_5 group has been OTP, then index_ vcom_5 = '1'.									
Restriction	-									

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11.11. RD_OTP_IND_INF: Read OTP Index Information (Page9-R67h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	67
Parameter 1	**R**	-	-	-	index_vcomr_5	index_vcomr_4	index_vcomr_3	index_vcomr_2	index_vcomr_1	00
Description	This command is used to read OTP index information. **index_vcomr_1** : if any register in vcomr_1 group has been OTP, then index_vcomr_1 = '1'. **index_vcomr_2** : if any register in vcomr_2 group has been OTP, then index_vcomr_2 = '1'. **index_vcomr_3** : if any register in vcomr_3 group has been OTP, then index_vcomr_3 = '1'. **index_vcomr_4** : if any register in vcomr_4 group has been OTP, then index_vcomr_4 = '1'. **index_vcomr_5** : if any register in vcomr_5 group has been OTP, then index_vcomr_5 = '1'.									
Restriction	-									

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11.12. RD_OTP_IND_INF: Read OTP Index Information (Page9-R68h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	0	68
Parameter 1	**R**	index_dphy_rterm	index_cp_ver	index_dabc	index_04	index_vcom	index_pg8_dummy	index_vcom2	index_cp_id	00
Description	This command is used to read OTP index information. **index_cp_id** : if any register in cp_id group has been OTP, then index_cp_id = '1'. **index_vcom2** : if any register in vcom2 group has been OTP, then index_vcom2 = '1'. **index_pg8_dummy** : if any register in pg8_dummy group has been OTP, then index_pg8_dummy = '1'. **index_vcom** : if any register in vcom group has been OTP, then index_vcom = '1'. **index_04** : if any register in 04 group has been OTP, then index_04 = '1'. **index_dabc** : if any register in dabc group has been OTP, then index_dabc = '1'. **index_cp_ver** : if any register in cp_ver group has been OTP, then index_cp_ver = '1'. **index_dphy_rterm** : if any register in dphy_rterm group has been OTP, then index_dphy_rterm= '1'.									
Restriction	-									

11.13. RD_OTP_IND_INF: Read OTP Index Information (Page9-R69h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	1	69
Parameter 1	**R**	-	-	-	-	-	-	index_tch_sram	index_bist	00
Description	This command is used to read OTP index information. **index_tch_sram** : if any register in tch_sram group has been OTP, then index_tch_sram_id = '1'. **index_bist** : if any register in bist2 group has been OTP, then index_bist = '1'.									
Restriction	-									

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11.14. SET_REG_IND_INF: Set Register Index Information (Page9-R70h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	0	70
Parameter 1	**R/W**	index_gip	index_ce	index_le	index_dgamma_set	index_r_dgamma	index_g_dgamma	index_b_dgamma	index_dither	00
Description	This command is used to set register index information. **index_dither** : if any register in dither group has been written, then index_dither = '1'. **index_b_dgamma** : if any register in b_dgamma group has been written, then index_b_dgamma = '1'. **index_g_dgamma** : if any register in g_dgamma group has been written, then index_g_dgamma = '1'. **index_r_dgamma** : if any register in r_dgamma group has been written, then index_r_dgamma = '1'. **index_dgamma_set** : if any register in dgamma_set group has been written, then index_dgamma_set = '1'. **index_le** : if any register in le group has been wrritten, then index_le = '1'. **index_ce** : if any register in ce group has been written, then index_ce = '1'. **index_gip** : if any register in gip group has been written, then index_gip = '1'.									
Restriction	-									

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11.15. SET_REG_IND_INF: Set Register Index Information (Page9-R71h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	0	1	71
Parameter 1	**R/W**	index_mipi	index_pg7_dummy	index_corestate	index_regen	index_glue	index_framing	index_group	index_esd	00
Description	This command is used to set register index information. **index_esd** : if any register in esd group has been written, then index_esd = '1'. **index_group** : if any register in group has been written, then index_group = '1'. **index_framing** : if any register in framing group has been written, then index_framing = '1'. **index_glue** : if any register in glue group has been written, then index_glue = '1'. **index_regen** : if any register in regen group has been written, then index_regen = '1'. **index_corestate** : if any register in corestate group has been written, then index_corestate = '1'. **index_pg7_dummy** : if any register in pg7_dummy group has been written, then index_pg7_dummy = '1'. **index_mipi** : if any register in mipi group has been written, then index_mipi = '1'.									
Restriction	-									

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11.16. SET_REG_IND_INF: Set Register Index Information (Page9-R72h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	1	0	72
Parameter 1	**R/W**	index_sync_ctrl	index_repaire	index_sram_ctrl	index_clk_gen	index_tm	index_pw	index_gas	index_tsen	00
Description	This command is used to set register index information. **index_tsen** : if any register in tsen group has been written, then index_tsen = '1'. **index_gas** : if any register in gas group has been written, then index_gas = '1'. **index_pw** : if any register in pw group has been written, then index_pw = '1'. **index_tm** : if any register in tm group has been written, then index_tm = '1'. **index_ clk_gen** : if any register in clk_gen group has been written, then index_clk_gen = '1'. **index_ sram_ctrl** : if any register in sram_ctrl group has been written, then index_sram_ctrl = '1'. **index_ repaire** : if any register in repaire group has been written, then index_repaire = '1'. **index_ sync_ctrl** : if any register in sync_ctrl group has been written, then index_sync_ctrl = '1'.									
Restriction	-									

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11.17. SET_REG_IND_INF: Set Register Index Information (Page9-R73h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	0	1	1	73
Parameter 1	**R/W**	index_pw_tp	index_dphy_ana	index_misc1	index_misc2	index_misc3	index_misc4	index_vddd	index_vcipr	**00**
Description	This command is used to set register index information. **index_vcipr** : if any register in vcipr group has been written, then index_ vcipr = '1'. **index_vddd** : if any register in vddd group has been written, then index_vddd = '1'. **index_misc4** : if any register in misc4 group has been written, then index_misc4 = '1'. **index_misc3** : if any register in misc3 group has been written, then index_misc3 = '1'. **index_misc2** : if any register in misc2 group has been written, then index_misc2 = '1'. **index_misc1** : if any register in misc1 group has been written, then index_misc1 = '1'. **index_dphy_ana** : if any register in dphy_ana group has been written, then index_dphy_ana = '1'. **index_pw_tp** : if any register in pw_tp group has been written, then index_pw_tp = '1'.									
Restriction	-									

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11.18. SET_REG_IND_INF: Set Register Index Information (Page9-R74h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	0	0	74
Parameter 1	**R/W**	index_pw_ana1	index_dcdc_ana1	index_sd_ana	index_gam	index_sram	index_afe	index_trimming1	index_trimming2	00
Description	This command is used to set register index information. **index_trimming2** : if any register in trimming2 group has been written, then index_trimming2 = '1'. **index_trimming1** : if any register in trimming1 group has been written, then index_trimming1 = '1'. **index_afe** : if any register in afe group has been written, then index_afe = '1'. **index_sram** : if any register in sram group has been written, then index_sram = '1'. **index_gam** : if any register in gam group has been written, then index_gam = '1'. **index_sd_ana** : if any register in sd_ana group has been written, then index_sd_ana = '1'. **index_dcdc_ana1** : if any register in dcdc_ana1 group has been written, then index_dcdc_ana1 = '1'. **index_pw_ana1** : if any register in pw_ana1 group has been written, then index_ pw_ana1 = '1'.									
Restriction	-									

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11.19. SET_REG_IND_INF: Set Register Index Information (Page9-R75h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	0	1	75
Parameter 1	**R/W**	index_sacode	index_icinfo	index_gamma	index_sd_dmy	index_ana_dmy	index_trimming3	index_trimming4	index_trimmingtp	00
Description	This command is used to set register index information. **index_trimmingtp** : if any register in trimmingtp group has been written, then index_trimmingtp = '1'. **index_trimming4** : if any register in trimming4 group has been written, then index_trimming4 = '1'. **index_trimming3** : if any register in trimming3 group has been written, then index_trimming3 = '1'. **index_ana_dmy** : if any register in ana_dmy group has been written, then index_ana_dmy = '1'. **index_sd_dmy** : if any register in sd_dmy group has been written, then index_sd_dmy = '1'. **index_gamma** : if any register in gamma group has been written, then index_gamma = '1'. **index_icinfo** : if any register in icinfo group has been written, then index_icinfo = '1'. **index_sacode** : if any register in sacode group has been written, then index_sacode = '1'.									
Restriction	-									

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11.20. SET_REG_IND_INF: Set Register Index Information (Page9-R76h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	1	0	76
Parameter 1	**R/W**	-	-	-	index_vcom_5	index_vcom_4	index_vcom_3	index_vcom_2	index_vcom_1	00
Description	This command is used to set register index information. **index_vcom_1** : if any register in vcom_1 group has been written, then index_ vcom_1 = '1'. **index_vcom_2** : if any register in vcom_2 group has been written, then index_ vcom_1 = '1'. **index_vcom_3** : if any register in vcom_3 group has been written, then index_ vcom_3 = '1'. **index_vcom_4** : if any register in vcom_4 group has been written, then index_ vcom_4 = '1'. **index_vcom_5** : if any register in vcom_5 group has been written, then index_ vcom_5 = '1'.									
Restriction	-									

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11.21. SET_REG_IND_INF: Set Register Index Information (Page9-R77h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	0	1	1	1	77
Parameter 1	**R/W**	-	-	-	index_vcomr_5	index_vcomr_4	index_vcomr_3	index_vcomr_2	index_vcomr_1	00
Description	This command is used to set register index information. **index_vcomr_1** : if any register in vcomr_1 group has been written, then index_vcomr_1 = '1'. **index_vcomr_2** : if any register in vcomr_2 group has been written, then index_vcomr_2 = '1'. **index_vcomr_3** : if any register in vcomr_3 group has been written, then index_vcomr_3 = '1'. **index_vcomr_4** : if any register in vcomr_4 group has been written, then index_vcomr_4 = '1'. **index_vcomr_5** : if any register in vcomr_5 group has been written, then index_vcomr_5 = '1'.									
Restriction	-									

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11.22. SET_REG_IND_INF: Set Register Index Information (Page9-R78h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	1	0	0	0	78
Parameter 1	**R/W**	index_dphy_rterm	index_cp_ver	index_dabc	index_04	index_vcom	index_pg8_dummy	index_vcom2	index_cp_id	00
Description	This command is used to set register index information. **index_cp_id** : if any register in cp_id group has been written, then index_cp_id = '1'. **index_vcom2** : if any register in vcom2 group has been written, then index_vcom2 = '1'. **index_pg8_dummy** : if any register in pg8_dummy group has been written, then index_pg8_dummy = '1'. **index_vcom** : if any register in vcom group has been written, then index_vcom = '1'. **index_04** : if any register in 04 group has been written, then index_04 = '1'. **index_dabc** : if any register in dabc group has been written, then index_dabc = '1'. **index_cp_ver** : if any register in cp_ver group has been written, then index_cp_ver = '1'. **index_dphy_rterm** : if any register in dphy_rterm group has been written, then index_dphy_rterm = '1'.									
Restriction	-									

11.23. SET_REG_IND_INF: Set Register Index Information (Page9-R79h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	1	1	0	0	1	79
Parameter 1	**R**	-	-	-	-	-	-	index_tch_sram	index_bist	00
Description	This command is used to read OTP index information. **index_tch_sram** : if any register in tch_sram group has been written, then index_tch_sram_id = '1'. **index_bist** : if any register in bist2 group has been written, then index_bist = '1'.									
Restriction	-									

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12. Page A Command Description

12.1. SET_SD_INF: Set Source Driver Information (PageA-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	SD_SLPOUT	SD_GND	SD_EN	SD_OP_EN	-	SD_OUT_FIX	SD_POL_LEVEL_FIX[1:0]		B0
Description	This command is used to set source output information. **SD_POL_LEVEL_FIX[1:0]**: //2b'00: Normal POL operation //2b'01: Normal POL operation //2b'10: Fixed @ AD_SD_POL_LEAD/LAG=0 => SOUT (2n+1)= negative ; SOUT (2n+2)= positive //2b'11: Fixed @ AD_SD_POL_LEAD/LAG=1 => SOUT (2n+1)= positive ; SOUT (2n+2)= negative **SD_OUT_FIX**: //0: Normal operation //1: keep Latch data **SD_OP_EN**: SOURCE OP enable. //0: SD OP disable. //1: SD OP enable. **SD_ENX**: SOURCE with BIAS Enable. //0: SD diable. //1: SD Enable. (BIAS + OP enable) **SD_GND**: Source ground. //0: S[X] Shorted to GND. //1: S[X] Normal. **SD_SLPOUT**: SOURCE BIAS Enable. //0: Sleep-in. (Use to define POL @ slpin mode) //1: SD Sleep-out. (Only SD_BIAS_SEL enable)									
Restriction	-									

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12.2. SET_SD_INF: Set Source Driver Information (PageA-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	SD_OP_EN_SEL[7:0]								FF
Description	This command is used to set source output information. **SD_OP_EN_SEL[7:0]**: set source output channel. //00h: all source channel disable. //01h: 240RGB. (S1~S360 enable / S2041~S2400 enable) //03h: 320RGB. (S1~S480 enable / S1921~S2400 enable) //07h: 400RGB. (S1~S600 enable / S1801~S2400 enable) //0Fh: 480RGB. (S1~S720 enable / S1681~S2400 enable) //1Fh: 560RGB. (S1~S840 enable / S1561~S2400 enable) //3Fh: 640RGB. (S1~S960 enable / S1441~S2400 enable) //7Fh: 720RGB. (S1~S1080 enable / S1321~S2400 enable) //FFh: 800RGB. (S1~S24000 enable)									
Restriction	-									

12.3. SET_SD_INF: Set Source Driver Information (PageA-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	-	SD_SDT[6:0]							08

Description	This command is used to set source output information.
	
	SD_SDT[6:0]: HYNC to Hi-Z timing.
	//00h: 0ns.
	//01h: 25ns.
	//08h: 200ns.
	//09h: 250ns.
	//FEh: 1525ns.
	//FFh: 1550ns.									

Restriction	-									

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12.4. SET_SD_INF: Set Source Driver Information (PageA-R3Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	0	1	1	3B
Parameter 1	**R/W**	SD_CRT[7:0]								00

Description	This command is used to set source output information.
	
	SD_CRT[7:0]: Charge recycle timing@DOT inversion.
	//00h: 0ns.
	//01h: 50ns.
	//08h:400ns.
	//09h: 450ns.
	//FEh: 6250ns.
	//FFh: 6300ns.									

Restriction	-									

12.5. SET_SD_INF: Set Source Driver Information (PageA-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	SD_EQT[7:0]								0C

Description	This command is used to set source output information.
	
	SD_EQT[7:0]: Equalizer timing@DOT inversion.
	//00h: 0ns.
	//01h: 50ns.
	//0Ch:600ns.
	//0Dh: 650ns.
	//FEh: 6250ns.
	//FFh: 6300ns.									

Restriction	-									

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12.6. SET_SD_INF: Set Source Driver Information (PageA-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	SD_PCT[7:0]								0C

Description	This command is used to set source output information.
	
	SD_PCT[7:0]: Pre-charge timing@DOT inversion.
	//00h: 0ns.
	//01h: 50ns.
	//0Ch:600ns.
	//0Dh: 650ns.
	//FEh: 6250ns.
	//FFh: 6300ns.									

Restriction	-									

12.7. SET_SD_INF: Set Source Driver Information (PageA-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	SD_PCST[7:0]								18

Description	This command is used to set source output information.
	
	SD_PCST[7:0]: Pre-charge timing@COL inversion.
	//00h: 0ns.
	//01h: 50ns.
	//18h:1200ns.
	//19h: 1250ns.
	//FEh: 6250ns.
	//FFh: 6300ns.									

Restriction	-									

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12.8. SET_SD_INF: Set Source Driver Information (PageA-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3F
Parameter 1	**R/W**	SD_POLT[1:0]		SD_DRVT[1:0]		-	SD_OVDT[2:0]			53
Description	This command is used to set source output information. **SD_OVDT[2:0]**: SD overdrive timing. //00h: disable. //01h: 300ns. //03h:500ns. //04h: 600ns. //06h: 800ns. //07h: 900ns. **SD_DRVT[1:0]**: OP driving nonoverlap. //2b'00: 25ns. //2b'01: 50ns. //2b'10: 75ns. //2b'11: 100ns. **SD_POLT[1:0]**: POL switching time. //2b'00: 25ns. //2b'01: 50ns. //2b'10: 75ns. //2b'11: 100ns.									
Restriction	-									

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12.9. SET_SD_INF: Set Source Driver Information (PageA-R40h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	SD_TP_PS_SEL[1:0]		SD_CHOP_FRAME_OPT[1:0]		-	SD_CHOP_LINE_OPT[2:0]			10
Description	This command is used to set source output information. **SD_CHOP_LINE_OPT[2:0]**: SD frame line selection. //00h: Smart chopper. //01h: No line chop. //02h: 1-line. //03h: 2-line. //04h: 4-line. //05h: 1+2-line. //06h: Fixed AS_SD_CHOP=0. //07h: Fixed AS_SD_CHOP=1. **SD_CHOP_FRAME_OPT[1:0]**: SD frame chopper selection. //2b'00: No frame chop. //2b'01: 1-frame.. //2b'10: 2-frame. //2b'11: no-used. **SD_TP_PS_SEL[1:0]**: PS function selection. //2b'00: TP: All on. //2b'01: OP_EN=0. //2b'10: OP_EN=0 + BIAS_EN=0. //2b'11: OP_EN=0 + BIAS_EN=0 +GA_EN=0.									
Restriction	-									

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12.10. SET_SD_INF: Set Source Driver Information (PageA-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	SD_CP_TEST_EN	SD_CP_TEST_PHASE1[2:0]			-	-	SD_CP_TEST_PHASE2[1:0]		00
Description	This command is used to set source output information. **SD_CP_TEST_PHASE2[1:0]**: CP_PHASE2. //2b'00: reserved //2b'01: Measure S[4n+1]/S[4n+2]. //2b'10: Measure S[4n+3]/S[4n+4]. //2b'11: reserved **SD_CP_TEST_PHASE1[2:0]**: CP_PHASE1. //01h: Measure S[6n+1]/S[6n+2]. //02h: Measure S[6n+3]/S[6n+4]. //04h: Measure S[6n+5]/S[6n+6]. Others setting are reserved **SD_CP_TEST_EN**: CP_TEST enable for reduce channel measurement. //0: disable //1: enable									
Restriction	-									

12.11. SET_SD_INF: Set Source Driver Information (PageA-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	-	SD_BIAS_SEL_SRE[2:0]			-	SD_BIAS_SEL_N[2:0]			33
Description	This command is used to set source output information. **SD_BIAS_SEL_N[2:0]**: BIAS_SEL to SD_OP_BIAS under Normal Display. //00h: 0.25uA. //03h: 1.0uA. //07h: 2.0uA. **SD_BIAS_SEL_SRE[2:0]**: BIAS_SEL to SD_OP_BIAS under Slew Rate enhance. //00h: 0.25uA. //03h: 1.0uA. //07h: 2.0uA.									
Restriction	-									

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12.12. SET_SD_INF: Set Source Driver Information (PageA-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	-	SD_BIAS_SEL_TP[2:0]			-	SD_BIAS_SEL_VPH[2:0]			33
Description	This command is used to set source output information. **SD_BIAS_SEL_VPH[2:0]**: BIAS_SEL to SD_OP_BIAS under V-Proch. //00h: 0.25uA. //03h: 1.0uA. //07h: 2.0uA. **SD_BIAS_SEL_TP[2:0]**: BIAS_SEL to SD_OP_BIAS under TP. //00h: 0.25uA. //03h: 1.0uA. //07h: 2.0uA.									
Restriction	-									

12.13. SET_SD_INF: Set Source Driver Information (PageA-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	-	SD_COMPR[2:0]			-	SD_BIAS_SEL_SRE_TD[2:0]			70
Description	This command is used to set source output information. **SD_BIAS_SEL_SRE_TD[2:0]**: Slew-Rate Enhancement Timing. //00h: disable. //01h: 100ns. //02h: 200ns. //03h: 300ns. //04h: 400ns. //05h: 500ns. //06h: 600ns. //07h: 700ns. **SD_COMPR[2:0]**: Zero-R compensation for SD OP. //00h: Largest output MUX RES. (m=1) //07h: Smallest output MUX RES. (m=6)									
Restriction	-									

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12.14. SET_SD_INF: Set Source Driver Information (PageA-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	-	-	-	SD_PS_EN	SD_CTOT_TD[1:0]		SD_CTOT_LT[1:0]		15
Description	This command is used to set source output information. **SD_CTOT_LT[1:0]**: CTOT Latency. //2b'00: 0ns. //2b'01: 25ns. //2b'10: 50ns. //2b'11: 100ns. **SD_CTOT_TD[1:0]**: CTOT Delay Time. //2b'00: 0ns. //2b'01: 25ns. //2b'10: 50ns. //2b'11: 100ns. **SD_PS_EN**: Power-Saving (PS) enable signal @ TP. //0: disable //1: enable									
Restriction	-									

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12.15. SET_SD_INF: Set Source Driver Information (PageA-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	-	-	-	-	SD_TP_OUT_SEL[1:0]		SD_TP_OUT_DLY[1:0]		08
Description	This command is used to set source output information. **SD_TP_OUT_DLY[1:0]**: TP_OUTPUT_DELAY. //2b'00: 0ns. //2b'01: 1us. //2b'10: 2us. //2b'11: 4us. **SD_TP_OUT_SEL[1:0]**: TP_OUTPUT_SEL. //2b'00: GND. //2b'01: No-used. //2b'10: Keep. //2b'11: Hi-Z.									
Restriction	-									

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12.16. SET_SD_INF: Set Source Driver Information (PageA-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	-	SD_GAS_OPT	SD_DLY_SEL[1:0]		SD_DRV_SEL	SD_SLPIN_OPT	SD_V_PORCH_OPT[1:0]		20
Description	This command is used to set source output information. **SD_V_PORCH_OPT[1:0]**: SD output during the V_porch period. //2b'00: GND. //2b'01: No-used. //2b'10: Normal driving. //2b'11: Hi-Z. **SD_SLPIN_OPT**: SD OUTPUT@SLPIN. //0: GND. //1: Hi-Z. **SD_DRV_SEL**: OP driving selection. //0: S1=>S2=>S3=>S4=>S5=>S6. //1: S1/S2=>S3/S4=>S5/S6. **SD_DLY_SEL[1:0]**: CLK delay time. //2b'00: 0ns. //2b'01: 1ns. //2b'10: 2ns. //2b'11: 3ns. **SD_GAS_OPT**: SD OUTPUT@GAS. //0: GND. //1: Hi-Z									
Restriction	-									

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12.17. SET_SD_INF: Set Source Driver Information (PageA-R48h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	zdata[7:0]								80
Description	This command is used to set source output information. **zdata[7:0]**:									
Restriction	-									

12.18. SET_SD_INF: Set Source Driver Information (PageA-R49h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	sd_test_en	r_frm_line_tog_clr	invsel[2:0]			zz_2pl	zz_lr	zz_eo	13
Description	This command is used to set source output information. **zz_eo**: ziazag data maaping at odd or even line **zz_lr**: ziazag dummy channnel at left or right side. **zz_2pl**: 2 line ziazag enable **invsel[2:0]**: //00h: column //01h: 1L1D //02h: 1+2L2D //03h: 4L1D //04h: 2L1D //05h: Zigzag **r_frm_line_tog_clr**: //0: disable //1: enable **sd_test_en**: //0: disable //1: enable									
Restriction	-									

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12.19. SET_SD_INF: Set Source Driver Information (PageA-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	-	-	-	-	-	nbw_inv	data_clr_blk	sd_dls_always_on	00
Description	This command is used to set source output information. **sd_dls_always_on**: **data_clr_blk**: **nbw_inv**:									
Restriction	-									

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13. Page B Command Description

13.1. SET_DSTB_INF: Set Deep Standby Information (PageB-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	-	-	-	-	-	-	-	DSTBY	00
Description	This command is used to set IC deep standby mode. **DSTBY**: DSTBY enable. //0: disable. //1: enble.									
Restriction	-									

13.2. SET_INT_OTP_INF: Set Internal OTP Information (PageB-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	OTP_INT_VPP	VPP_VGHO_S[6:0]							8B
Description	This command is used to set OTP information. **VPP_VGHO_S[6:0]**: internal VPP voltage selection. //00h: 7.3V. //01h: 7.4V. //4Dh: 15V. //4Eh: 15.1V. //7Eh: 19.9V. //7Fh: 20V. **OTP_INT_VPP**: OTP program voltage internal/external selection. //0: external OTP voltage. //1: internal OTP voltage.									
Restriction	-									

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13.3. SET_INT_VCOM_INF: Set Internal VCOM Information (PageB-R33h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	1	33
Parameter 1	**R/W**	-	-	-	-	-	-	-	VCOM_SEL[8]	00
Parameter 2	**R/W**	VCOM_SEL[7:0]								64
Description	This command is used to set VCOM voltage information. **VCOM_SEL[8:0]**: VCOM voltage select when panel is normal scan. //000h: 0V. //001h: -0.01V. //064h:-1V. //065h:-1.01V. //12Ch: -3V. //12Dh: -3V. . . //1FFh: -3V. //1FFh: -3V.									
Restriction	-									

13.4. SET_INT_VCOM_INF: Set Internal VCOM Information (PageB-R3Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	0	3C
Parameter 1	**R/W**	-	-	-	-	-	-	-	VCOMR_SEL[8]	00
Parameter 2	**R/W**	VCOMR_SEL[7:0]								64
Description	This command is used to set VCOM voltage information. **VCOMR_SEL[8:0]**: VCOM voltage select when panel is reverse scan. //000h: 0V. //001h: -0.01V. //064h:-1V. //065h:-1.01V. //12Ch: -3V. //12Dh: -3V. . . //1FFh: -3V. //1FFh: -3V.									
Restriction	-									

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13.5. SET_INT_IGZO_VCOM_INF: Set Internal IGZO VCOM Information (PageB-R3Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	0	1	3D
Parameter 1	**R/W**	-	-	-	-	-	-	-	VCOM2_SEL[8]	00
Parameter 2	**R/W**	VCOM2_SEL[7:0]								64
Description	This command is used to set IGZO VCOM voltage information. **VCOM2_SEL[8:0]**: VCOM voltage select for IGZO panel. //000h: 0V. //001h: -0.01V. //064h:-1V. //065h:-1.01V. //12Ch: -3V. //12Dh: -3V. . . //1FFh: -3V. //1FFh: -3V.									
Restriction	-									

13.6. SET_POS_Gamma_INF: Set Positive Gamma Information (PageB-R3Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	0	3E
Parameter 1	**R/W**	-	GA_VHP0[6:0]							00
Parameter 2	**R/W**	-	GA_VHP1[6:0]							04
Parameter 3	**R/W**	-	GA_VHP2[6:0]							0B
Parameter 4	**R/W**	-	GA_VHP3[6:0]							11
Parameter 5	**R/W**	-	GA_VHP4[6:0]							19
Description	This command is used to set gamma information. **GA_VHP0[6:0]**: set positive gamma voltage for grayscale 255. (white pattern) **GA_VHP1[6:0]**: set positive gamma voltage for grayscale 254. **GA_VHP2[6:0]**: set positive gamma voltage for grayscale 252. **GA_VHP3[6:0]**: set positive gamma voltage for grayscale 250. **GA_VHP4[6:0]**: set positive gamma voltage for grayscale 247.									
Restriction	-									

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13.7. SET_POS_Gamma_INF: Set Positive Gamma Information (PageB-R3Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	1	1	1	1	3F
Parameter 1	**R/W**	GA_VMP0[7:0]								2A
Parameter 2	**R/W**	GA_VMP1[7:0]								42
Parameter 3	**R/W**	GA_VMP2[7:0]								4A
Parameter 4	**R/W**	GA_VMP3[7:0]								53
Parameter 5	**R/W**	GA_VMP4[7:0]								51
Parameter 6	**R/W**	GA_VMP5[7:0]								6E
Parameter 7	**R/W**	GA_VMP6[7:0]								77
Parameter 8	**R/W**	GA_VMP7[7:0]								81
Parameter 9	**R/W**	GA_VMP8[7:0]								93
Parameter 10	**R/W**	GA_VMP9[7:0]								95
Parameter 11	**R/W**	GA_VMP10[7:0]								A0
Parameter 12	**R/W**	GA_VMP11[7:0]								AA
Parameter 13	**R/W**	GA_VMP12[7:0]								BD
Parameter 14	**R/W**	GA_VMP13[7:0]								BB
Description	This command is used to set gamma information. **GA_VMP0[7:0]**: set positive gamma voltage for grayscale 243. **GA_VMP1[7:0]**: set positive gamma voltage for grayscale 235. **GA_VMP2[7:0]**: set positive gamma voltage for grayscale 227. **GA_VMP3[7:0]**: set positive gamma voltage for grayscale 215. **GA_VMP4[7:0]**: set positive gamma voltage for grayscale 203. **GA_VMP5[7:0]**: set positive gamma voltage for grayscale 179. **GA_VMP6[7:0]**: set positive gamma voltage for grayscale 155. **GA_VMP7[7:0]**: set positive gamma voltage for grayscale 127. **GA_VMP8[7:0]**: set positive gamma voltage for grayscale 99. **GA_VMP9[7:0]**: set positive gamma voltage for grayscale 75. **GA_VMP10[7:0]**: set positive gamma voltage for grayscale 51. **GA_VMP11[7:0]**: set positive gamma voltage for grayscale 39. **GA_VMP12[7:0]**: set positive gamma voltage for grayscale 27. **GA_VMP13[7:0]**: set positive gamma voltage for grayscale 19.									
Restriction	-									

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13.8. SET_POS_Gamma_INF: Set Positive Gamma Information (PageB-R40Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	0	40
Parameter 1	**R/W**	-	GA_VLP0[6:0]							59
Parameter 2	**R/W**	-	GA_VLP1[6:0]							5B
Parameter 3	**R/W**	-	GA_VLP2[6:0]							5F
Parameter 4	**R/W**	-	GA_VLP3[6:0]							65
Parameter 5	**R/W**	-	GA_VHN0[6:0]							00
Parameter 6	**R/W**	-	GA_VHN1[6:0]							04
Parameter 7	**R/W**	-	GA_VHN2[6:0]							0B
Parameter 8	**R/W**	-	GA_VHN3[6:0]							11
Parameter 9	**R/W**	-	GA_VHN4[6:0]							19
Description	This command is used to set gamma information. **GA_VLP0[6:0]**: set positive gamma voltage for grayscale 12. **GA_VLP1[6:0]**: set positive gamma voltage for grayscale 8. **GA_VLP2[6:0]**: set positive gamma voltage for grayscale 4. **GA_VLP3[6:0]**: set positive gamma voltage for grayscale 0. **GA_VHN0[6:0]**: set negative gamma voltage for grayscale 255. **GA_VHN1[6:0]**: set negative gamma voltage for grayscale 254. **GA_VHN2[6:0]**: set negative gamma voltage for grayscale 252. **GA_VHN3[6:0]**: set negative gamma voltage for grayscale 250. **GA_VHN4[6:0]**: set negative gamma voltage for grayscale 247.									
Restriction	-									

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13.9. SET_NEG_Gamma_INF: Set Negative Gamma Information (PageB-R41h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	0	1	41
Parameter 1	**R/W**	GA_VMN0[7:0]								2A
Parameter 2	**R/W**	GA_VMN1[7:0]								42
Parameter 3	**R/W**	GA_VMN2[7:0]								4A
Parameter 4	**R/W**	GA_VMN3[7:0]								53
Parameter 5	**R/W**	GA_VMN4[7:0]								51
Parameter 6	**R/W**	GA_VMN5[7:0]								6E
Parameter 7	**R/W**	GA_VMN6[7:0]								77
Parameter 8	**R/W**	GA_VMN7[7:0]								81
Parameter 9	**R/W**	GA_VMN8[7:0]								93
Parameter 10	**R/W**	GA_VMN9[7:0]								95
Parameter 11	**R/W**	GA_VMN10[7:0]								A0
Parameter 12	**R/W**	GA_VMN11[7:0]								AA
Parameter 13	**R/W**	GA_VMN12[7:0]								BD
Parameter 14	**R/W**	GA_VMN13[7:0]								BB
Description	This command is used to set gamma information. **GA_VMN0[7:0]**: set negative gamma voltage for grayscale 243. **GA_VMN1[7:0]**: set negative gamma voltage for grayscale 235. **GA_VMN2[7:0]**: set negative gamma voltage for grayscale 227. **GA_VMN3[7:0]**: set negative gamma voltage for grayscale 215. **GA_VMN4[7:0]**: set negative gamma voltage for grayscale 203. **GA_VMN5[7:0]**: set negative gamma voltage for grayscale 179. **GA_VMN6[7:0]**: set negative gamma voltage for grayscale 155. **GA_VMN7[7:0]**: set negative gamma voltage for grayscale 127. **GA_VMN8[7:0]**: set negative gamma voltage for grayscale 99. **GA_VMN9[7:0]**: set negative gamma voltage for grayscale 75. **GA_VMN10[7:0]**: set negative gamma voltage for grayscale 51. **GA_VMN11[7:0]**: set negative gamma voltage for grayscale 39. **GA_VMN12[7:0]**: set negative gamma voltage for grayscale 27. **GA_VMN13[7:0]**: set negative gamma voltage for grayscale 19.									
Restriction	-									

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13.10. SET_NEG_Gamma_INF: Set Negative Gamma Information (PageB-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	-	GA_VLN0[6:0]							5E
Parameter 2	**R/W**	-	GA_VLN1[6:0]							67
Parameter 3	**R/W**	-	GA_VLN2[6:0]							70
Parameter 4	**R/W**	-	GA_VLN3[6:0]							73
Description	This command is used to set gamma information. **GA_VLN0[6:0]**: set negative gamma voltage for grayscale 12. **GA_VLN1[6:0]**: set negative gamma voltage for grayscale 8. **GA_VLN2[6:0]**: set negative gamma voltage for grayscale 4. **GA_VLN3[6:0]**: set negative gamma voltage for grayscale 0.									
Restriction	-									

13.11. SET_GAMMA_REF_VOL: Set Gamma Reference Voltage (PageB-R43h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	1	43
Parameter 1	**R/W**	VGM_EN_M	VGM_EN_S	VGMP_S[5:0]						AB
Description	This command is used to set gamma positive reference voltage information. **VGMP_S[5:0]**: VGMP voltage selection. //00h: 3.05V. //01h: 3.1V. //2Bh:5.2V. //2Ch: 5.25V. //3Eh: 6.15V. //3Fh: 6.2V. **VGM_EN_S**: Slave VGM enable. //0: disable. //1: enable. **VGM_EN_M**: Master VGM enable. //0: disable. //1: enable.									
Restriction	-									

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13.12. SET_GAMMA_REF_VOL: Set Gamma Reference Voltage (PageB-R44h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	0	44
Parameter 1	**R/W**	-	-	VGMN_S[5:0]						2B
Description	This command is used to set gamma negative reference voltage information. **VGMN_S[5:0]**: VGMP voltage selection. //00h: -3.05V. //01h: -3.1V. //2Bh: -5.2V. //2Ch: -5.25V. //3Eh: -6.15V. //3Fh: -6.2V.									
Restriction	-									

13.13. SET_INT_VGHO_VOL: Set Internal VGHO Voltage (PageB-R45h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	0	1	45
Parameter 1	**R/W**	-	VGHO_S[6:0]							4D
Description	This command is used to set VGHO voltage information. **VGHO_S[6:0]**: VGHO voltage for normal display. //00h: 7.3V. //01h: 7.4V. //4Dh: 15V. //4Eh: 15.1V. //7Eh: 19.9V. //7Fh: 20V.									
Restriction	-									

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13.14. SET_INT_VGLO_VOL: Set Internal VGLO Voltage (PageB-R46h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	0	46
Parameter 1	**R/W**	-	VGLO_S[6:0]							2F
Description	This command is used to set VGLO voltage information. **VGLO_S[6:0]**: VGLO voltage for normal display. //00h: -5.3V. //01h: -5.4V. //2Fh: -10V. //30h: -10.1V. //7Eh: -17.9V. //7Fh: -18V.									
Restriction	-									

13.15. SET_INT_PWR_INF: Set Internal Power Information (PageB-R47h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	1	1	1	47
Parameter 1	**R/W**	-	-	-	-	VCL_EN	VCI_EN	VGL_EN	VGH_EN	0F
Description	This command is used to set internal power information. **VGH_EN**: VGH enable. //0: disable. //1: enable. **VGL_EN**: VGL enable. //0: disable. //1: enable. **VCI_EN**: VCIenable. //0: disable. //1: enable. **VCL_EN**: VCL enable. //0: disable. //1: enable.									
Restriction	-									

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13.16. SET_INT_VGH_VOL: Set Internal VGH Voltage (PageB-R48h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	0	48
Parameter 1	**R/W**	-	VGH_S[6:0]							2D
Description	This command is used to set VGH voltage information. **VGH_S[6:0]**: VGH voltage for normal display. //00h: 7V. //01h: 7.2V. //2Dh: 16V. //2Eh: 16.2V. //7Eh: 19.4V. //7Fh: 19.6V.									
Restriction	-									

13.17. SET_INT_VGL_VOL: Set Internal VGL Voltage (PageB-R49h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	0	1	49
Parameter 1	**R/W**	-	-	VGL_S[5:0]						19
Description	This command is used to set VGL voltage information. **VGL_S[6:0]**: VGL voltage for normal display. //00h: -7V. //01h: -7.2V. //19h: -12V. //1Ah: -12.2V. //7Eh: -15.4V. //7Fh: -15.6V.									
Restriction	-									

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13.18. SET_INT_PWR_INF: Set Internal Power Information (PageB-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	VGH_NC	VGL_NC	VGH_RT[2:0]			VGL_RT[2:0]			1B
Description	This command is used to set CP version information. **VGL_RT[2:0]**: VGL Ratio selection. //00h: VCL-AVDD. (1.5x) //01h: AVEE-AVDD. (2x) //02h: AVEE+VCL-AVDD. (2.5x) //03h: 2*AVEE-AVDD. (3x) (Default=-12V) //04h: inhibit. //05h: inhibit. //06h: inhibit. //07h: inhibit. **VGH_RT[2:0]**: VGH Ratio selection. //00h: VSP+VCI. (1.5x) //01h: 2*VSP. (2x) //02h: 2*VSP+VCI. (2.5x) //03h: 2*VSP-VSN. (3x) //04h: (VSP-VSN)+VSP+VCI. (3.5x) //05h: (VSP-VSN)+VSP+VSP. (4x) //06h: (2*VSP)-(2*VSN)+VCI. (4.5x) //07h: (2*VSP)-(2*VSN)+VSP. (5x) **VGL_NC**: VGL no clamp selection. //0: clamp //1: No clamp **VGH_NC**: VGH no clamp selection. //0: clamp //1: No clamp									
Restriction	-									

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13.19. SET_GAS_INF: Set GAS Information (PageB-R4Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	1	4B
Parameter 1	**R/W**	GAS_DEBOUNCE_SEL[1:0]		GAS_DELAY_OPT[1:0]		GAS_EN	GAS_VDDI_EN	GAS_VSP_EN	GAS_VSN_EN	0F
Description	This command is used to set GAS function information. **GAS_VSN_EN** : VSN GAS function enable. //0: disable. //1: enable. **GAS_VSP_EN** : VSP GAS function enable. //0: disable. //1: enable. **GAS_VDDI_EN** : VDDI GAS function enable. //0: disable. //1: enable. **GAS__EN** : GAS function enable. //0: disable. //1: enable. **GAS_DELAY_OPT[1:0]**: GAS delay time option from abnormal power-off detect to VGH/VGL discharge. //00h: 0ms. //01h: 10ms. //02h: 20ms. //03h: 30ms. **GAS_DEBOUNCE_SEL[1:0]**: GAS debounce time from GAS detected to internal GAS logic function operation. //00h: 0.2us. //01h: 1us. //02h: 3us. //03h: 4.5us.									
Restriction	-									

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13.20. SET_GAS_INF: Set GAS Information (PageB-R4Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	0	0	4C
Parameter 1	**R/W**	GAS_SEL_VDDI [1:0]		GAS_SEL_VSP[2:0]			GAS_SEL_VSN[2:0]			9B
Parameter 2	**R/W**	-	-	-	-	-	-	-	rst_hw_ gas	00
Description	This command is used to set GAS voltage detect range information. **GAS_SEL_VSN[2:0]**: GAS select bits for VSN adjust GAS trigger voltage. //00h: -2.8V. //01h: -3V. //02h: -3.2V. //03h: -3.4V. //04h: -3.6V. //05h: -3.8V. //06h: -4.0V. //07h: -4.2V. **GAS_SEL_VSP[2:0]**: GAS select bits for VSP adjust GAS trigger voltage. //00h: 2.8V. //01h: 3V. //02h: 3.2V. //03h: 3.4V. //04h: 3.6V. //05h: 3.8V. //06h: 4.0V. //07h: 4.2V. **GAS_SEL_VDDI[1:0]**: GAS select bits for VDDI adjust GAS trigger voltage. //00h: 1.2V. //01h: 1.3V. //02h: 1.4V. //03h: 1.5V. **rst_hw_gas**: GAS state(abnormal power off) only can be clear by hardware reset.									
Restriction	-									

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13.21. SET_IC_TSEN_INF: Set IC T-sen Information (PageB-R4Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	1	0	4E
Parameter 1	**R/W**	-	-	TEMP_VCOM_EN	TEMP_VCOMR_EN	TEMP_VGHO_EN	TEMP_VGLO_EN	TEMP_VGH_EN	TEMP_VGL_EN	3F
Description	This command is used to set IC T-sen control information. **TEMP_VGL_EN**: T-sen control enable for VGL. //0: disable. //1: enable. **TEMP_VGH_EN**: T-sen control enable for VGH. //0: disable. //1: enable. **TEMP_VGLO_EN**: T-sen control enable for VGLO. //0: disable. //1: enable. **TEMP_VGHO_EN**: T-sen control enable for VGHO. //0: disable. //1: enable. **TEMP_VCOMR_EN**: T-sen control enable for VCOMR. //0: disable. //1: enable. **TEMP_VCOM_EN**: T-sen control enable for VCOM. //0: disable. //1: enable.									
Restriction	-									

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13.22. SET_IC_TSEN_INF: Set IC T-sen Information (PageB-R4Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	1	1	4F
Parameter 1	**R/W**	THH[7:0]								96
Description	This command is used to set IC T-sen information. **THH[7:0]**: T-sen control for high level select. //00h : -90C //01h : -89C //55h : -5C //73h : 25C //78h : 30C //96h : 60C //FFh : 165C									
Restriction	-									

13.23. SET_IC_TSEN_INF: Set IC T-sen Information (PageB-R50h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	0	0	0	50
Parameter 1	**R/W**	THL[7:0]								55
Description	This command is used to set IC T-sen information. **THL[7:0]**: T-sen control for low level select. //00h : -90C //01h : -89C //55h : -5C //73h : 25C //78h : 30C //96h : 60C //FFh : 165C									
Restriction	-									

13.24. SET_IC_TSEN_INF: Set IC T-sen Information (PageB-R55h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	0	1	55
Parameter 1	**R/W**	-	-	TEMP_IS[5:0]						20
Description	This command is used to set IC T-sen information. **TEMP_IS[5:0]**: T-sen control for low level select.									
Restriction	-									

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13.25. SET_VCOM_TSEN_INF: Set VCOM T-sen Information (PageB-R56h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	0	56
Parameter 1	**R/W**	-	HT_VCOM_OS[6:0]							40
Description	This command is used to set VCOM voltage select at high temp. **HT_VCOM_OS[6:0]**: For Different TEMP to setting VCOM + Delta V. //00h: 0V. //01h: -0.01V. //40h:-0.4V. //41h:-0.41V. //7Eh: -1.27V. //7Fh: -1.28V.									
Restriction	-									

13.26. S SET_VCOM_TSEN_INF: Set VCOM T-sen Information (PageB-R57h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	0	1	1	1	57
Parameter 1	**R/W**	-	LT_VCOM_OS[6:0]							40
Description	This command is used to set VCOM voltage select at low temp. **LT_VCOM_OS[6:0]**: For Different TEMP to setting VCOM + Delta V. //00h: 0V. //01h: -0.01V. //40h:-0.4V. //41h:-0.41V. //7Eh: -1.27V. //7Fh: -1.28V.									
Restriction	-									

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13.27. SET_VCOMR_TSEN_INF: Set VCOMR T-sen Information (PageB-R58h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	0	58
Parameter 1	**R/W**	-	HT_VCOMR_OS[6:0]							40
Description	This command is used to set VCOMR voltage select at high temp. **HT_VCOMR_OS[6:0]**: For Different TEMP to setting VCOMR + Delta V. //00h: 0V. //01h: -0.01V. //40h:-0.4V. //41h:-0.41V. //7Eh: -1.27V. //7Fh: -1.28V.									
Restriction	-									

13.28. SET_VCOMR_TSEN_INF: Set VCOMR T-sen Information (PageB-R59h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	0	1	59
Parameter 1	**R/W**	-	LT_VCOMR_OS[6:0]							40
Description	This command is used to set VCOMR voltage select at low temp. **LT_VCOMR_OS[6:0]**: For Different TEMP to setting VCOMR + Delta V. //00h: 0V. //01h: -0.01V. //40h:-0.4V. //41h:-0.41V. //7Eh: -1.27V. //7Fh: -1.28V.									
Restriction	-									

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13.29. SET_VGHO_TSEN_INF: Set VGHO T-sen Information (PageB-R5Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	0	5A
Parameter 1	**R/W**	-	TSENH_VGHO_S[6:0]							4D
Description	This command is used to set VGHO voltage select at high temp. **TSENH_VGHO_S[6:0]**: VGHO voltage for normal display when TSEN enable. //00h: 7.3V. //01h: 7.4V. //4Dh: 15V. //4Eh: 15.1V. //7Eh: 19.9V. //7Fh: 20V.									
Restriction	-									

13.30. SET_VGHO_TSEN_INF: Set VGHO T-sen Information (PageB-R5Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	0	1	1	5B
Parameter 1	**R/W**	-	TSENL_VGHO_S[6:0]							4D
Description	This command is used to set VGHO voltage select at low temp. **TSENL_VGHO_S[6:0]**: VGHO voltage for normal display when TSEN enable. //00h: 7.3V. //01h: 7.4V. //4Dh: 15V. //4Eh: 15.1V. //7Eh: 19.9V. //7Fh: 20V.									
Restriction	-									

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13.31. SET_VGLO_TSEN_INF: Set VGLO T-sen Information (PageB-R5Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	0	5C
Parameter 1	**R/W**	-	TSENH_VGLO_S[6:0]							2F
Description	This command is used to set VGLO voltage select at high temp. **TSENH_VGLO_S[6:0]**: VGLO voltage for normal display when TSEN enable. //00h: -5.3V. //01h: -5.4V. //2Fh: -10V. //30h: -10.1V. //7Eh: -17.9V. //7Fh: -18V.									
Restriction	-									

13.32. SET_VGLO_TSEN_INF: Set VGLO T-sen Information (PageB-R5Dh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	0	1	5D
Parameter 1	**R/W**	-	TSENL_VGLO_S[6:0]							2F
Description	This command is used to set VGLO voltage select at low temp. **TSENL_VGLO_S[6:0]**: VGLO voltage for normal display when TSEN enable. //00h: -5.3V. //01h: -5.4V. //2Fh: -10V. //30h: -10.1V. //7Eh: -17.9V. //7Fh: -18V.									
Restriction	-									

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13.33. SET_VGL_TSEN_INF: Set VGL T-sen Information (PageB-R5Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	0	5E
Parameter 1	**R/W**	-	-	TSENH_VGL_S[5:0]						19
Description	This command is used to set VGL voltage select at high temp. **TSENH_VGL_S[6:0]**: VGL voltage for normal display when TSEN enable. //00h: -7V. //01h: -7.2V. //19h: -12V. //1Ah: -12.2V. //7Eh: -15.4V. //7Fh: -15.6V.									
Restriction	-									

13.34. SET_VGL_TSEN_INF: Set VGL T-sen Information (PageB-R5Fh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	1	5F
Parameter 1	**R/W**	-	-	TSENL_VGL_S[5:0]						19
Description	This command is used to set VGL voltage select at low temp. **TSENL_VGL_S[6:0]**: VGL voltage for normal display when TSEN enable. //00h: -7V. //01h: -7.2V. //19h: -12V. //1Ah: -12.2V. //7Eh: -15.4V. //7Fh: -15.6V.									
Restriction	-									

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13.35. SET_VGH_TSEN_INF: Set VGH T-sen Information (PageB-R60h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	0	60
Parameter 1	**R/W**	-	TSENH_VGH_S[6:0]							2D
Description	This command is used to set VGH voltage select at high temp. **TSENH_VGH_S[6:0]**: VGH voltage for normal display when TSEN enable. //00h: 7V. //01h: 7.2V. //2Dh: 16V. //2Eh: 16.2V. //7Eh: 19.4V. //7Fh: 19.6V.									
Restriction	-									

13.36. SET_VGH_TSEN_INF: Set VGH T-sen Information (PageB-R61h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	1	61
Parameter 1	**R/W**	-	TSENL_VGH_S[6:0]							2D
Description	This command is used to set VGH voltage select at low temp. **TSENL_VGH_S[6:0]**: VGH voltage for normal display when TSEN enable. //00h: 7V. //01h: 7.2V. //2Dh: 16V. //2Eh: 16.2V. //7Eh: 19.4V. //7Fh: 19.6V.									
Restriction	-									

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13.37. SET (PageB-R62h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	0	62
Parameter 1	**R/W**	H10000_TEMP_Q[7:0]								00
Description	This command is used to set **H10000_TEMP_Q[7:0]**: //00h : -90C //01h : -89C //55h : -5C //73h : 25C //78h : 30C //96h : 60C //FFh : 165C									
Restriction	-									

13.38. SET (PageB-R63h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	1	63
Parameter 1	**R/W**	H11111_TEMP_Q[7:0]								00
Description	This command is used to set **H11111_TEMP_Q[7:0]**: //00h : -90C //01h : -89C //55h : -5C //73h : 25C //78h : 30C //96h : 60C //FFh : 165C									
Restriction	-									

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13.39. SET_INT_OTP_PWR: Set Internal OTP Power (PageB-R66h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	0	66
Parameter 1	**R/W**	-	VPP_VGH_S[6:0]							07
Description	This command is used to set internal OTP voltage information. **VPP_VGH_S[6:0]**: OTP voltage selection. //00h: 7V. //01h: 7.2V. //07h: 8.4V. //08h: 8.6V. //7Eh: 19.4V. //7Fh: 19.6V.									
Restriction	-									

13.40. SET_VGH_RAT_TSEN_INF: Set VGH Ratio T-sen Information (PageB-R67h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	67
Parameter 1	**R/W**	-	TSENH_VGH_RT[2:0]			-	TSENL_VGH_RT[2:0]			33
Description	This command is used to set VGH ratio select at high and low temp. **TSENL_VGH_RT[2:0]**: VGH Ratio selection for TSENL. //00h: VSP+VCI. (1.5x) //01h: 2*VSP. (2x) //02h: 2*VSP+VCI. (2.5x) //03h: 2*VSP-VSN. (3x) //04h: (VSP-VSN)+VSP+VCI. (3.5x) //05h: (VSP-VSN)+VSP+VSP. (4x) //06h: (2*VSP)-(2*VSN)+VCI. (4.5x) //07h: (2*VSP)-(2*VSN)+VSP. (5x) **TSENH_VGH_RT[2:0]**: VGH Ratio selection for TSENH. //00h: VSP+VCI. (1.5x) //01h: 2*VSP. (2x) //02h: 2*VSP+VCI. (2.5x) //03h: 2*VSP-VSN. (3x) //04h: (VSP-VSN)+VSP+VCI. (3.5x) //05h: (VSP-VSN)+VSP+VSP. (4x) //06h: (2*VSP)-(2*VSN)+VCI. (4.5x) //07h: (2*VSP)-(2*VSN)+VSP. (5x)									
Restriction	-									

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13.41. SET_VGL_RAT_TSEN_INF: Set VGL Ratio T-sen Information (PageB-R68h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	0	0	68
Parameter 1	**R/W**	-	TSENH_VGL_RT[2:0]			-	TSENL_VGL_RT[2:0]			33
Description	This command is used to set VGL ratio select at high and low temp. **TSENL_VGL_RT[2:0]**: VGH Ratio selection for TSENL. //00h: VCL-AVDD. (1.5x) //01h: AVEE-AVDD. (2x) //02h: AVEE+VCL-AVDD. (2.5x) //03h: 2*AVEE-AVDD. (3x) (Default=-12V) //04h: inhibit. //05h: inhibit. //06h: inhibit. //07h: inhibit. **TSENH_VGL_RT[2:0]**: VGH Ratio selection for TSENH. //00h: VCL-AVDD. (1.5x) //01h: AVEE-AVDD. (2x) //02h: AVEE+VCL-AVDD. (2.5x) //03h: 2*AVEE-AVDD. (3x) (Default=-12V) //04h: inhibit. //05h: inhibit. //06h: inhibit. //07h: inhibit.									
Restriction	-									

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14. Page C Command Description

14.1. SET_DPHY_INF:Set DPHY Information(PageC-R31h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	0	1	31
Parameter 1	**R/W**	-	-	-	DPHY_HSRX_RT_HALF	-	DPHY_HSRX_PRE_ISEL	DPHY_HSRX_PST_ISEL	DPHY_HSRX_D2S_ISEL	A0
Description	This command is used to set DPHY information. **DPHY_HSRX_RT_HALF:** if=1,HS-RX terminator resistance value double. **DPHY_HSRX_PRE_ISEL:** HS-RX pre-amplifier bias current adjustment. **DPHY_HSRX_PST_ISEL:** HS-RX post-amplifier bias current adjustment. **DPHY_HSRX_D2S_ISEL:** HS-RX diff-to-single amplifier bias current.									
Restriction	-									

14.2. SET_DPHY_INF:Set DPHY Information(PageC-R32h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	0	1	1	0	0	1	0	32
Parameter 1	**R/W**	0	DPHY_HSRX_ALL_ISEL[2:0]			0	DPHY_FF_1ST_EN_SEL	DPHY_HSRX_STL[1:0]		12
Description	This command is used to set DPHY information. **DPHY_HSRX_ALL_ISEL[2:0]:** HS-RX pre-amp/post-amp/diff-to-single amp bias current adjustment **DPHY_FF_1ST_EN_SEL:** DPHY 1st flip-flop initial setting to HS_FF_EN or HS_SETTLE_DT (for debug or experiment only) **DPHY_HSRX_STL[1:0]:** Adjustment setting for HS-RX settling timer									
Restriction	-									

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14.3. SET_Analog_INF:Set Analog Information(PageC-R42h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	0	0	1	0	42
Parameter 1	**R/W**	BGR_TEST_SEL[1:0]		BGR_PRE_EN	BGR_BUF_EN	BGR_LPWG_OPTION	BGR_SHORT_EN	BGR_CAS_EN_M	BGR_CAS_EN_S	31
Description	This command is used to set Analog information. **BGR_TEST_SEL[1:0]:** Test port select **BGR_PRE_EN:** IOVCC_REG regulator enable //0: disable //1: enable **BGR_BUF_EN:** Buffer enable signal for BGR_VDDI_TOP //0: disable //1: enable **BGR_LPWG_OPTION:** VREF_BGR_1D8V always on option //0: follow LPWG_EN //1: always on **BGR_SHORT_EN:** Internal short enable **BGR_CAS_EN_M:** Cascade mode select, MASL=1=[1], MASL=0=[0] **BGR_CAS_EN_S:** Cascade mode select									
Restriction	-									

14.4. SET_Analog_INF:Set Analog Information(PageC-R4Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	0	4A
Parameter 1	**R/W**	THH_HYS_S[1:0]		THL_HYS_S[1:0]		TEMP_EN_BLANK	TEMP_DIM[2:0]			5B
Description	This command is used to set Analog information. **THH_HYS_S[1:0]:** 遲滯溫度選擇(Step=3℃) **THL_HYS_S[1:0]:** 遲滯溫度選擇(Step=3℃) **TEMP_EN_BLANK:** BLANK區開始做偵測 **TEMP_DIM[2:0]:** 當偵測到溫度進到另一組溫度區間，溫補項目每幾個Frame改變一個code，直到達到位置									
Restriction	-									

14.5. SET_Analog_INF:Set Analog Information(PageC-R4Bh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	0	1	1	4B
Parameter 1	**R/W**	TSEN_EN_M	TSEN_EN_S	TEMP_RST2CLK[1:0]		TEMP_DLY_LINE[3:0]				13
Description	This command is used to set Analog information. **TSEN_EN_M:** TSEN開關 **TSEN_EN_S:** TSEN開關 **TEMP_RST2CLK[1:0]:** RST至CLK delay((Step=8us) **TEMP_DLY_LINE[3:0]:** 決定TSEN_EN要在blank後delay幾個4us再發,default=12us)									
Restriction	-									

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14.1. SET_Analog_INF:Set Analog Information(PageC-R5Eh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	1	1	1	1	0	5E
Parameter 1	**R/W**	VCOM_GND_HIZ_GAS	VCOM2_EN	0	0	0	0	0	0	00
Description	This command is used to set Analog information. **VCOM_GND_HIZ_GAS:** Control VCOM output state during GAS area. 0: HIZ 1: GND **VCOM2_EN:** Control VCOM2 enable //0: disable //1: enable									
Restriction	-									

14.2. SET_Analog_INF:Set Analog Information(PageC-R60h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	0	60
Parameter 1	**R/W**	VCOM_EN_M	VCOM_EN_S	VCOM_SSTB_TIME	VCOM_TEST	VCOM_EXTSEL	VCOM_FBSEL	VCOM_FB_SHORT_M	VCOM_FB_SHORT_S	80
Description	This command is used to set Analog information. **VCOM_EN_M:** Master VCOM Enable **VCOM_EN_S:** Slaver VCOM Enable **VCOM_SSTB_TIME:** VCOM SST Time **VCOM_TEST:** VCOM Test Mode **VCOM_EXTSEL:** VCOM External Ref select **VCOM_FBSEL:** VCOM Feedback Select **VCOM_FB_SHORT_M:** Master VCOM Feedback Short **VCOM_FB_SHORT_S:** Slaver VCOM Feedback Short									
Restriction	-									

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14.3. SET_Analog_INF:Set Analog Information(PageC-R61h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	0	1	61
Parameter 1	**R/W**	VCOM_DA[1:0]		VCOM_HD_LPC[2:0]			VCOM_LPC[2:0]			C0
Description	This command is used to set Analog information. **VCOM_DA[1:0]:** VCOM Driving setting **VCOM_HD_LPC[2:0]:** VCOM High Driving Timing Setting **VCOM_LPC[2:0]:** VCOM Internal Slew Rate Enhancement									
Restriction	-									

14.4. SET_Analog_INF:Set Analog Information(PageC-R63h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	0	1	1	63
Parameter 1	**R/W**	VDDD_EXTRA_VDDD_EN	VDDD_VDDD_S[2:0]			VDDD_SRE_EN	VDDD_MILLER_EN	VDDD_POUT_S[1:0]		35
Description	This command is used to set Analog information. **VDDD_EXTRA_VDDD_EN:** Enable extra VDDD mode **VDDD_VDDD_S[2:0]:** Select IOVCC_VDDD's voltage (SLPIN).									
Restriction	-									

VDDD_VDDD_S[2:0]	VDDD voltage (SLPIN)
000	1.17
001	1.22
010	1.27
011	1.32
100	1.37
101	1.42
11x	1.45

VDDD_SRE_EN: Sunlight readability enhancement
VDDD_MILLER_EN: Miller compensation enable control
VDDD_POUT_S[1:0]: SLPIN VDDD_REG pass transistor size select

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14.5. SET_Analog_INF:Set Analog Information(PageC-R64h)

<table>
<tr><th>CMD/PAs</th><th>R/W</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>R/W</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>64</td></tr>
<tr><td>Parameter 1</td><td>R/W</td><td>0</td><td colspan="3">VDDD_VDDD_S_R[2:0]</td><td colspan="2">VDDD_ISEL_R[1:0]</td><td colspan="2">VDDD_NOUT_S_R[1:0]</td><td>3D</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to set Analog information.

VDDD_VDDD_S_R[2:0]: Select VCCD_VDDD's voltage (Right side)
<table>
<tr><th>VDDD_VDDD_S_R[2:0]</th><th>VDDD voltage
(SLPOUT @ right side)</th></tr>
<tr><td>000</td><td>1.17</td></tr>
<tr><td>001</td><td>1.22</td></tr>
<tr><td>010</td><td>1.27</td></tr>
<tr><td>011</td><td>1.32</td></tr>
<tr><td>100</td><td>1.37</td></tr>
<tr><td>101</td><td>1.42</td></tr>
<tr><td>11x</td><td>1.45</td></tr>
</table>
VDDD_ISEL_R[1:0]: Bias current selection (Right side)
VDDD_NOUT_S_R[1:0]: SLPOUT VDDD_REG pass transistor size select (Right side)</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
</table>

14.6. SET_Analog_INF:Set Analog Information(PageC-R65h)

<table>
<tr><th>CMD/PAs</th><th>R/W</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>R/W</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>65</td></tr>
<tr><td>Parameter 1</td><td>R/W</td><td>0</td><td>0</td><td>VDDD_VDDD_IOVCC_EN_L</td><td>VDDD_EXTRA_VDDD_EN_L</td><td>VDDD_DSI_ON_EN_L</td><td>VDDD_VDDD_IOVCC_EN_R</td><td>VDDD_EXTRA_VDDD_EN_R</td><td>VDDD_DSI_ON_EN_R</td><td>2D</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to set Analog information.

VDDD_VDDD_IOVCC_EN_L: Enable IOVCC_VDDD_GEN (left side)
VDDD_EXTRA_VDDD_EN_L: Enable extra VDDD mode (left side)
VDDD_DSI_ON_EN_L: Enable DSI VDDD mode (left side)
VDDD_VDDD_IOVCC_EN_R: Enable IOVCC_VDDD_GEN (right side)
VDDD_EXTRA_VDDD_EN_R: Enable extra VDDD mode (right side)
VDDD_DSI_ON_EN_R: Enable DSI VDDD mode (right side)</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
</table>

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14.7. SET_Analog_INF:Set Analog Information(PageC-R66h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	0	66
Parameter 1	**R/W**	0	VDDD_VDDD_S_L[2:0]			VDDD_ISEL_L[1:0]		VDDD_NOUT_S_L[1:0]		3D

Description

This command is used to set Analog information.

VDDD_VDDD_S_L[2:0]: Select VCCD_VDDD's voltage (left side)

VDDD_VDDD_S_L[2:0]	VDDD voltage (SLPOUT @ left side)
000	1.17
001	1.22
010	1.27
011	1.32
100	1.37
101	1.42
11x	1.45

VDDD_ISEL_L[1:0]: Bias current selection (left side)
VDDD_NOUT_S_L[1:0]: SLPOUT VDDD_REG pass transistor size select (left side)

Restriction: -

14.8. SET_Analog_INF:Set Analog Information(PageC-R67h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	0	1	1	1	67
Parameter 1	**R/W**	VGLO_DISCH_EN	VGLO_EN	VGHO_DISCH_EN	VGHO_EN	VCIPR_SLPIN_EN	VCIPR_S[2:0]			53

Description

This command is used to set Analog information.

VGLO_DISCH_EN: VGLO power off dischage to VGL during dischage range
//0: no discharge
//1: discharge to VGL
VGLO_EN: VGLO enable control
//0: disable
//1: enable
VGHO_DISCH_EN: GHO power off dischage to VGH during dischage range
//0: no discharge
//1: discharge to VGH
VGHO_EN: VGHO enable control
//0: disable
//1: enable
VCIPR_SLPIN_EN: VCIPR(OTP VDD2) enable control at sleep in
VCIPR_S[2:0]: VCIPR voltage select

Restriction: -

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14.9. SET_Analog_INF:Set Analog Information(PageC-R6Ah)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	1	0	1	0	1	0	6A
Parameter 1	**R/W**	CPCLK_TP[2:0]			CHG2AVDD_EH	VGL_POS_DISCH	VGH_POS_DISCH	VCI_NC	VCL_NC	7C
Description	This command is used to set Analog information. **CPCLK_TP[2:0]:** VGH/VGL pump clock frequency selction at TP mode **CHG2AVDD_EH:** VGL discharge control **VGL_POS_DISCH:** VGL discharge control **VGH_POS_DISCH:** VGH discharge control **VCI_NC:** VCI No Clamp mode selection **VCL_NC:** VCL No Clamp mode selectionn									
Restriction	-									

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15. Page D Command Description

15.1. SET_GMA_INF:Set Gamma Information(PageD-R4Ch)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	**R/W**	0	1	0	0	1	1	0	0	4C
Parameter 1	**R/W**	-	VGM_MC_EN	VGM_BUF_EN_SEL_TP[1:0]		VGM_TIE_SEL_M[1:0]		VGM_TIE_SEL_S[1:0]		34
Description	This command is used to set Gamma information. **VGM_MC_EN:** VGM_MC enable **VGM_BUF_EN_SEL_TP[1:0]:** VGM_BUF_EN_SEL @ TP //00: all disable // 01: VGMN_BUF enable // 10: VGMP_BUF enable // 11: all enable **VGM_TIE_SEL_M[1:0]:** Master setting //00: Hi-Z // 01: GND // 10: PW // 11: No-used **VGM_TIE_SEL_S[1:0]:** Slave setting //00: Hi-Z // 01: GND // 10: PW // 11: No-used									
Restriction	-									

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16. OTP Program Flow

16.1. ALL OTP program Flow

When OTP programming must display off and TP_RESX pull GND.

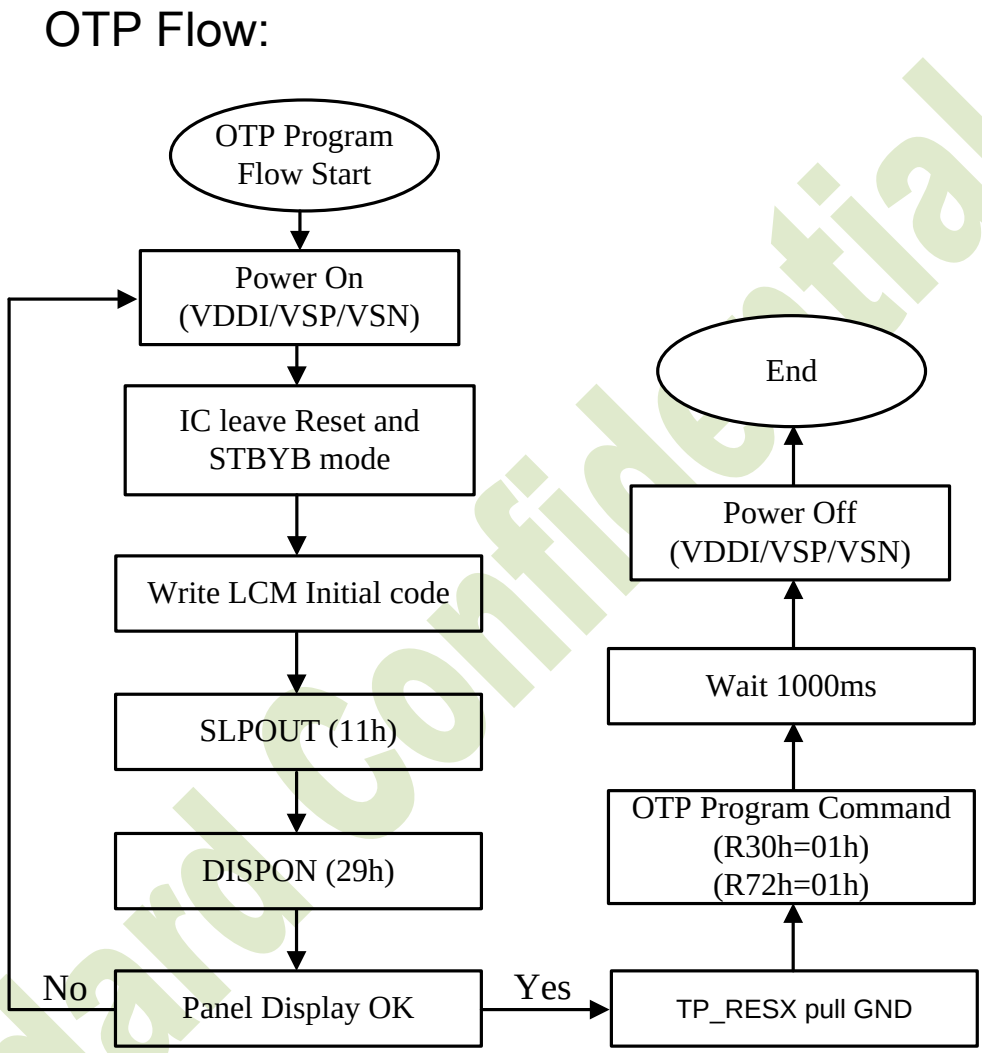


Figure 16.1: ALL OTP Program Flow

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16.2. OTP program for VCOM only

When OTP programming must display off and TP_RESX pull GND.

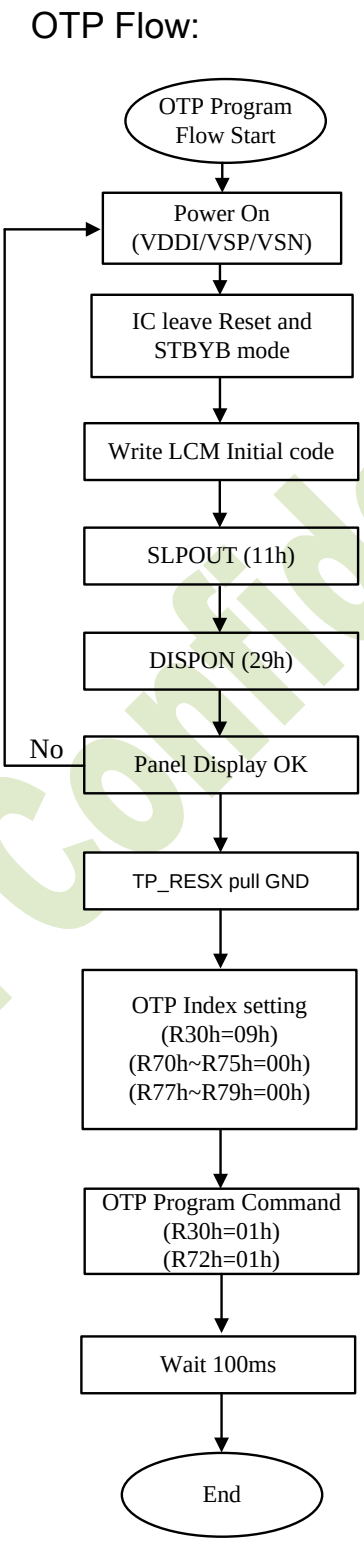


Figure 16.2: OTP Program for VOCM only

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